

**Android-based Inventory Management System for Tita Doc Pediatric and
FamiLee Dental Clinic**

**A Capstone Project
Presented to the Faculty of the
Information and Communications Technology Program
STI College Ortigas-Cainta**

**In Partial Fulfilment
of the Requirements for the Degree
Bachelor of Science in Information Technology**

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NOVEMBER 2025

ENDORSEMENT FORM FOR FINAL DEFENSE

TITLE OF CAPSTONE: **Android-based Inventory Management
System for Tita Doc Pediatric and FamiLee
Dental Clinic**

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for the degree Bachelor of Science in Information Technology
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Approval Sheet

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ABSTRACT

Title of capstone project: **Android-based Inventory Management System for Tita Doc Pediatric and FamiLee Dental Clinic**

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This study presents the development and evaluation of an Android-based Inventory Management System designed for Tita Doc Pediatric and FamiLee Dental Clinic to address the inefficiencies of manual inventory tracking. The clinic previously relied on handwritten notes and visual inspection, which often led to stock shortages, overstocking, and difficulty in monitoring expiration dates. To resolve these challenges, the developers designed a centralized digital system that enables real-time stock monitoring, expiry tracking, and basic purchase order management using Flutter and Supabase for cloud synchronization and secure data handling.

The system was developed using the Agile methodology, allowing iterative revisions and client feedback to improve usability and system accuracy. Core modules include inventory management, stock deductions, purchase order monitoring, activity logging, backup and restore, and role-based access control. Alpha testing based on ISO/IEC 25010 software quality standards yielded a 95% overall conformance level, 86% in functionality, 93% in reliability, 96% in usability, and 100% in both performance efficiency and security indicating that the system is of high quality and fully meets both functional and non-functional requirements.

The study concludes that the system significantly improves operational efficiency by minimizing human error, inventory updates, and ensuring timely replenishment of supplies. It provides a reliable and user-friendly solution tailored for small-scale dental clinics, demonstrating its potential for broader adoption in similar healthcare settings.

Table of Contents

	Page
Title Page	i
Endorsement form for Oral Defense	ii
Approval Sheet	iii
Acknowledgement	iv
Abstract	v
Table of Contents	vi
List of Figures	vii
<i>Figure 1. Agile Methodology</i>	24
<i>Figure 2. Capstone 1 Gantt Chart</i>	37
<i>Figure 3. Capstone 2 Gantt Chart</i>	38
<i>Figure 4.1. Owner & Admin Flowchart</i>	45
<i>Figure 4.2. Staff Flowchart</i>	46
<i>Figure 5. Login Flowchart</i>	47
<i>Figure 6. Inventory Management Flowchart</i>	48
<i>Figure 7. Purchase Order Flowchart</i>	49
<i>Figure 8. Stock Deduction Flowchart</i>	50

<i>Figure 9. Settings</i>	51
<i>Figure 10. Context Diagram</i>	52
<i>Figure 11. Top-Level Diagram</i>	53
<i>Figure 12. Middle-Level Diagram</i>	54
<i>Figure 13. Entity Relationship Diagram</i>	55
<i>Figure 14.1. Use-Case Owner</i>	56
<i>Figure 14.2. Use-Case Admin</i>	57
<i>Figure 14.3. Use-Case Staff</i>	58
<i>Figure 15.1. Login & Forgot Password</i>	80
<i>Figure 15.2. Email Confirmation & New Password</i>	80
<i>Figure 16. Dashboard</i>	81
<i>Figure 17.1. Inventory Owner</i>	82
<i>Figure 17.2. Add Supply</i>	83
<i>Figure 17.3. Archived Supplies</i>	83
<i>Figure 17.4. Add Category</i>	84
<i>Figure 18.1. Purchase Order Owner</i>	85
<i>Figure 18.2. Create Purchase Order</i>	86
<i>Figure 18.3. Add Supply in PO</i>	86
<i>Figure 19. Stock Deduction Owner</i>	87
<i>Figure 20. Activity Logs Owner</i>	88
<i>Figure 21.1. Settings Owner</i>	89

<i>Figure 21.2. Edit Profile</i>	90
<i>Figure 21.3. Employee List</i>	90
<i>Figure 21.4. Backup & Restore</i>	91
<i>Figure 22.1. Inventory Admin</i>	92
<i>Figure 22.2. Add Supply</i>	93
<i>Figure 22.3. Archived Supplies</i>	93
<i>Figure 22.4. Categories</i>	94
<i>Figure 23.1. Purchase Order Admin</i>	95
<i>Figure 23.2. Create Purchase Order</i>	96
<i>Figure 23.3. Add Supply in PO</i>	96
<i>Figure 24. Stock Deduction Admin</i>	97
<i>Figure 25. Activity Logs Admin</i>	98
<i>Figure 26.1. Settings Admin</i>	99
<i>Figure 26.2. Edit Profile</i>	100
<i>Figure 26.3. Employee List</i>	100
<i>Figure 27.1. Inventory Staff</i>	101
<i>Figure 27.2. Add Supply</i>	102
<i>Figure 28.1. Purchase Order Staff</i>	103
<i>Figure 28.2. Create Purchase Order</i>	104
<i>Figure 29. Stock Deduction Staff</i>	105
<i>Figure 30.1. Settings Staff</i>	106

<i>Figure 30.2. Edit Profile</i>	107
<i>Figure 31. Consent Form</i>	177
<i>Figure 32. Acknowledgement and Consent</i>	178
<i>Figure 33. Consent Form Ancheta</i>	179
<i>Figure 34. Acknowledgement and Consent Ancheta</i>	180
<i>Figure 35. Consent Form Navarez</i>	181
<i>Figure 36. Acknowledgement and Consent Navarez</i>	182
<i>Figure 37. Consent Form Nakpil</i>	183
<i>Figure 38. Acknowledgement and Consent Nakpil</i>	184
<i>Figure 39. Beta Test Consent Form</i>	185
<i>Figure 40. Acknowledgement and Consent Form</i>	186
<i>Figure 41. User Acceptance Letter</i>	187
<i>Figure 42.1. Accomplishment and Consultation Form (week 5)</i>	189
<i>Figure 42.2 Accomplishment and Consultation Form (Week 6)</i>	190
<i>Figure 42.3 Accomplishment and Consultation Form (Week 8)</i>	191
<i>Figure 42.4 Accomplishment and Consultation Form (Week 9)</i>	192
<i>Figure 42.5 Accomplishment and Consultation Form (Week 11)</i>	193
<i>Figure 42.6 Accomplishment and Consultation Form (Week 14)</i>	194
<i>Figure 42.7 Accomplishment and Consultation Form (Week 17)</i>	195
<i>Figure 42.8. Accomplishment and Consultation Form (Week 17)</i>	196
<i>Figure 42.9. Accomplishment and Consultation Form (Week 18)</i>	197

<i>Figure 42.10. Accomplishment and Consultation Form (Week 2)</i>	198
<i>Figure 42.11. Accomplishment and Consultation Form (Week 3)</i>	199
<i>Figure 42.12. Accomplishment and Consultation Form (Week 7)</i>	200
<i>Figure 42.13. Accomplishment and Consultation Form (Week 10)</i>	201
<i>Figure 42.14. Accomplishment and Consultation Form (Week 13)</i>	202
List of Tables	viii
<i>Table 1. Hardware Requirements</i>	39
<i>Table 2. Functionality Testing</i>	70
<i>Table 3. Reliability Testing</i>	71
<i>Table 4. Usability Testing</i>	72
<i>Table 5. Performance Efficiency Testing</i>	72
<i>Table 6. Security Testing</i>	73
<i>Table 7. Beta Testing</i>	78
Introduction	1
Project Context	1
Purpose and Description	3
Objectives	4
Scope and Limitations	5
Review of Related Literature/Studies/Systems	13

Methodology	24
Technical Background	33
Technologies to be Used	33
Calendar of Activities	37
Resources	39
Requirements Analysis	40
Requirements Documentation	41
Design of Software, System, Product, and/or Processes	45
Development	59
Results and Discussion	68
Testing	68
Description of Prototype	79
Implementation Plan	108
Conclusion	109
References	113
Appendices	116
Appendix A. Transcript of Interview	117

Appendix B. User's Guide	158
Appendix C. Resource Persons	171
Appendix D. Alpha Tester Curriculum Vitae	203
Appendix E. Personal Technical Vitae	207

INTRODUCTION

Project Context

Dental clinics have helped people care for their teeth for thousands of years. Ancient tools were used to treat tooth decay (Edwards & Long, 2023). In the Middle Ages, barbers and doctors pulled teeth, but dentistry became a real profession in the 18th century with new techniques from Pierre Fauchard (Morris, 2023). Today, modern clinics use advanced tools to treat patients. Dental clinics are important because oral health affects overall health. Poor dental care can lead to infections and gum disease. Regular checkups help catch issues early (American Academy of Pediatric Dentistry, 2024). However, many small clinics face problems managing supplies. Some still use paper or visual checks, which can cause errors, shortages, or overstocking. Tracking expiration dates is also a challenge, risking the use of expired materials (Cavity Free at Three, n.d.; Singh et al., 2022).

One example of a clinic facing these challenges in managing their inventory and the tracking of expiry is Tita Doc Pediatric and FamiLee Dental Clinic. Established in 2021, the clinic is a sole proprietorship specializing in pediatric dental care and treatment for patients with special needs. It operates with a small, dedicated team and is typically open from Tuesday to Sunday. On average, the clinic serves around six patients per day, but on particularly busy days, they can accommodate up to nineteen people. To handle this number of patients, the clinic prioritizes scheduled appointments while remaining flexible for walk-in patients, especially when someone has a dental emergency or when there are open time slots. The dental assistants are responsible for preparing and handling all materials, tools, and consumables needed for each procedure. They set up the procedure room before each appointment, check and adjust supplies throughout the day, and clean up after every patient. The dentist focuses solely on performing the procedures. Efficient coordination, experience with pediatric and special needs cases, and quick turnover between patients help the clinic maintain steady operations despite the limited staff.

Even though the clinic runs smoothly in many ways, it still faces some serious problems with how it manages its supplies. The clinic currently uses MyMedsPH, a cloud-based dental practice management system used by clinics to handle their operational and patient-related processes. Clinics that want to use MyMedsPH typically begin by inquiring through the platform's official website to learn about its subscription plans and account setup requirements. After coordinating with MyMedsPH, the clinic receives login credentials that allow access through their web portal. Once an account is activated, the clinic can use MyMedsPH to manage several tasks, including patient records, appointments, billing, and financial reports. It helps the clinic organize patient information, create invoices, manage treatment histories, and monitor income and expenses. Because access requires proper registration and a subscription, MyMedsPH is not publicly open and is intended specifically for licensed dental clinics and practitioners who coordinate directly with the provider. However, the system lacks in managing dental supplies such as recording stock levels of boxes, bottles, and vials, which are essential for the clinic's operations. Because of these, the staff still rely on traditional methods to manage inventory. Currently, the staff simply take what they need from the storage before and while doing procedures. They usually find out that something is running low only when they happen to check the stock or when they suddenly need an item and realize it is almost gone. This way of managing supplies often leads to mistakes. Sometimes, they end up buying too much, which takes up space and wastes money. Most of the time, the assistants just tell Dr. Lee what items are missing or need to be reordered. Occasionally, they write reminders on a small whiteboard hanging on a cabinet. To verify expiration dates, the client must examine the labels on each item or packaging unit, such as boxes of face masks or vials, before obtaining a general assessment of their condition. This takes time and can be easy to forget, especially when the clinic is busy. Since there is no organized system or central list of what supplies they have and when items expire, it is difficult to keep track of everything. This can lead to accidents like missing the chance to reorder items in time. It also means that dental assistants spend extra time checking supplies and giving updates, which takes time away from helping Dr. Lee and assisting patients. These ongoing issues show that the clinic needs a better and more effective way to manage its supplies, so it can continue providing safe and high-quality dental care.

To address these issues, this project proposes an application that will help Tita Doc Pediatric and FamiLee Dental Clinic manage their inventory more efficiently. The clinic currently handles over 600 supplies across different types of materials, and the existing tracking methods are time-consuming and burdensome for both the staff and Dr. Lee. The proposed application will serve as a centralized tool where staff can record and update inventory data after checking the supplies on hand. While they still need to physically verify the items in storage, the app simplifies how this information is recorded, organized, and reviewed. It provides a clear overview of stock levels, expiration dates, and ordered items, making it easier to monitor supplies and prevent issues such as overstocking, missed expirations, or unexpected shortages.

Purpose and Description

This project aims to develop an Android-based inventory management application specifically tailored to the needs of Tita Doc Pediatric and FamiLee Dental Clinic. The system is designed to improve the clinic's efficiency in handling supplies by organizing inventory data, monitoring expirations, and providing timely insights into stock availability. Through its user-friendly interface, the application allows staff to manage items more accurately and ensures that Dr. Lee has reliable, updated information for better decision-making. Android was chosen as the target platform due to its accessibility and familiarity among the clinic's staff.

The system keeps track of stock levels, restocking activities, and user actions to maintain proper organization and accountability. It is composed of three main modules. The Inventory Management module organizes and monitors supplies, including their quantity and expiration dates. The Purchase Order module manages the restocking process by allowing users to create and approve purchase orders, providing the clinic with an organized record of ordered items. The Stock Deduction module handles the reduction of inventory by recording deducted supplies.

The application will be developed using Flutter and Dart, with Visual Studio Code as the main code editor. Android Studio will be used mainly for testing the app on Android devices. Supabase will handle real-time storage and syncing of data. Before building the app, the design will be created in Figma to make sure it is easy to use and organized. By using this application, the clinic

will be able to avoid inventory issues and keep track of expiry dates. These improvements will lead to more reliable operations, helping the clinic continue delivering quality dental care.

Objectives

General Objective

- **To develop an Android-based application for Tita Doc Pediatric and FamiLee Dental Clinic to improve inventory handling.**

The system aims to provide an organized way for the clinic to manage its supplies. It will allow users to track stock levels, update inventory data, record ordered and deducted items. The application will also help prevent shortages and the use of expired materials by showing notifications for low-stock and near-expiry items. Overall, the system is designed to make inventory management more accurate and convenient for the clinic's daily operations.

Specific Objectives

- **To develop a centralized inventory that allows users to monitor, organize, and update clinic supplies.**

This objective focuses on creating a centralized system that provides users with a view of the clinic's supplies. The inventory will enable users to add, edit, and manage items while ensuring that updates are reflected across all connected accounts.

- **To implement a reporting feature that generates printable summaries of inventory data for documentation and external use.**

This objective focuses on developing a feature that allows users to generate comprehensive reports containing essential inventory information. The reports will be formatted and can be printed or exported for documentation and external reference.

- **To implement an expiration monitoring feature that tracks expiry dates, segments batches, and alerts users of expiring supplies.**

This objective focuses on developing an expiration monitoring feature that helps the clinic prevent the use of outdated materials. The system will record expiry dates for each batch of supplies, segment multiple expiry groups, and generate alerts for expiring or expired items. This ensures timely disposal or usage of supplies and reduces risks associated with expired medical materials.

Scope and Limitations

I. Scope

- **Login**

The user is required to enter their valid account credentials, specifically their given E-mail/username, and password to access the system.

- **Credentials Verification**

The system will verify if the username/e-mail and password entered are valid or not.

- **Forgot Password**

Allows the user to change their password once they have forgotten. This includes a 6-digit token that is sent to the user's email.

- **Remember Me**

Allows the user to select this option in order to avoid frequently logging in whenever the app is relaunched.

- **Dashboard Screen**

The dashboard provides an overview of inventory statuses such as In Stock, Low Stock, Out of Stock, Expiring, Expired, and purchase order statuses including Open, Partial,

Approval, and Closed which can be downloaded either .CSV or PDF file. The file includes tables with essential supply details such as Supply and Type Name, Stock quantity, Packaging Unit, Packaging Content, Brand name, Supplier name, Cost, Expiry Date, and Status. The purchase order table includes ID, supplier name, number of supplies received inside the PO, expiry date for each supply, date created, date received, PO status, receipt number, recipient, and remarks.

- **Inventory Check**

A graphical representation based on stock levels of the inventory. This contains the percentage of the following: In stock, Low stock, and Out of stock. It also displays the total supplies in stock, the number of low stocks, and the number of out of stock.

- **Expiry Alerts**

Displays the number of supplies needed to be consumed/disposed of. This is shown by the number of expiring and expired supplies.

- **Usage Speed**

Displays a summary of the speed of items that are consumed within a certain period. These interpretations include Fast, Normal, and Slow consumptions which helps the user to determine if their current stock is appropriate to their consumptions. Fast usage speed indicates that the stock for a specific supply is understocked and may pose running out of stock quickly. While slow means that the user overstocked a specific supply since its consumption is slow and may pose unused supplies. Normal usage indicates that the stock for that supply is neither overstocked or understocked. Once exported, all items that are deducted (weekly, monthly, yearly) are displayed.

- **Purchase Order Summary**

This displays an overview of the current purchase order statuses such as Open, Partial, Approval, and Closed.

- **Inventory Management**

The inventory management system allows users to efficiently track and manage clinic supplies.

- **Category Management**

Allows the owner/admin to organize items into predefined categories. The owner/admin can also add, edit, and archive categories as needed.

- **Edit item details**

Enables users to update item details such as name, specs, brand name, expiry date, category, image, cost, packaging unit, packaging content, and supplier name/company.

- **Add and Archive items**

Allows the user to add new inventory items and include the name, specs, brand name, expiry date, category, image, quantity, cost, packaging unit, packaging content, and supplier name/company. The owner/admin can archive items if they are no longer needed, which will then be stored in the archived supplies.

- **Archived Supplies**

This stores all archived items that allows the owner/admin to view, restore, and delete said items permanently; it can also be deleted after 30 days without restoration.

- **Expired Section**

This contains all expired items for tracking and disposing of said items.

- **Expiry Segmentation**

Items with the closest expiry date are displayed accordingly. Additionally, items stocked with different expiry dates are treated separately according to their date of expiration.

- **Purchase Orders**

This section enables users to track materials for restocking purposes. This compiles inventory materials that need to be restocked. Each PO contains supplies from a specific supplier. To update stock levels, the owner/admin must approve a purchase order; either complete or partially delivered.

- **Create Purchase Order**

This allows the user to list items that's going to be restocked. This includes information such as supplier name, items to be restocked, calculated price for each item, and total price. The details of a supply such as type, supplier, brand, cost, quantity of packaging unit, packaging content, and expiry date are editable. The expiry can be separated depending on different expiry dates given by the supplier on a specific supply.

- **PO List**

Shows the list of purchase orders according to their date created.

- **Flag Supplier Status**

This allows the user to change the supply status inside the PO from "Pending" to "Received." After all supplies are marked as "received," the user shall input the reference number of the issued receipt of their purchase, as well as the name of the user who flagged the supplier. An image shall also be uploaded containing the receipt mentioned. Additional remarks can also be included. The user can also partially receive

a supply; If a PO still contains pending supplies when saved, the PO will be flagged as “Partially Received.” The quantity of the supplies depends on what the user inputs; it will then update the supply stock based on that quantity once approved.

- **Approve PO**

Once all the supplies are flagged received or partially received, the owner/admin shall approve the PO and stock of the listed items will be updated.

- **Stock Deduction**

This section serves to reduce the stock quantity of supplies inside the inventory. This section follows First Expiry First Out (FEFO) to ensure the usage of supplies that are closer to expiration. Additionally, the user can select a purpose on why the supply/supplies will be deducted from the inventory. Once the submitted stock deduction list is evaluated and approved, the number of stocks from the listed supplies will update the inventory. The deduction log serves as the history of deducted supplies to ensure auditability of supply consumption.

- **Activity Logs**

This consists of all the activities a user has done that caused changes within the modules.

- **Settings**

This allows users to customize the application’s behavior to fit their preferences.

- **General**

This contains features relating to the appearance and notifications on the application.

- **Appearance**

Lets the user choose between light mode and dark mode.

➤ **Notifications**

This feature lets users choose if they want to receive notifications. The app will ask for permission to send alerts, like reminders when supplies are running low. Users can turn these notifications on or off anytime.

▪ **Inventory**

This includes toggling notifications for updating stocks, low stock, expired, near expiry, and out of stock alerts.

▪ **Purchase Order Approval**

This includes toggling notifications for POs that are ready for approval.

○ **Account Management**

This is the section where the user can edit their profile details. While the owner can manage all accounts. Admin accounts can also manage staff accounts but can't manage other admin accounts.

➤ **Owner Role**

Can access the inventory, purchase order, activity logs, stock deduction, staff list, and backup and restore while having full permissions in managing accounts.

➤ **Admin Role**

Can access the inventory, purchase order, activity logs, stock deduction, and staff list. This role can only manage staff accounts, and other admin accounts are not visible.

➤ **Staff Role**

It has access only to inventory, purchase order, and stock deduction. This role can only manage their own profile.

➤ **Edit Profile**

This module allows the user to edit their name, username, e-mail, and manage their password.

➤ **Staff List**

This allows the owner to manage all account roles. On the other hand, admin accounts can only manage staff accounts.

➤ **Reset Password**

This allows both owner and admin roles to reset the password on their responsible management permissions. Once an account's password is reset, their default password will now be "familee2021."

○ **System**

➤ **Backup and Restore**

This feature allows the owner to create a copy of the data in the application, store it in the database, and can also use this copy to restore lost data.

➤ **Automatic Backup**

When enabled, it automatically backs up data daily at 11:59PM.

➤ **Backup Files**

This contains the created backups and are stored based on their created dates from latest to oldest.

● **Search Bar**

Allows the user to enter keywords related to a specific module for quick retrieval of data or items.

- **Sort By**

This feature arranges items to A-z, Z-a, soonest or latest expiry dates, lowest to highest stock, or highest to lowest stock.

- **Filter**

This group displays items by category, brand name, supplier, stock status, expiry date, cost range, and per unit.

- **Log Out**

Allows the user to sign out of the application.

II. Limitations

- **Available to Android Devices Only**

The application will only be available on Android operating systems.

- **No Patient Data Integration**

The application operates independently and does not integrate with the clinic's existing system, MyMedsPH.

- **No Automated Reordering from Suppliers**

The app does not contact suppliers when a purchase order is made.

Review of Related Literature/Studies/Systems

International Sources

Inventory Control: Its principles and application (March 2022)

The essence of efficient inventory control is the availability of good quality stock, in right quantity, at right place and time and at right cost, which in turn is mandatory for smooth functioning and service delivery of any health care (Singh, A. et. al, 2022). Materials and equipment are essential components of the inputs used in health facilities. Material management is a scientific technique focused on planning, organizing, and controlling the flow of materials from initial purchase through internal operations to their distribution at the service point. According to the study, Inventory control is a method to maintain the stock level at its lowest possible without interference with supply. Inventory can be defined as a complete list of usable resources such as drugs, syringes etc. in stock. The diverse types, costs, and complexities of materials required in hospitals such as consumables (e.g., drugs, surgical sutures, medical gases), disposable items (e.g., syringes, needles, reagents, kits), semi-durables (e.g., surgical instruments, linen), and durables (e.g., machinery, lab instruments) pose a significant challenge for effective inventory control. Moreover, maintaining large stockpiles increases carrying costs and ties up funds that could otherwise be allocated more efficiently for other purposes. In essence, effective inventory control will not only provide efficient stock management but will also help clinics allocate the money in excess stocks to other needs. This study aligns with our objective to deliver smooth inventory management by determining the purpose of the system.

Through this, the proponents will focus on its objective to track inventory levels to lessen supply shortages. It is also essential to determine the categories that a clinic/hospital uses for its inventory, the complexity of these can pose a significant challenge in inventory control. Maintaining large stockpiles is also mentioned in the study, the system will help

prevent this challenge by helping the clinic to keep track of their stock and giving them the insight they need for better purchasing techniques.

How to Effectively Manage Dental Inventory Like A Pro (2022)

Inventory management begins by conducting a thorough audit on the current supplies. This involves counting every item on the shelf to check the quantity and conditions of the said items. Cataloging inventory provides a clear understanding of what's available and what may be of short supply. Moreover, tracking the usage of each product will help identify which items are consumed faster than others, while determining which ones that sit on the shelf longer; This prevents shortage of items and excessively overstocking. Lastly, cost analysis will help better understand the financial impact in developing cost-effective ordering strategy. Implementing a software for inventory management can play a significant role in utilizing cutting-edge applications, automate tracking, reduce human errors, and save time when doing manual stock recording. A software should be able to: track inventory levels, receive alerts when supplies run low, monitor expiration dates to reduce waste, and analyze data for better decision-making. According to Cavity Free At Three (n.d.), adopting a consistent system for organizing inventory is essential, this might include categorizing items by type, frequency of use, or supplier. Color-coding and labeling can also help the staff to quickly locate items, keeping the focus on patient care, not hunting for supplies. To promote best practices, the article highlights the importance of inventory management in enhancing patient care. By showing how effective inventory systems impact patient services, staff gain a clearer understanding of their broader role. Regular team meetings provide a forum to address challenges and exchange improvement ideas, reinforcing the value of a collaborative approach. Additionally, implementing a strategy of assigning inventory tasks based on each team member's strengths. Detail-oriented staff manage complex data entry, while those skilled in organization handle the physical setup of supplies. Clearly defining roles ensure that everyone knows their responsibilities in keeping the inventory well-maintained. This article covers what is included and the processes in the system that the proponents will develop. Inventory catalogue provides

better decision-making by tracking supply levels, this reduces and saves time when doing manual stock tracking. Supply alerts will also be integrated to the proposed system to provide quick insights on stock levels preventing shortage of items.

As the study listed, the proponents will integrate the following functions: track inventory levels, receive alerts when supplies run low, monitor expiration dates to reduce waste, and analyze data for better decision-making. Furthermore, the system will be designed to adapt to the evolving needs of dental clinics, ensuring it remains responsive and scalable. The goal is not only to improve operational efficiency but also to enhance patient satisfaction by ensuring that essential supplies are always available when needed, thereby supporting smooth, uninterrupted service delivery.

Enhancing Inventory Management in Dental Clinics through Agile Methodology: A Practical Approach (2024)

Dental clinics, as healthcare facilities specializing in oral and dental care, manage a wide range of inventory, including medications, medical equipment, disposable items, and general clinic supplies. This complex inventory often presents challenges in clinic operations. Common issues include a lack of transparency in stock tracking, difficulties in planning timely procurement, and the potential for lost information about expiration dates or item conditions. Given the continually evolving healthcare landscape and varying patient demands, effective inventory management is essential. This affects not only the clinic's operational and financial performance but also the quality of care delivered to patients. Efficient inventory management helps dental clinics prevent unexpected stockouts, make better use of resources, minimize waste and losses from unused items, and streamline service processes. Therefore, a deep understanding of the critical role inventory management plays is essential for developing a responsive, integrated, and sustainable system that ensures smooth operations and high-quality patient care. A major challenge is the lack of integrated and adaptable systems to support change. Dental clinics often struggle with accurately tracking stock, planning timely purchases, and maintaining

complete and reliable inventory records. These difficulties can disrupt smooth operations, reduce resource efficiency, and negatively impact the quality of patient care. Furthermore, the inflexibility of existing systems remains a persistent problem, making it hard for clinics to quickly adapt their inventory management processes in response to shifting needs or changing market conditions. To overcome these challenges, dental clinics need suitable and innovative inventory management solutions. The use of information technology and integrated inventory systems offers promising ways to streamline the process. By adopting systems that automate stock tracking, send alerts for low-stock items, and deliver precise, real-time inventory reports, clinics can manage their supplies more efficiently and proactively. As a result, dental clinics can boost operational performance, minimize the chances of stockouts or waste, and enhance patient satisfaction through more organized and higher-quality services.

Dental clinics face persistent challenges due to complex inventories ranging from medications to equipment and often struggle with transparent stock monitoring, timely procurement, and data reliability. These issues are compounded by outdated, inflexible systems that hinder the clinics' ability to adapt to changing demands. The study highlights innovative, technology-driven solutions, emphasizing integrated systems that automate stock tracking, provide real-time alerts, and generate accurate inventory reports. By implementing such agile systems, dental clinics can enhance resource utilization, reduce waste, prevent stockouts, and ultimately improve service delivery and patient satisfaction.

Requirements for Inventory Management System (August 2020)

The key inventory management system specifications involve providing a way to store, organize, manage and analyze inventory data (Jenkins, A. 2020). The system should have a user-friendly interface that doesn't require advanced training, support or documentation. Manual processes of business functions related to inventory management are eliminated. A dependable and secure database provides accurate, real-time information, ensuring efficient inventory management. High-performance capabilities enable swift and effective monitoring and control. The system is designed to be scalable, allowing administrators to

easily add software modules with minimal configuration. Finally, seamless software integrations and automated features help reduce the need for manual inventory updates or data entry, enhancing overall efficiency. Inventory management systems provide a range of features to help oversee, regulate, track, and plan inventory effectively. Inventory control is responsible for managing the products already in stock at the warehouse, playing a key role in supply chain management. It includes tools that categorize products by type, location, and SKU, audit data, generate real-time reports, and enable searching, filtering, and viewing of products. Inventory management, on the other hand, handles data from systems like inventory control and manages business processes before inventory arrives at the warehouse and how it is distributed to other destinations. This feature includes tools for managing multiple warehouse locations and integrating with other software or ERP platforms, such as generating digital invoices and purchase orders. Inventory tracking is essential for implementing inventory controls, automating manual tasks like generating tracking numbers for receipts or invoices. Integrating with third-party logistics providers and email solutions allows businesses to improve customer relations by providing real-time stock updates and shipping times. Inventory optimization takes planning to the next level by offering automated reports, inventory trends, and insights into supply chain changes, enabling better alignment of supply with demand. Lastly, inventory alerts trigger automated notifications for low stock levels and delays in shipments or the supply chain, helping businesses reduce waste, optimize inventory finances, and manage customer expectations, all while enhancing operational control across sales forecasting, materials planning, shipping logistics, and supplier management.

The proponents will focus on having a user-friendly interface and smooth software integrations to reduce manual work and increase automation. It will include reliable functions for inventory control, management, tracking, optimization, and alerts, all backed by a secure, high-performing database that provides real-time and accurate data. The system will also be scalable and modular, making it easy to add new features as the business grows. By combining what the proponents learned from research with practical features,

the goal is to improve overall efficiency, simplify inventory processes, and create a reliable system for managing and monitoring inventory effectively.

Inventory Management Practices in Private Healthcare Facilities in Nairobi County (February 2022)

Universal health coverage (UHC) is a key component of the United Nations sustainable development goals, and the private sector plays a crucial role in making UHC a reality, as it often serves as the main source of essential medicines for many populations (Karamshetty, V, et. al., 2022). However, in low- and middle-income countries (LMICs), private healthcare facilities frequently face challenges with medicine availability; they often run out of essential drugs while at the same time holding excessive stocks of less-needed medications, which can become obsolete. This points to weaknesses in their inventory control systems. To recommend effective solutions, it is essential to first gain a thorough understanding of the root causes behind these inventory management issues. This study seeks to explore a key question within the context of low- and middle-income countries; Why is there a gap between demand and supply in private healthcare facilities? Addressing this imbalance is a major development goal, and gaining a clear understanding of its root causes is essential for identifying effective solutions. Ensuring the availability of essential medicines affects not only health and well-being but also supports broader progress toward other sustainable development goals. Achieving universal health coverage can help reduce poverty and social inequality, while improvements in health are closely tied to economic growth and productivity. Imbalances between demand and supply, which result in overstocking or stockouts, can be managed by implementing an effective inventory control system. This involves the use of proper tools, policies, and procedures to guide ordering decisions. Improving inventory control requires maintaining accurate records of stock purchased, received, and sold, and using this data to inform decisions. These practices are important regardless of whether the system is manual or computerized. Some facility managers check their inventory levels in person only when they are about to place an order, choosing to restock the medicines they believe are most urgently needed. Others maintain an ongoing list of medicines they consider to be running low, which they

periodically turn into an order. The timing of these orders is often opportunistic, based on factors like available time, budget, and the immediate need to restock such as when a patient is waiting for a specific drug. The exact quantities to order are sometimes decided at the last minute.

The study mentioned that these facilities give little consideration to formal inventory control systems, which may explain why they frequently experience both overstocking and stockouts. Relying on restocking only when the need arises increases the risk of running out of essential medicines, potentially leading to interruptions in patient care and lower quality services. This reactive approach also makes it difficult to predict demand accurately, manage budgets effectively, and prevent medicine wastage due to expiration. Without a structured system in place, facilities are more likely to face inefficiencies that compromise their ability to provide consistent and reliable healthcare services.

Local Sources

Sales and Inventory Management System with Mobile Application

A problem this study addresses is that their client doesn't have any system that will support and save time for their daily operations and still struggle to keep track of their sales and profits on a daily basis. According to the study, to provide quality service to customers, a sales and inventory management system must eliminate the problems regarding manual inventory process and lack of documented reports. The system provides an efficient and convenient service for businesses, especially in taking orders, monitoring sales, and managing their inventory faster. Through this, a business will take part in the progress and innovation of technologies for the enhancement and better operation within the company.

Web Based Ordering and Online Appointment with Sales and Inventory for PrimozCycles

This study addresses the problem of the client not having a proper system to handle online orders, appointments, and inventory, which causes delays and errors in their daily operations. To provide better service to customers, the study highlights the need for a web-based system that can eliminate manual recording, reduce security risks, and improve inventory tracking. With the system in place, the business can manage sales and appointments more efficiently and focus more on growing their operations with the help of technology.

A Design of Web-Based Dental Information Management System with SMS Notification and Decision Support System for Idagdag Tooth Care Clinic

This study addresses the problem of the client relying on manual processes to manage patient records, schedule appointments, and send out notifications, leading to slower and less organized service. To improve patient care and clinic operations, the study proposes a web-based management system that simplifies record-keeping, automates appointment setting, and provides SMS reminders. Having an organized and accessible system allows the clinic to run smoother and adapt to more modern ways of delivering healthcare services.

Related Studies/Systems

Effective Dentist-Patient Communication

Effective communication between dentists and patients is crucial for delivering high-quality dental care. This interaction involves exchanging and interpreting information through both verbal and non-verbal channels. In today's global dental environment, communication can occur through traditional in-person consultations or remotely using telecommunication technologies (Ho, J. et. al., 2024). Clear and effective communication between dentists and patients plays a vital role in providing quality oral healthcare. It ensures that patients receive proper treatment, understand their diagnoses and available

options, and feel supported throughout their care. Conversely, poor communication can lead to misunderstandings, inadequate treatment, and a lack of support during the treatment process. Moreover, an article on "Managing Dental Practice Inventory and Supplies: A Comprehensive Guide" stresses the importance of having straightforward inventory management techniques that ensure essential supplies are always available. These methods are especially beneficial for small clinics, where simple systems are preferred to avoid overcomplicating operations while maintaining stock levels that meet the clinic's needs (My Dentist Hub, 2023). Furthermore, an article on "8 Best Practices for Clinic Inventory Management Systems" provides actionable insights for dental clinics to better manage their supplies. The study advocates for easy-to-implement inventory control systems that help avoid stockouts and excess inventory, fitting the needs of a straightforward system designed for small-scale practices (DrPro, 2025).

Material Costing for dental clinics

A significant portion of a dental practice's expenses is dedicated to the regular purchase of dental supplies. Essential items such as gloves, masks, and sterilization products are crucial for maintaining high hygiene standards and ensuring patient safety, as seen at SmileCraft Dental Studio. In addition to these daily necessities, consumables like fillings, crowns, and braces play a vital role in delivering quality dental treatments and supporting advanced procedures. These materials can account for approximately 10–15% of a clinic's total budget. Effective inventory management is essential to avoid both shortages and excess stock, with data-driven tracking systems helping practices optimize usage and reduce waste, sometimes cutting costs by up to 20%. Moreover, negotiating with suppliers for bulk purchases and fostering long-term partnerships can further lower expenses, potentially saving clinics 15% or more on equipment and supplies (Business Plans, 2025).

Inventory Management System: Features, Requirements & Modules (2024)

Features to consider in inventory management include Mobile Accessibility, User-Friendly Interface, Cloud Infrastructure, Reporting & Analytics, and Real-Time Reporting; These features streamline operations by providing valuable insights, improve efficiency, and

reduce manual efforts (Team Tranzact, 2024). Requirements include real-time updates to prevent stock-outs and overstock situations, scalability when the business expands and handling increased data volume.

Effective inventory management software assists business owners by automating inventory tracking, planning, and manufacturing processes (Stevens, C. 2022). Inefficient inventory management can result in overstocking, shortages, higher expenses, and interruptions in patient care. On the other hand, a well-structured inventory system ensures essential supplies are readily available, controls costs, and keeps operations running seamlessly (The Dental CFO, 2024).

By assessing your requirements, adopting a strong inventory management system, setting up an organized reordering process, performing regular audits, training staff, tracking expiration dates, optimizing storage, and keeping up with industry trends, you can ensure a well-managed and cost-effective inventory. This ensures continuous patient care while also enhancing the success and profitability of your dental practice. By applying these strategies, you can maintain an efficient inventory system.

Synthesis

The international resources emphasize that efficient inventory control is essential in healthcare settings to ensure the availability of necessary supplies at the right time, place, quantity, and cost (Singh, A. et al., 2022). Effective inventory management helps minimize waste, reduce costs, and improve service delivery by maintaining appropriate stock levels and avoiding overstocking. Studies highlight that automated inventory management systems improve accuracy, save time, track stock levels, monitor expiration dates, and generate insights for better decision-making (Cavity Free At Three, n.d.). Moreover, the quality of care delivered to patients is affected by efficient inventory management; This helps dental clinics prevent unexpected stockouts, make better use of resources, minimize waste and losses from unused items, and streamline service processes (Sihombing, D. J. C. 2024). Key system requirements include real-time tracking, user-friendly interfaces, database reliability, scalability, and seamless integration with other processes (Jenkins, A.

2020). Private healthcare facilities frequently face challenges with medicine availability; they often run out of essential drugs while at the same time holding excessive stocks of less-needed medications, which can become obsolete (Karamshetty, V. et al., 2022).

Local studies support these findings by demonstrating the value of integrating sales, inventory, and appointment systems into a single digital platform. For instance, web-based systems have proven effective for small businesses and clinics, improving operational efficiency and customer/patient engagement through features like online ordering, appointment setting, SMS notifications, and real-time stock tracking (Atanacio et al., 2023; Academia.edu, 2022). Additionally, related studies underline the importance of clear dentist-patient communication and simple inventory management practices for small clinics. Straightforward systems that prevent stockouts and overcomplication are favored for maintaining efficiency and service quality (Ho et al., 2024; My Dentist Hub, 2023; DrPro, 2025). Moreover, good inventory management also translates into better financial management and continuous, high-quality patient care (Stevens, 2022; The Dental CFO, 2024).

Overall, the reviewed literature and studies consistently advocate for automated, user-friendly, and integrated systems in managing inventory, significantly enhancing operational efficiency, cost-effectiveness, and patient satisfaction in healthcare settings. These studies further validate the impact of integrated systems, such as those that combine sales processing with inventory tracking and reporting functions. Regardless, the studies emphasize that integrating these systems will provide efficient, user-friendly digital systems is essential for improving healthcare operations, cost-effectiveness, and patient satisfaction.

METHODOLOGY

Agile Methodology

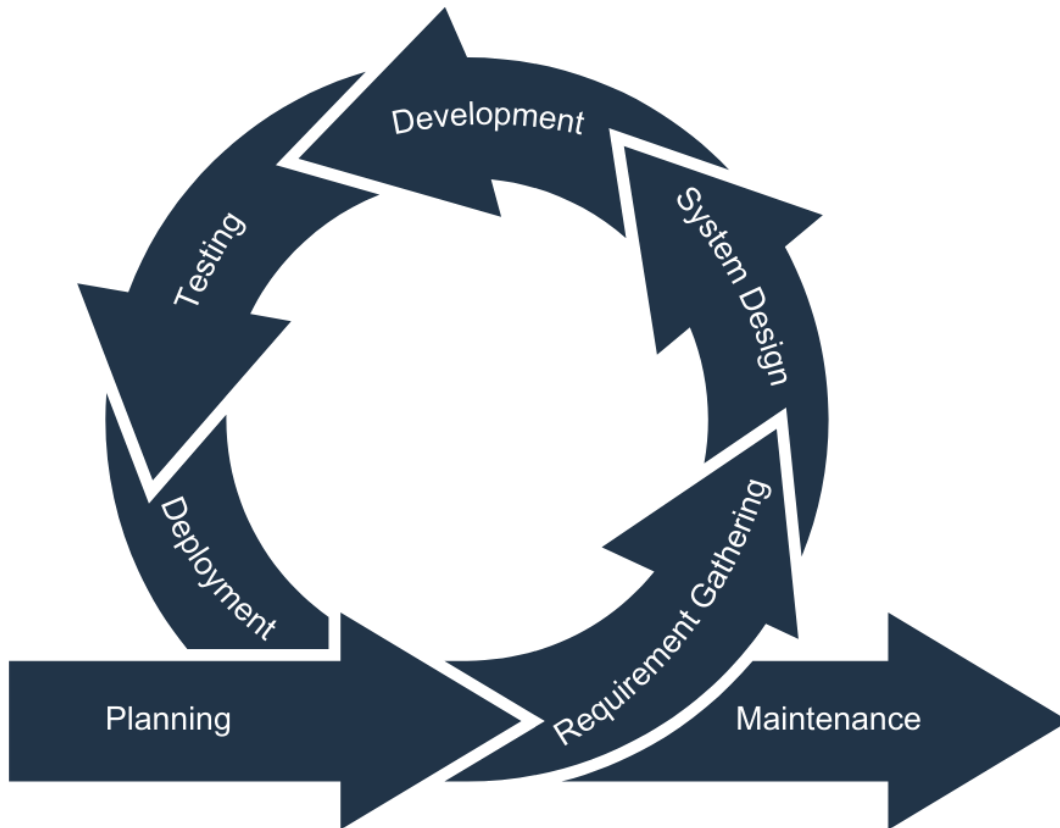


Figure 1. Agile Methodology

Figure 1 shows the Agile Methodology. In preparing the documentation for the application, the developers adopted the Agile approach by organizing the planning process into small, manageable sprints. After each sprint, the results, such as identified features, user requirements, and interface designs, were reviewed and refined based on feedback. This iterative process helped shape essential features like real-time stock monitoring, expiry date tracking, and structured user access for owner, admin, and staff. By following this method, the developers ensured that the system will be well aligned with the needs of Tita Doc Pediatric and FamiLee Dental Clinic, supporting more accurate inventory management.

Planning

This phase covers the discussion and brainstorming of the developers, wherein functions and key features are defined. By determining the application's scope, target users, and key features, the developers were able to establish a clear vision for the project. The application was purposefully limited to focusing solely on inventory, ensuring a smooth development process and addressing the most critical needs of the clinic.

During this phase, the developers explored various system ideas, including inventory management, appointment scheduling, and school navigation. After evaluating multiple fields such as education, OB-GYN, and dental, the project was focused on the dental sector due to its evident need for improved inventory management. The developers finalized an Android-based system for dental clinics, centered on real-time stock monitoring, expiry tracking, a basic purchase order feature for managing restocks, and user activity logging to enhance supply management and daily operations.

Requirements Gathering

This phase of the project focuses on identifying the specific needs of the dental clinic to ensure that the system supports their daily operations. The phase emphasizes close collaboration with the client and developers to understand their existing challenges in inventory management, allowing the system to cater to their needs. The developers prepared interview questions and conducted discussions with the clinic owner to gain detailed insights into their current problems and key processes.

By conducting these consultations, the developers identified key system features such as stock level alerts and expiry management. In addition, features such as a basic purchase order module were included to help staff manage restocking more effectively, along with activity logging to improve transparency. User roles were defined to establish a structured

and secure access system, assigning specific functions to Admin and Staff roles. By outlining these critical features and access levels, this phase lays the foundation for a system that is both functional and user-friendly.

Functional Requirements

1.1 Dashboard

REQ001. The dashboard shall show an overview of inventory statuses.

REQ002. The dashboard shall show the current status of purchase orders.

REQ003. The dashboard shall display the number of expiring and expired items.

REQ004. The dashboard shall show a graph of usage speed items by selected time period.

REQ005. The owner or admin shall be able to download reports in CSV or PDF format.

1.2 Inventory

REQ006. The user shall be able to add new items to the inventory.

REQ007. The user shall be able to edit existing item details.

REQ008. The owner or admin shall be able to manage item categories.

REQ009. The owner or admin shall be able to archive, restore, or permanently delete items and categories.

REQ010. The system shall delete archived items after 30 days.

REQ011. The user shall be able to view expired items in a separate section.

REQ012. The system shall display items according to their expiry dates.

REQ013. The system shall follow the First-Expiry-First-Out method for expirable items.

1.3 Purchase Orders

REQ014. The user shall be able to create new purchase orders.

REQ015. The user shall be able to edit purchase order details before approval.

REQ016. The system shall allow multiple expiry dates for the same item within a purchase order.

REQ017. The user shall be able to view all purchase orders by creation date.

REQ018. The user shall be able to update supplier status in a purchase order.

REQ019. The user shall be able to attach receipts and add remarks to a purchase order.

REQ020. The system shall mark incomplete deliveries as partially received.

REQ021. The owner or admin shall approve completed or partially received purchase orders.

REQ022. The system shall update inventory stocks after purchase order approval.

1.4 Stock Deduction

REQ023. The user shall be able to deduct supplies.

REQ024. The system shall record all deducted supplies in a deduction log.

1.5 Activity Log

REQ025. The system shall record all user activities that change data or settings.

REQ026. The owner or admin shall be able to view the list of all activity logs.

1.6 Settings

REQ027. The user shall be able to change the application theme between light and dark mode.

REQ028. The user shall be able to enable or disable inventory notifications.

REQ029. The user shall be able to enable or disable purchase order notifications.

REQ030. The user shall be able to edit their profile information.

REQ031. The owner shall be able to manage all user roles.

REQ032. The owner shall be able to create new admin and staff accounts.

REQ033. The admin shall be able to manage staff accounts only.

REQ034. The owner or admin shall be able to reset passwords of managed accounts.

REQ035. The staff shall be able to edit only their own account information.

REQ036. The owner shall be able to create data backups of the system.

REQ037. The system shall automatically perform daily backups when enabled.

REQ038. The owner shall be able to restore the system using saved backup files.

1.7 Alerts and Notifications

REQ039. The system shall notify users when items are low, out of stock, expiring, or expired.

REQ040. The system shall notify users when purchase orders are ready for review or approval.

REQ041. The system shall send alerts and notifications based on user preferences.

1.8 Authenticate User

REQ042. The owner shall log into the application by providing an e-mail/username and password.

REQ043. The admin shall log into the application by providing an e-mail/username and password.

REQ044. The staff shall log into the application by providing an e-mail/username and password.

REQ045. The system shall verify user credentials.

REQ046. The system shall provide a “Remember Me” option for login.

REQ047. The user shall be able to change their password using the “Forgot Password”.

Non-Functional Requirements

2.1 Operational Requirements

REQ048. The application shall be accessible on Android devices.

REQ049. The application shall function correctly both online and offline depending on available features.

REQ050. The application shall integrate with Supabase services for real-time data handling and cloud integration.

2.2 Performance Requirements

REQ051. The application shall respond to user interactions within 2 seconds under normal operating conditions.

REQ052. The application shall synchronize data with the cloud in real-time when internet connectivity is available.

REQ053. The application shall store and manage data for inventory items without performance degradation.

REQ054. The application shall support up to 10 concurrent users accessing or syncing data without crashing or excessive lag.

2.3 Security Requirements

REQ055. The application shall require all users to log in using secure credentials.

REQ056. The application shall restrict access to features and data based on user roles (admin vs. staff).

REQ057. The application shall allow admin to reset password securely if they forget it.

REQ058. The application shall encrypt the .JSON file for backups.

2.4 Cultural and Political Requirements

REQ059. The application shall be in English to ensure accessibility for users in the clinic.

REQ060. The application shall avoid using any culturally offensive symbols, terminology, or imagery that could cause harm or discomfort to users.

REQ061. The application shall comply with the Data Privacy Act of 2012 (Republic Act No. 10173), ensuring that all user and clinic-related data is handled with the highest standards of privacy and security.

System Design

This phase focuses on implementing the gathered requirements into a structured and visual representation of the application's layout and architecture. Following the client's request from the project, ensuring that the UI/UX design is intuitive, user-friendly, and the system's structure is prepared for efficient development. To visualize the system's functionalities, the developers created detailed wireframes using Figma, serving as a prototype to visualize how the user will interact with the application.

Alongside wireframing, UI mockups were designed to reflect the visual aesthetics and usability of the final product, helping to align the design with user expectations and preferences to improve system accessibility. Database structure planning is also conducted to organize and visualize essential data flow. Additionally, this phase will also include Client approval in cases of revisions and changes in the UI.

Development

This phase focuses on translating the system design into a fully functional application that meets the specific needs of FamiLee dental clinic. The development phase aims to build a reliable system by combining both frontend and backend development, ensuring that all UI/UX and backend logic work harmoniously. The backend logic will be developed to handle core functionalities and processes, while the Supabase database will be integrated to provide real-time data storage and retrieval for inventory records.

Alongside backend integration, the frontend interfaces will be designed and developed for different user roles such as Owner, Admin, and Staff, ensuring that each user type has access to features relevant to their responsibilities. Key system features that will be

implemented during this phase include low-stock alerts, expiry management, and restocking.

Testing

This phase focuses on ensuring that the system performs as intended, is free of major bugs, and fully meets the defined requirements. The developers will be conducting unit testing to verify that each individual module, such as adding inventory items or triggering stock alerts functions correctly in isolation. Once individual components are verified, integration testing will be performed to ensure smooth interaction between different system modules.

To validate the system from the end user's perspective, a User Acceptance Test (UAT) will be carried out, allowing clinic staff to assess the system's usability and effectiveness in real-world scenarios. Lastly, performance testing will be conducted to evaluate how well the system performs under expected workloads, while security testing ensures that sensitive clinic data are properly protected.

Deployment

This phase focuses on transitioning the developed system into FamiLee Dental Clinic's live environment, marking the official release of the application. The developers will ensure that the deployment process will be smooth and well-coordinated, allowing the clinic to begin using the system effectively. The deployment involves the official installation of the system on the clinic's devices, ensuring compatibility and proper configuration for daily use.

In preparation for full adoption, the developers will conduct data migration to transfer existing records into the new system without data loss or inconsistencies. To support the clinic staff, user tutorials shall be carried out in the initialization of the application to familiarize them with system features, workflows, and best practices. Once all components

are in place and users are confident in navigating the system, the developers will proceed with the Go Live stage, officially launching the system for operational use.

Maintenance

The maintenance phase focuses on providing ongoing support to ensure the system remains reliable, up-to-date, and responsive to the evolving needs of FamiLee Dental Clinic. After deployment, the developers will be committed to addressing any unexpected issues or bugs that may arise during actual use. Regular monitoring will be conducted to ensure that the system performs optimally and that users experience minimal disruptions.

Technical Background

- **Technologies to be Used**

- **Figma**

- An online design tool that helps in creating user interface (UI) layouts and prototypes. Before starting the development of the app, the developers will use Figma to design wireframes that show how the app screens will look. This ensures that the application is easy to navigate and meets the users' needs.

- **Android Studio**

- An integrated development environment (IDE) commonly used for building Android applications. In this project, the developers will primarily use Android Studio for its built-in emulator feature, which allows the application to be tested on different screen sizes and Android devices without the need for a physical phone.

- **Flutter**

- An open-source software development toolkit created by Google. It allows developers to build applications for both Android and iOS using a single codebase;

Which in this case, we will focus on developing android devices. This makes the development process faster and more efficient. The developers will use Flutter to design and build the user interface of the application, ensuring that it looks and works well across screen sizes.

- **Dart Programming Language**

The programming language used in Flutter. It is object-oriented and easy to understand, making it suitable for mobile app development. The developers will use Dart to write the logic behind the app's features use.

- **TypeScript**

TypeScript was used specifically to create the automatic daily backup system that runs in the cloud. The developers used it to write a serverless function that creates encrypted backups of all our inventory data every day at 11:59 PM. The function automatically saves data from all our database tables and stores it securely in cloud storage. TypeScript's error-checking features helped ensure the backup system works reliably without manual intervention. This automated system helps protect the business' data by creating regular backups that can be restored if needed.

- **HTML**

HTML was used to design the password reset email template in the Supabase dashboard. The developers used HTML to create a professional-looking email that users receive when they request to reset their password. The template includes proper formatting, company branding, and clear instructions for users to follow. HTML helped make sure the email displays correctly across different email programs and devices, providing a consistent user experience when users need to reset their passwords.

- **Visual Studio Code**

A lightweight code editor that supports many programming languages. The developers will use it as the main tool for writing and editing the source code of the application, especially when working with Flutter and Dart. It helps organize the code and makes it easier to detect and fix errors.

- **Canva**

An online graphic design platform used for creating a variety of visual content. The developers will use Canva to create diagrams, flowcharts, and other visual aids for the documentation and presentation of the project. It is a user-friendly tool that supports clear and professional design, making technical illustrations easier to understand.

- **Supabase**

An open-source backend-as-a-service (BaaS) platform that provides a scalable database, authentication, and storage services. In this project, the developers will use Supabase to manage user authentication, store app-related data, and handle backend operations. Its real-time capabilities and easy integration with Flutter make it an effective choice for ensuring that the app's data remains secure, consistent, and instantly updated across devices.

- **Cursor AI**

An AI-assisted coding tool used to make development easier and faster. The developers used Cursor AI to get code suggestions, fix errors, and handle complex logic during the development. It helped improve productivity and made the development process more efficient.

- **Visual Paradigm**

A visual modeling and diagramming tool that supports software design and project management. The developers will use Visual Paradigm to create Unified Modeling Language (UML) diagrams, data flow diagrams, and other system modeling artifacts. This ensures that the application's architecture and workflow are well-planned and documented, providing a clear guide for both development and future maintenance.

- Calendar of Activities

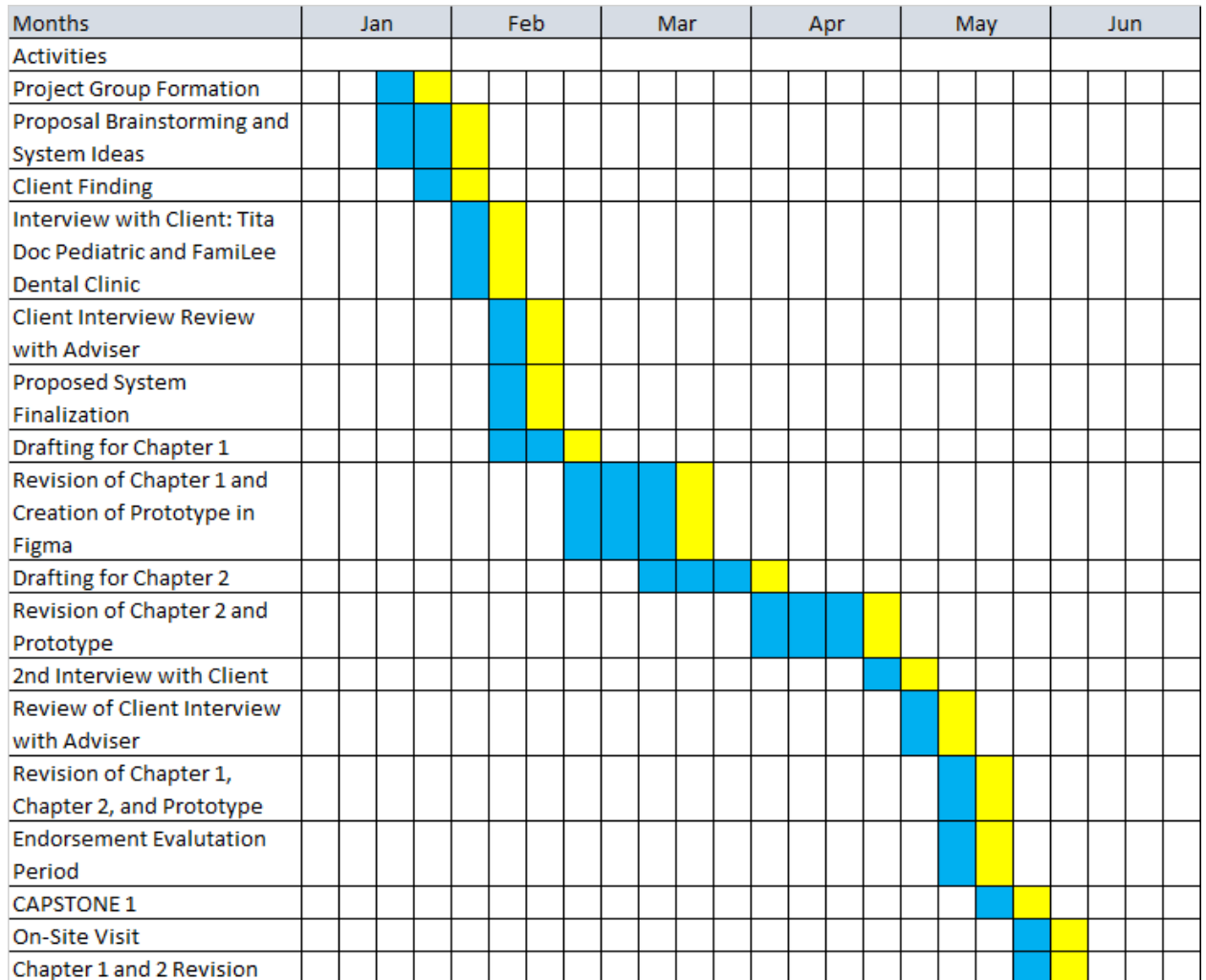


Figure 2. Capstone 1 Gantt Chart

Months	Jul				Aug				Sept				Oct				Nov			
Activities																				
Login and Inventory Module																				
Dashboard and PO Module																				
Stock Deduction and Activity Log Module																				
PO Module Revision																				
Notifications																				
Inventory Module Revision																				
Settings Module																				
Backup and Restore																				
PO Module Refinement																				
Dashboard Module Refinement																				
Alpha Testing																				
App Refinements																				
Beta Testing																				
App Refinements																				
Conclusion																				
Endorsement Evaluation Period																				
Capstone 2 Defense																				
Documentation and App Revision																				
Alpha and Beta Testing																				
App Refinements																				
Capstone 2 Redefense																				

Figure 3. Capstone 2 Gantt Chart

 In Progress  Finish

- **Resources**

To successfully develop the Android-based Inventory Management System for Tita Doc Pediatric and FamiLee Dental Clinic, the following hardware and software tools will be needed to support the application’s design, development, testing, and implementation.

Hardware	Minimum Requirements
Processor	Intel Core i5 10th Gen / AMD Ryzen 5 4500U
RAM	8GB
Storage	100GB
Display	13” or larger (1080p resolution)
Android Device	Android smartphone (Android 8.0 or later)

Table 1. Hardware Requirements

Software Requirements

Design and Prototyping:

- **Figma** – For wireframing and user interface design.
- **Canva** – For flowcharts, diagrams, and presentation materials.
- **Visual Paradigm** – For Unified Modeling Language (UML) diagrams, data flow diagrams, and other system modeling artifacts.

Development Tools:

- **Flutter** – For user interface design.
- **Dart** – Programming language used with Flutter
- **Visual Studio Code** – Primary code editor for writing and debugging code.
- **Android Studio** – IDE with emulator support for Android app testing.

Backend and Database Services:

- **Supabase** – For real time database management, user authentication, cloud storage, and backend hosting.

Requirements Analysis

The proposed application aims to address the operational challenges faced by Tita Doc Pediatric and FamiLee Dental Clinic by providing a computing solution tailored to the clinic's daily needs. The main users are the clinic's dentist and dental assistants, who handle inventory and deliver dental services. The focus is on managing dental supplies

These tasks take place in a dental clinic, where efficient stock management and accurate monitoring of supplies are essential. Currently, these tasks are done traditionally using handwritten logs and visual checks. This leads to errors such as supply shortages, overstocking, and missed expiration dates.

These activities happen during the clinic's operating hours, with staff managing supplies alongside patient care. Without a reliable system, staff must traditionally check inventory, which takes time and increases the chance of making mistakes.

The proposed application will improve these procedures by providing a digital solution that simplifies inventory management. Staff can update stock in real time using Android devices, receive alerts for low or expiring items, and keep a detailed log of user actions for accountability. The system will also include a basic purchase order feature that allows the creation and management of orders during restocking. When a purchase order is marked as received, the items will be added to the inventory. These features will improve workflow, accuracy and overall organization.

Requirements Documentation

The proposed system is an Android-based Inventory Management Application for Tita Doc Pediatric and FamiLee Dental Clinic. It aims to solve challenges in tracking stock levels and managing expiring supplies.

- **Functional Requirements**

- **User Login and Access Control**

The app allows users to log in using a registered email or username and password. The system verifies user credentials before granting access. Each user role (Owner, Admin, or Staff) has specific access and permissions. The “Remember Me” option lets users stay signed in when reopening the app, while the “Forgot Password” feature allows them to reset their password through a 6-digit token sent via email.

- **Dashboard Overview**

The dashboard provides a summary of key information within the system. It displays inventory statuses such as in stock, low stock, out of stock, expiring, expired, and fast-moving items. It also shows the status of purchase orders including open, partial, approval, and closed. Users can view visual charts or graphs of stock levels and trends, and they can download summaries or reports in CSV or PDF format containing relevant inventory and purchase order details.

- **Inventory Management**

The inventory module allows users to efficiently add, edit, and organize supplies. The Owner and Admin can create, edit, archive, restore, and permanently delete both items and categories. Staff can only view and adjust stock quantities as needed. Archived items are deleted after 30 days if not restored. Expiry segmentation ensures that items with different expiration dates are displayed and managed separately. The system follows the First-Expiry-First-Out (FEFO) method to ensure proper handling of expirable items.

- **Purchase Order**

The purchase order module allows users to create, view, and manage purchase orders for restocking supplies. Users can list items that need to be restocked, input supplier details, and attach receipts with reference numbers and remarks. The system supports recording different expiry dates for the same item within a single order. Once all items in a purchase order are flagged as received or partially received, the Owner or Admin can approve the order. Upon approval, the system updates stock levels in the inventory. Purchase order summaries, including supplier name, quantity, cost, date received, and status, are displayed and updated in real time.

- **Stock Deduction**

The Stock Deduction module allows users to record the usage or disposal of supplies while ensuring accurate inventory tracking. It follows the First Expiry First Out (FEFO) method to prioritize the consumption of items nearing expiration. Users can specify the purpose of each deduction to maintain proper documentation. Once the submitted stock deduction list is reviewed and approved, the system updates the inventory and logs the transaction to maintain a clear record of supply consumption and accountability.

- **Alerts and Notification**

The system provides alerts and notifications to help users manage inventory efficiently. Notifications are sent for items that are low in stock, out of stock, expiring soon, or already expired. Users also receive notifications for purchase orders that are ready for approval. All notifications can be enabled or disabled in the settings according to user preferences.

- **Activity Logs**

The activity log records all actions performed by users that cause changes within the application. This includes updates, deletions, approvals, and adjustments made in any module. Only the Owner and Admin can access and review the activity logs for monitoring and auditing purposes.

- **Account Management and Settings**

The settings module allows users to customize their account and application preferences. Users can edit their personal details such as name, username, email, and password. Staff can edit only their own profile, while Admins can manage staff accounts, and the Owner can manage all user roles. The app also allows users to switch between light and dark mode and manage notification preferences. The settings include options to enable or disable alerts for low stock, expiry, and purchase order approvals.

- **Backup and Restore**

The backup and restore feature helps prevent data loss by allowing the Owner to create and restore backups of system data. The app automatically performs a daily backup at 11:59 PM when the automatic backup option is enabled. All backups are stored and arranged by date for easy retrieval. The Owner can restore data using any saved backup file when needed.

- **Non-Functional Requirements**

- **Platform Compatibility**

The application must be accessible and fully functional on Android mobile devices used within the clinic.

- **Real-Time Synchronization**

The system should update data in real time across all connected users and devices. All changes are stored and synchronized using Supabase for reliable cloud integration.

- **Usability**

The user interface should be clean, simple, and intuitive to ensure that clinic staff can navigate and use the application with minimal training or technical experience.

- **Security**

The system must include secure login and authentication features to protect sensitive information. Access to data and features shall depend on user roles to prevent unauthorized actions or data exposure.

- **Performance**

The application must remain responsive during regular operations and be able to support multiple users at the same time without significant delays, lag, or crashes.

- **Data Storage and Reliability**

The system must use a stable Supabase backend to securely store and manage all records. It should ensure that data remains consistent, backed up, and protected from loss or corruption.

Design of Software, System, Product, and/or Processes

Owner & Admin

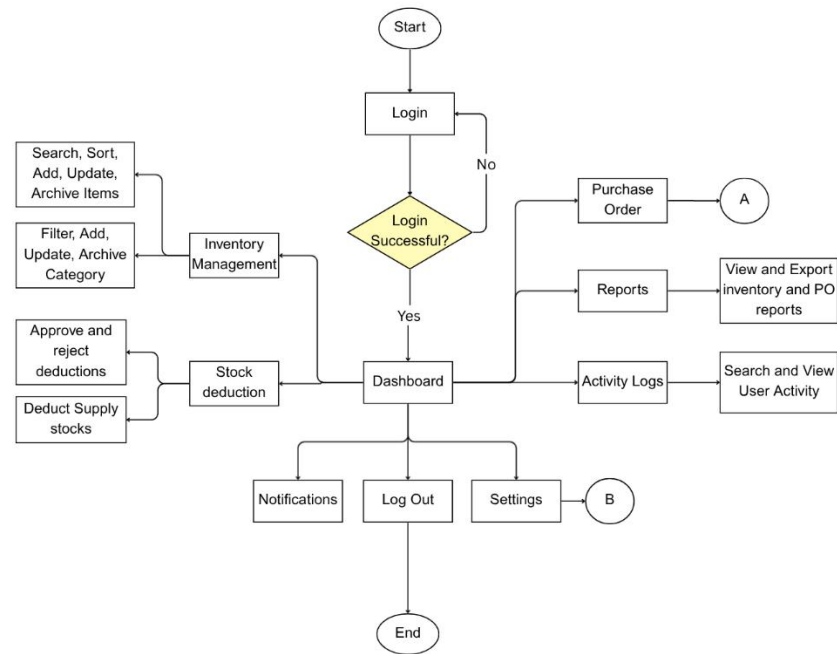


Figure 4.1. Owner & Admin Flowchart

Staff

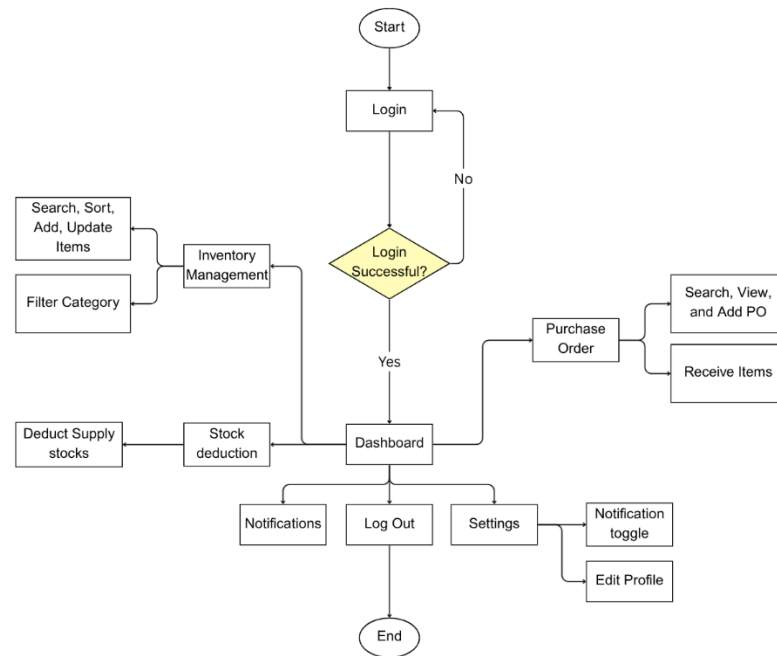


Figure 4.2. Staff Flowchart

The beginning process of the application requires the users: Owner, Admin, and Staff to enter their valid credentials to access the application. The landing page/dashboard displays brief summary of specific overviews: inventory, expired, and fast-moving supply which then the users can either access different parts of the system such as inventory, purchase order, stock deduction, activity log, notifications, settings, and the option to log out. Staff account is restricted from accessing activity logs and employee management, while having limited access to the inventory, stock deduction, and settings.

Login

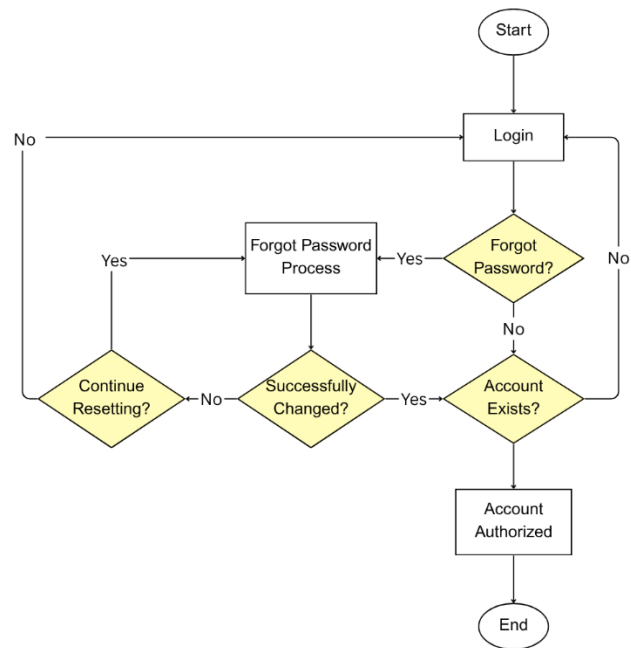


Figure 5. Login Flowchart

The login process includes the process of changing an account's password; This is accessible on the login page so that the user can have the option to change the password first before the account to be authorized. The application validates if the account being logged in by the user is authorized or not. The system also determines if the account credentials belong to specific account roles: Owner, Admin, and Staff to enforce hierarchy among the users.

Inventory Management

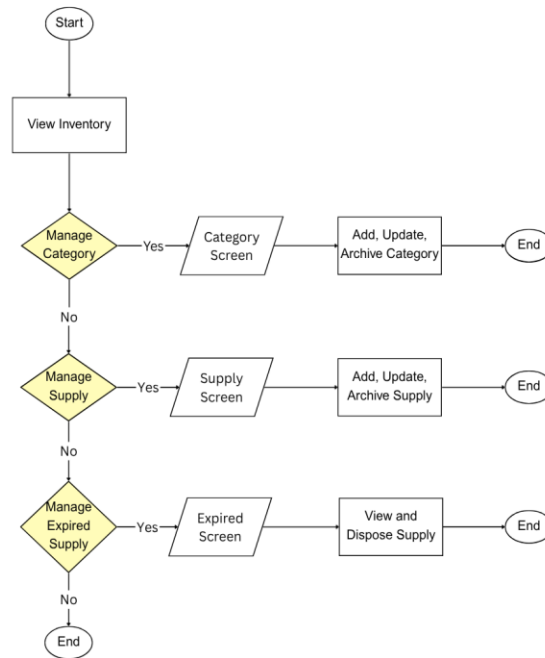


Figure 6. Inventory Management Flowchart

The process starts by viewing the inventory, which then provides the owner & admin an option to manage the category or supply. In the category screen, the owner/admin can Add, Update, and Archive a category which acts as the set holder of related items. Supply screen also allows the owner/admin to Add, Update, and Archive an item; All users can manage expired supply, view and dispose of said supply.

Purchase Order

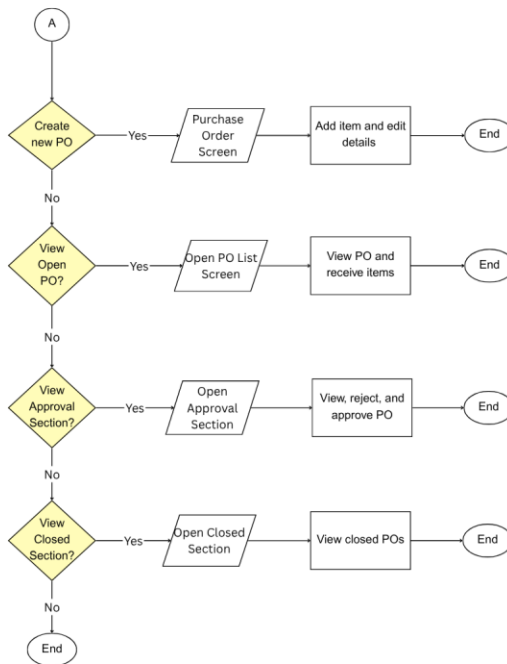


Figure 7. Purchase Order Flowchart

The process includes creating a new PO, viewing pending PO, approval, and closed POs. In the create new PO screen, the user is allowed to add and edit supply details needed for that supply. Once that's done, the created PO will be visible to the Open PO List screen, wherein the user can view and receive supplies. The approval section contains all POs that are not pending and are waiting for approval from the admin/owner. Lastly, closed PO contains all POs that are approved.

Stock Deduction

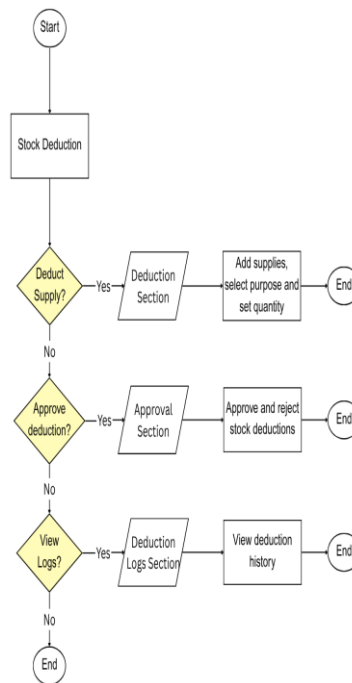


Figure 8. Stock Deduction Flowchart

This module allows users to easily deduct stocks of a specific supply from the inventory. The owner/admin can approve submitted stock deductions before updating the inventory. These deductions are saved in the deduction logs section.

Settings

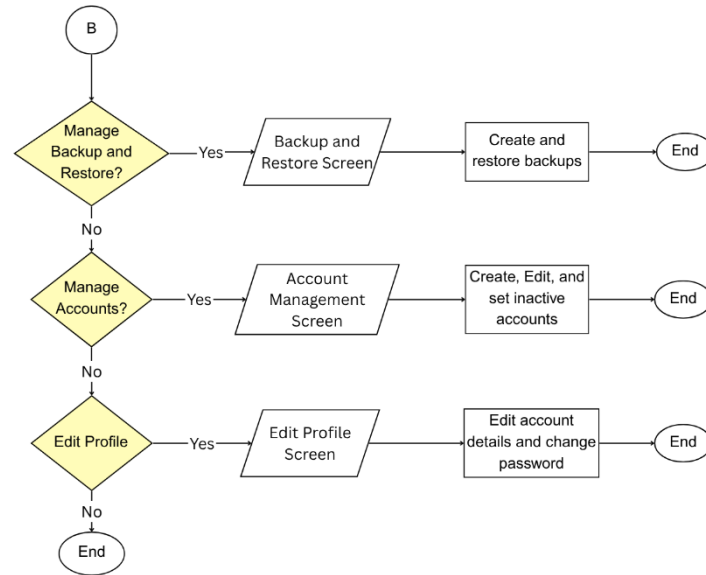


Figure 9. Settings

This module contains important features in managing inventory data, account and profile management. The backup and restore and account management are not accessible to the staff role; admin does not have access to the backup and restore, while the owner role retains all these features.

Context Diagram

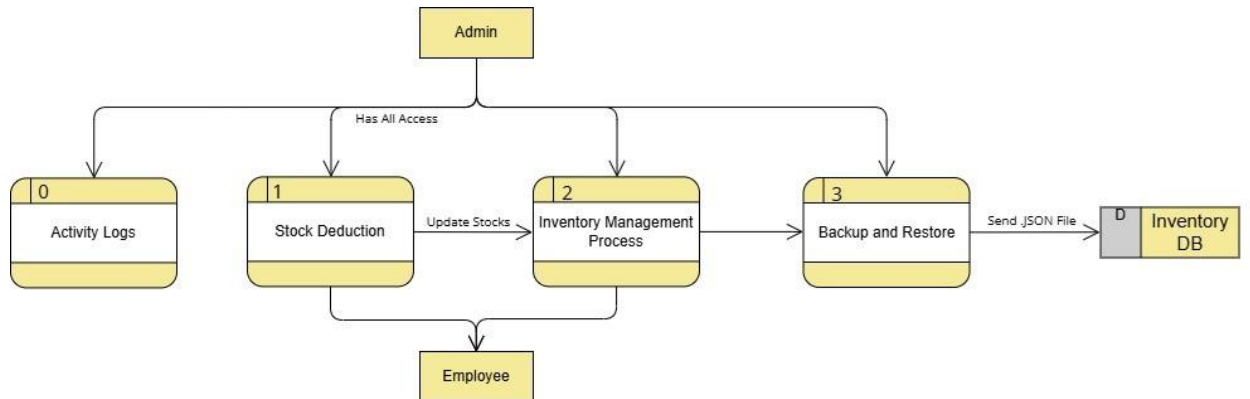


Figure 10. Context Diagram

The context diagram shows that the owner/admin and the staff have different access to certain main modules. The owner retains all access to processes: Stock Deduction, Activity Logs, Backup and Restore, and Inventory Management, the admin does not have access to backup and restore, while the staff can only update stocks through Stock Deduction. Inventory DB acts as the storage of all data that inventory management possesses.

Top-Level Diagram

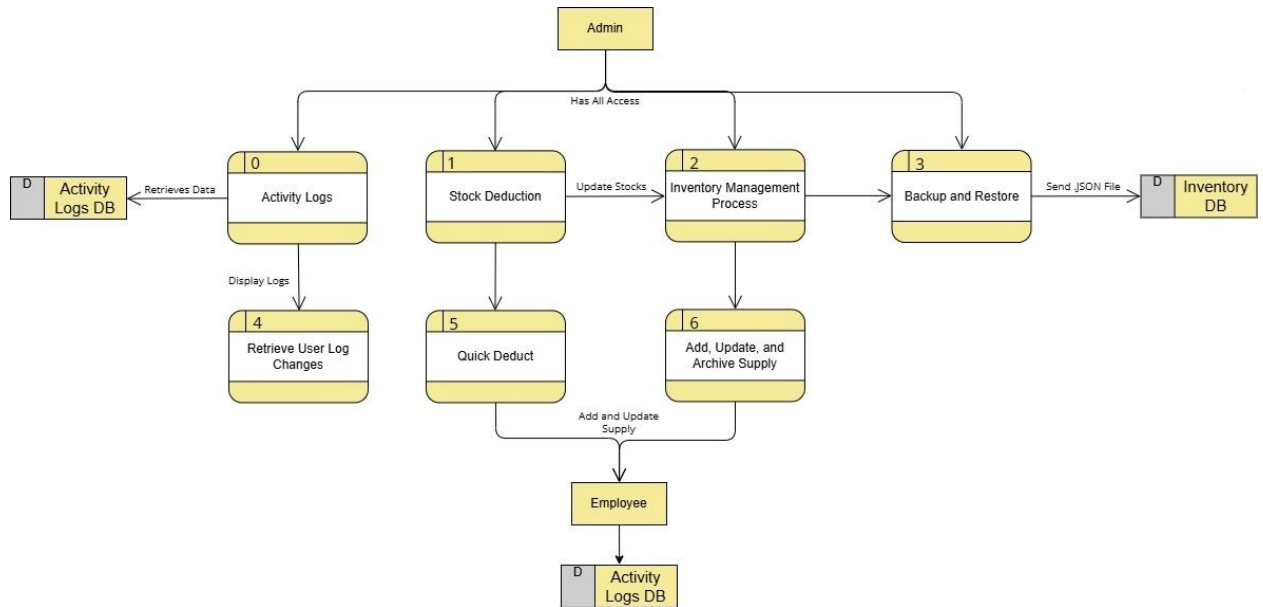


Figure 11. Top-Level Diagram

The top-level diagram showcases that each process has a designated process related to their functions. Process 0 gathers data from the application which will be sent to process 4 for generating user activity logs that can be used for viewing. Process 1 retrieves data from the inventory and will send stock updates when process 5 is used. Process 2 consists of all inventory process including process 6 which allows users to add, update, and for the owner/admin to archive supplies. Process 3 is connected to process 2 to collect the overall data of the inventory; Which allows the system to backup existing data and store it in the database, ready to make changes once the data is restored.

Middle-Level Diagram

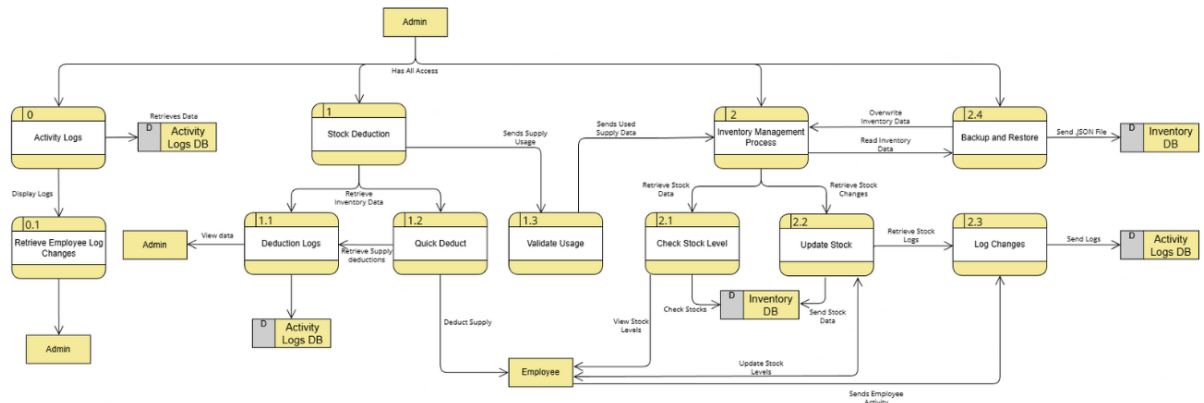


Figure 12. Middle-Level Diagram

This diagram showcases the detailed processes on how data is retrieved and sent from one process to another. Each main process is accessible by the owner/admin, while the staff only has access to a select process under Stock Deduction and Inventory Managements. Process 0 contains sub processes which showcase the user activity changes that are happening throughout the system, these activity log changes are only accessible to the owner/admin; The activity data are stored in Activity Logs DB. Process 1 serves as the point where stock decreases in the inventory, which includes processes 1.1 and 1.2 wherein the user can use to deduct supplies. Process 2 acts as the central storage of data within the system, where most data collected comes from this main process in order to display accurate and reliable data. Process 2 contains sub processes 2.1, 2.2, 2.3, and 2.4 which acts as the main processes of this feature. Process 2.1 retrieves stock data from the inventory DB to check the stock levels, the user can view this data. Process 2.2 retrieves stock changes and updates it in the database; This also includes supply detail changes in the inventory. Process 2.3 acts as a listener whenever a change in the inventory is made, and store it in the activity logs DB. Process 2.4 reads and copies inventory data into a .JSON file and stores it in the inventory DB. When called, the .JSON file will now overwrite existing inventory data to replace it, restoring the data inside that file.

Entity Relationship Diagram

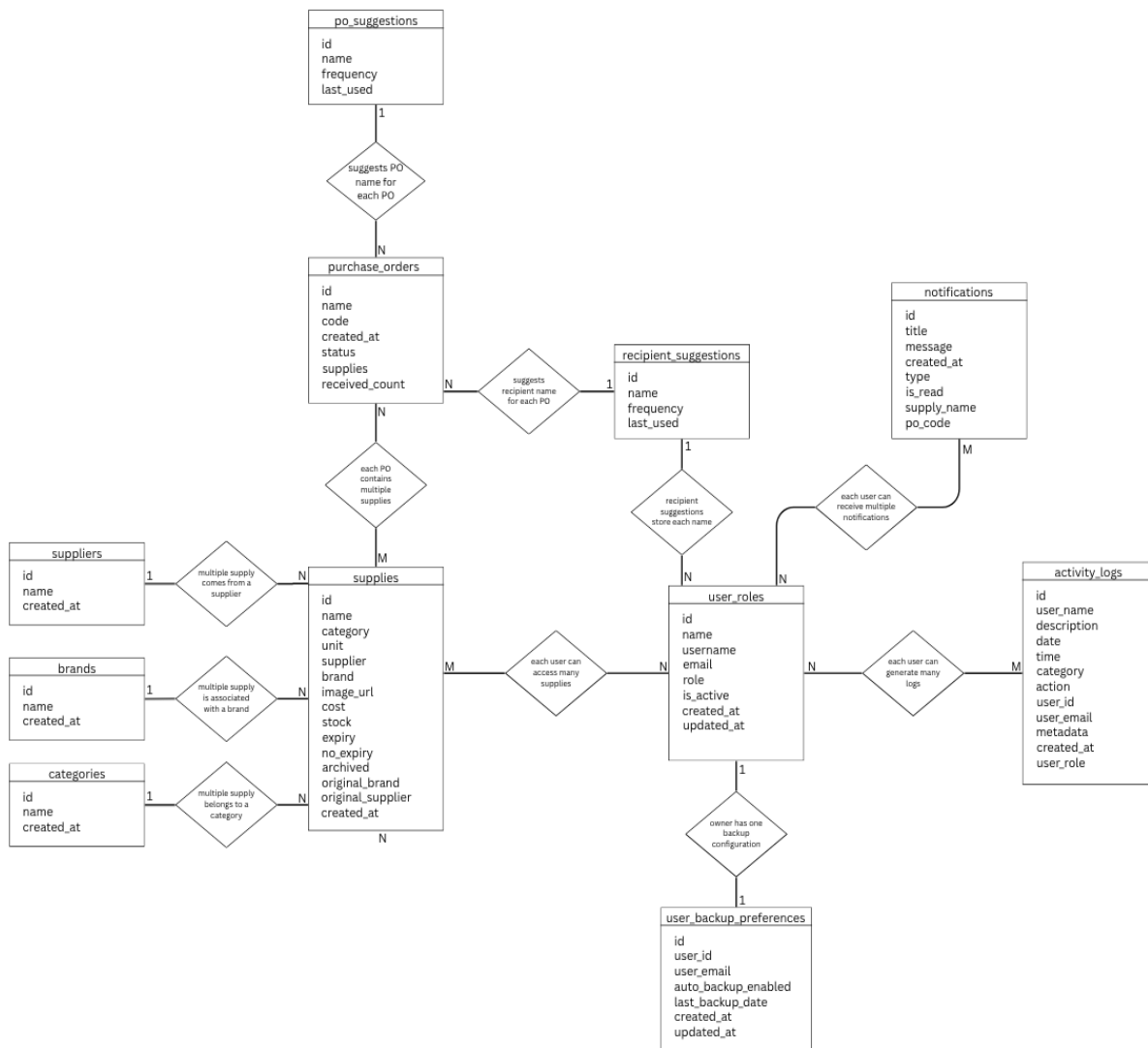


Figure 13. Entity Relationship Diagram

The diagram describes the connection between tables. The user_roles table is connected to 5 tables wherein each user can receive multiple notifications and generate many logs; Each users' name is stored inside the recipient suggestions, while the owner has one backup configuration. Additionally, each user can access many supplies which allow them to

manage the inventory. Now, purchase_orders table is connected to 3 tables wherein each PO contains multiple supplies from the supplies table; The table is also connected to the po_suggestions and recipient_suggestions table. The supplies table is connected to the suppliers, brands, and categories tables wherein multiple supply contains a supplier, brand, and belongs to a category within the inventory.

Use-case Diagrams

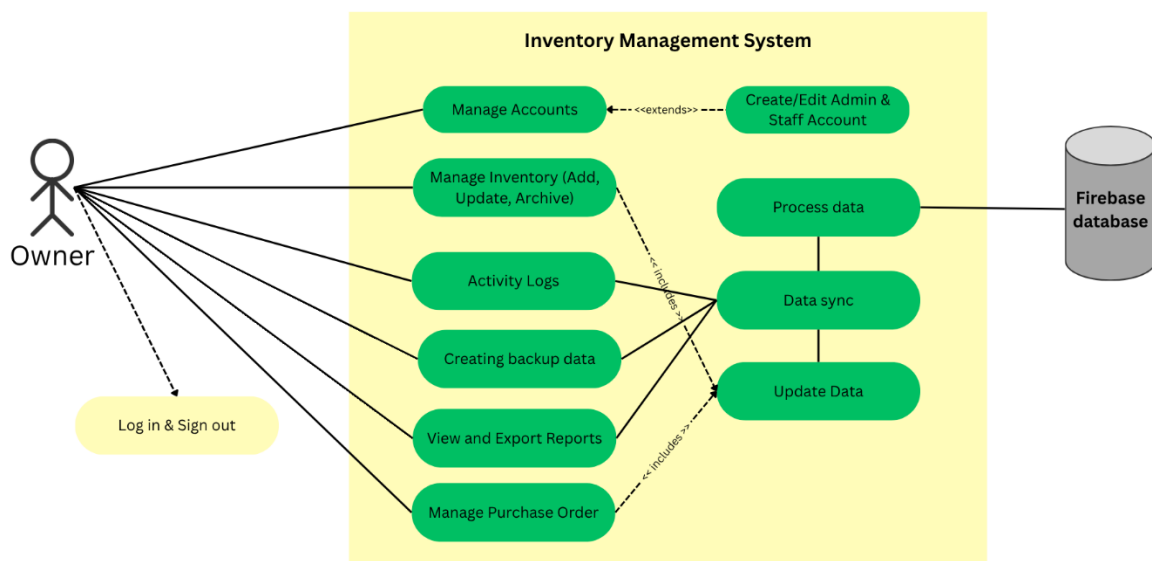


Figure 14.1. Use-case Owner

The owner use-case shows all modules that the owner can access. This account allows the management of both admin and staff account. Managing inventory, restocking, and stock deduction incorporates the behavior of update data which allows the database to retrieve the newly inputted data. Data sync distributes required information to procedures, create backup data, and reports.

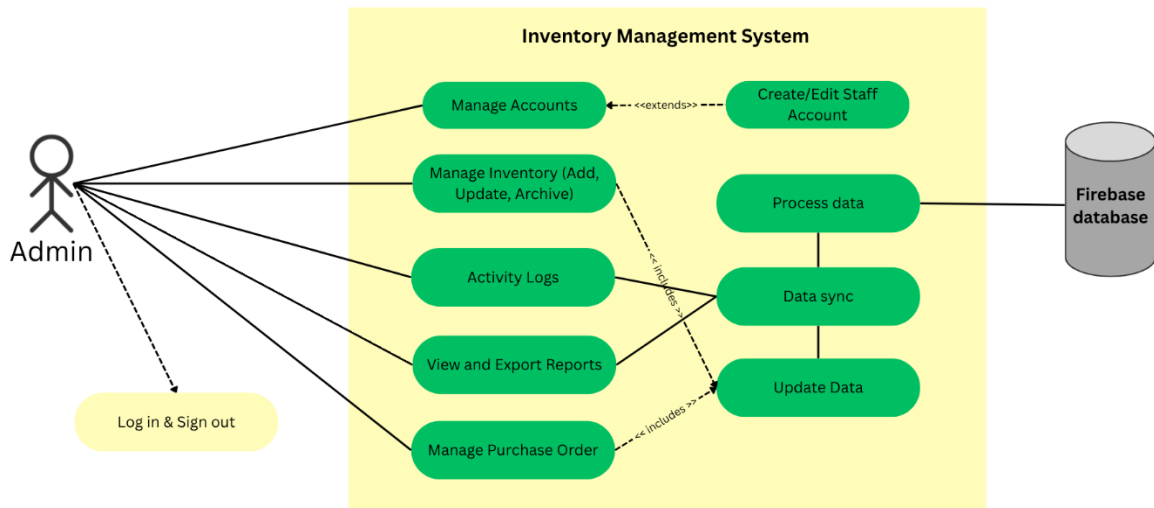


Figure 14.2. Use-case Admin

The admin use-case diagram shows the different features that the admin can access. These main features are accessible to the admin which includes: Manage Accounts, Manage Inventory, Activity Logs, and Purchase Order.

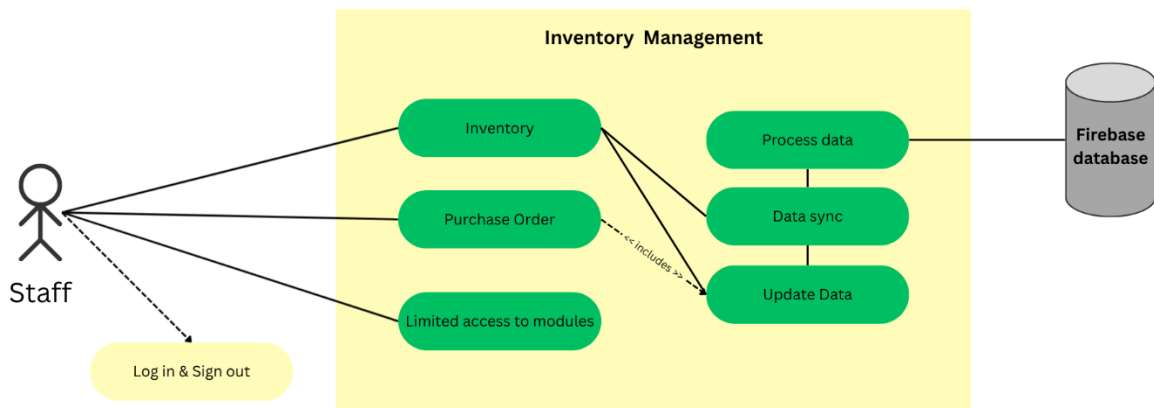


Figure 14.3. Use-case Staff

The staff use-case only includes the features Inventory and Purchase Order. The staff can only view and manage purchase orders by flagging items as “received”, which then proceeds to approval section before updating the inventory stocks. In the inventory, the staff can add and edit supplies and can dispose of expired supplies which are still connected to the database.

Development

The developers created the Android-based Inventory Management System for Tita Doc Pediatric and FamiLee Dental Clinic focusing on its key integral parts: Inventory, Purchase Order, Stock Deduction, Activity Logs, Role-based Access. Each module was developed in iterative sprints to efficiently focus on one (1) part of the system. During the development of each module, all functionalities are ensured to work as expected, the connection between the module and database was tested. The developers followed the Agile methodology approach, which allows planning, testing, and refine functionalities in iterative sprints based on continuous evaluation.

The mobile application was developed using Flutter, a user interface toolkit built on the Dart programming language. While Flutter is capable of cross-platform development, the application was developed specifically for the Android platform, which was aligned with the scope of this capstone. Development was done using Visual Studio Code (VS Code) as the main integrated development environment, while Cursor AI was used alongside it as an AI-assisted development tool. Cursor AI helped improve productivity especially when handling complex logic, resolving difficult errors, and generating code suggestions during implementation. Testing was conducted on both physical Android devices and virtual devices created using Android Studio's built-in emulator. For the frontend, Flutter was used to build the entire user interface and manage all user interactions. As for the backend, the application initially used Firebase for user authentication and database services. However, we later migrated from Firebase to Supabase, centralizing backend functionality including user login, data storage, and media file handling into one unified platform. This transition simplified the backend management by reducing the need to combine multiple tools and platforms. Supabase also provided clearer control over how data is structured, accessed, and stored, which better aligned with our app's modular setup. It allowed us to handle everything from login sessions to inventory records and uploaded files through one secure, consistent system that integrates well with the app's logic.

Several important packages and dependencies were included in the project, as listed in the `pubspec.yaml` configuration file. These packages helped add specific functionalities across the app. For example, `supabase_flutter` was used to connect with Supabase backend services, `flutter_dotenv`

allowed secure handling of sensitive environment variables and `shared_preferences` enabled local storage of user preferences and login status. Other packages such as `image_picker` were used for selecting files, `flutter_speed_dial` for custom floating action buttons, `encrypt` and `crypto` for secure backup handling, and `flutter_slidable` for adding swipe actions to list views. These dependencies were carefully chosen based on the project's feature requirements, and as more modules were developed, additional packages were integrated to meet the needs of each specific functionality.

The project follows a modular folder structure where each feature is organized into separate directories for screens, controllers, services, and reusable components, allowing the app to be maintainable and easy to expand. This structure was carefully planned from the beginning to make the codebase more organized and easier to navigate, especially during debugging or when adding new features. In addition to feature-specific folders, the project also includes a dedicated shared directory, which contains app-wide resources such as layout widgets, themes, utility functions, and global providers like user role management. This folder promotes code reusability and avoids duplication across modules. At the root of the `lib` folder is the `main.dart` file, which serves as the entry point of the application. It handles the initial setup by loading environment variables, initializing the Supabase client, and determining which page to display at launch based on the user's authentication state. This file also connects various services and feature modules, acting as the centralized starting point for launching the app and setting up its core dependencies. In addition, the `main.dart` file defines all the named routes used throughout the application for screen navigation, making it easier to manage page transitions in a consistent and centralized way. It also contains the configuration for the app's theme, including support for both dark mode and light mode, providing a flexible and user-friendly visual experience. Overall, the folder structure was intentionally designed to keep the project organized and make it easier for me to locate and work on specific parts of the code as needed.

The Login function was first developed and tested. This function ensured that only existing users with assigned account roles: owner, admin, and staff could access the system. During development, one of the initial challenges encountered was validating user input consistently across different devices. To address this, form validation was implemented on all login and password-related fields to prevent incomplete or incorrectly formatted data from being submitted. Another issue involved

the "Remember Me" functionality. Initially, even when the user checked the option, their email or username would not remain filled in after logging out and returning to the login screen. This was resolved by storing the user's email locally using `shared_preferences` when the "Remember Me" box is selected, so it could be automatically displayed the next time the app is opened. For account recovery, a Forgot Password feature was added, which originally relied on Supabase's default reset link behavior. However, this approach introduced issues during testing. One major problem occurred when users opened the reset link on a different device, for example, clicking the email link on a desktop while the app was installed on their phone. In such cases, the app could not detect that the reset process had been triggered, which interrupted the password reset flow and caused confusion. To solve this, the system was redesigned to use a custom OTP-based reset process instead. With this new approach, users receive a 6-digit code via email, which they enter directly into the app to verify their identity before setting a new password. This change not only made the process more intuitive but also kept the entire flow within the app itself. During the development of the OTP flow, we also encountered a critical issue with how the verification code was being consumed. Specifically, when users tried to change their password, a validation rule ensured that the new password had to be different from the old one. If this validation failed and the user attempted to submit again, the system had already used the OTP code, making it invalid for the next attempt. Since the code and password input were on the same page, the token was consumed too early in the process. To address this, adjustments were made to separate the validation from the token consumption, ensuring that the 6-digit code would only be marked as used after all validations had passed successfully. The email template used for the reset process was also customized using HTML to match the application's branding and provide clear instructions to users, improving both the usability and appearance of the recovery emails.

The Dashboard and Inventory modules were developed closely together, as the dashboard depends on real-time inventory data to display meaningful information. The dashboard was the first major screen designed after the login system, but during initial development, only its user interface was created. Since the dashboard relies on actual data from the Inventory, Purchase Order, and Stock Deduction modules, it could not be fully functional until those modules were built. Once the

interface was in place, development shifted to the Inventory module, which became one of the most important and data-heavy features of the app.

The developers focused on developing the main inventory system. This module was the most complex part of the system. It contains functionalities such as add & edit supply, add & edit categories, stock statuses, expired supplies, archived supplies, and expiry segmentation. The inventory acts as the central storage for supplies which other modules have access to. This module also automatically determines statuses for all supplies and displays them depending on stock levels for each supply. While testing the “Add Supply” functionality, a problem occurred where quickly pressing the save button multiple times caused duplicate records to be created. This was resolved by implementing a loading state and button disabling logic to prevent multiple form submissions during the saving process. Another issue emerged with how similar supplies were displayed. Initially, when adding the same supply name with different brand or supplier combinations, the system treated them as entirely separate supplies and created individual tiles for each variation. This cluttered the inventory list and made tracking difficult. To solve this, an “Other Supply Batches” section was added to each supply card. This allows all variants of the same supply, regardless of brand or supplier, to be grouped together under one main supply, improving the organization and visibility of related inventory. Additionally, supply types posed another challenge for the developers. At first, supplies that have types are technically unique supplies, but they are still the same supply. To display the stock statuses for these types, the developers created a stock status chip on the top-right part of a supply card; Red chip indicates how many types that have an out-of-stock status while orange is for low stock status. But this later posed some understanding issues since it is not intuitive and will pose confusion to the users. The developers later changed this into separate supply cards to better track statuses and provide intuitive inventory UI/UX.

The Purchase Order module was developed to help keep track of supply reordering. This also organizes purchase orders per supplier, which allows the user to upload an image of the physical receipt from a supplier. The user can also partially receive supplies and find a closed purchase order by using the receipt number. During testing of the Purchase Order module, several key issues were identified and addressed. Initially, to receive a supply, the user will flag each supply as received instead of managing the purchase order for a specific supplier itself. This way, it wasn't

aligned to the process our client has and posed no connection to their physical receipt. Offering no connectivity to the inventory and their physical storage. This problem was solved by implementing the grouping of supplies based on their suppliers and opened a path to uploading the physical issued receipt from a specific supplier. Another challenge occurred while implementing the feature for adding supplies to a purchase order. Initially, the process allowed users to input the quantity they wanted to order, but there was no option to assign different expiry dates to each unit of that quantity. This became a problem because in real-world cases, not all supplies delivered by suppliers have the same expiry date. To resolve this, a feature was added allowing users to input multiple expiry dates based on the quantity being ordered. An “Add another expiry date” button was introduced in the supply form, enabling the user to associate different expiry dates per batch of supply within the same order. This change allowed for more accurate tracking and ensured the system could reflect real inventory conditions more realistically. A related issue came during the integration of the Purchase Order module with other features, specifically when selecting supplies to be added to an order or deducted from stock. Expired supplies were still appearing in the selection lists, which could cause confusion or lead to errors in stock handling. To fix this, logic was added to automatically filter out expired supplies from these lists, while still giving users the ability to restore or manage them from a separate view if needed. In addition, a problem was identified in the Purchase Order Details screen where POs could be approved without any receipt information or remarks provided. This led to poor traceability and potential confusion, especially when reviewing order history. To address this, the process was updated so that once the ordered supplies were marked as received, the user would be prompted to enter essential receipt details including the receipt number, the recipient’s name, and a photo of the actual receipt. The developer also implemented a feature to capture or upload an image file was also added, supporting better documentation and transparency when receiving supplies. Lastly, during the development of partially receiving of supplies, the quantity of the partially received supply wasn’t updating the inventory which indicates poor connection of the partially receive feature and the inventory itself. But it was later resolved by enhancing the connection between the purchase order and inventory modules.

The Stock Deduction module serves to efficiently decrease stocks in the inventory module. This accesses inventory data which displays the number of stocks and the closest expiry date for each supply. During development and testing of the Stock Deduction module, several data handling issues were identified that directly impacted usability and inventory accuracy. One of the main problems was in the Add Supply interface, where expired supplies were still being displayed in the selection list just like in the Purchase Order. This led to confusion, as users could mistakenly select outdated supplies to be deducted. Additionally, the supply list did not reflect real-time data from the database. As a result, supplies that had already been removed or were no longer in stock would still appear on the list, causing deduction attempts to fail. To resolve this, the module was updated to filter out expired supplies by default and to fetch up-to-date inventory data in real-time. This ensured that only valid, in-stock supplies would appear when users add supplies to deduct. Another critical improvement involved the deduction logic itself. Originally, the system allowed users to select any available supply without considering expiry dates. However, this did not align with proper stock rotation practices, where supplies nearing expiry should be used first. The deduction flow was revised so that the system now automatically prioritizes supplies with the earliest expiry dates, ensuring that older batches are deducted before newer ones.

The Activity Logs module is developed to promote accountability. It records every significant user interaction within the system, including log ins, stock deductions, supply edits, and account modifications. Each log entry captures details such as the user's name, the changes that happened, action performed, and timestamp. This ensures that all activities are traceable, helping the clinic's owner/admin monitor system usage, identify errors, and maintain transparency in inventory operations. During testing, no major issues were encountered with this module, as most of its features relied on simple event recording and retrieval from the database. However, a few minor bugs were identified, such as incorrect timestamps and occasional display inconsistencies in the logs list. These were addressed early in development through debugging and small adjustments in how the data was formatted and fetched. Overall, the module performed reliably and required only minimal fixes, allowing it to integrate smoothly with the rest of the system.

The Settings tab includes functionalities such as dark mode, inventory & purchase order notifications, edit profile, employee list. This module allows users to personalize their experience

by switching between dark and light themes, enabling or disabling notifications based on their preferences, and updating personal information such as name, email, and password. The employee list section provides the owner with access to view, add, or manage staff and admin accounts within the system, ensuring organized account handling. During the development of the Settings module, a key enhancement was implementing dark mode support across the app. Instead of manually defining separate styles for each page, a global theme switcher was created using a `ValueNotifier<ThemeMode>` inside the shared theme configuration. This allowed the entire application to reactively toggle between light and dark themes with minimal code. Centralizing this logic avoided hardcoding styles on every screen and ensured consistency in the user interface regardless of the selected mode. While testing profile and account management features, we noticed missing input validations in several forms. For example, although the change password screen had field validation, the edit profile form lacked basic checks, such as preventing numbers in the name field. This oversight could lead to incorrect user data being saved. These validations were quickly added to prevent unintended input and maintain data integrity. Another issue emerged in the Add Employee feature, where the system failed to create new user accounts using the default hardcoded password. The cause was a mismatch with Supabase's password policy, which requires a mix of uppercase letters, lowercase letters, and digits. The default password originally used only lowercase letters and numbers, causing account creation to fail. This was resolved by updating the default password format to meet the required conditions and by adjusting Supabase's password settings to better match the system's intended user access policies. The Backup and Restore module also required fine-tuning. The automatic daily backup feature, which was intended to run every night at 11:59 PM, was not triggering as expected. To solve this, we integrated Supabase Edge Functions, which allowed scheduled execution on the backend. Using TypeScript within the Edge Function environment, we wrote custom logic to create a new encrypted backup file daily when the feature toggle was enabled. This ensured backups were reliably generated at the correct time without requiring the app to be open or active. To further protect sensitive data, we implemented encryption using the `encrypt` and `crypto` packages. Every backup file created by the system is encrypted before being stored, so even if someone downloads the file or gains access to the database, its contents would appear as unreadable characters. This

adds an extra layer of data protection, especially for clinics handling confidential inventory and purchase records.

The Notifications module was implemented to provide alerts for important supply-related events, such as out-of-stock or low-stock supplies, as well as purchase order approvals. One issue identified during development involved how the system was checking stock availability. Initially, notifications were triggered based on the quantity of individual supply batches rather than considering the total stock of a supply across all batches. This caused incorrect alerts to appear. For example, when one batch was out of stock, the system would notify the user that the entire supply was unavailable, even if other batches still had remaining stock. To correct this, the notification logic was updated to sum the total quantity of all batches belonging to the same supply. Instead of evaluating each batch in isolation, the system now calculates the combined stock and only triggers an out-of-stock or low-stock notification if the total quantity meets the defined threshold. This change ensures that notifications reflect the actual inventory status more accurately, reducing false alerts and helping users focus only on supplies that truly require attention.

Throughout development, the team encountered multiple challenges involving data synchronization, authentication logic, and backend integration. One of the earliest issues occurred during the login and authentication phase, where validating user input across different devices caused inconsistencies. The “Remember Me” and “Forgot Password” features also presented difficulties, particularly when password reset links were opened on different devices. This was resolved by implementing a custom OTP-based reset process to streamline password recovery directly within the app. In the Inventory and Dashboard modules, the most complex challenges involved real-time data handling and preventing duplicate records. Pressing the save button repeatedly caused redundant entries, which were fixed by adding a button-loading state and disabling multiple submissions. Another issue arose with how similar supplies were displayed; grouping identical supplies under an “Other Supply Batches” section improved organization and reduced clutter. The Purchase Order module also presented several technical and structural challenges. Initially, receiving supplies supply by supply did not align with real-world processes, so it was redesigned to group supplies per supplier and include receipt uploads for better traceability. Handling supplies with multiple expiry dates was another key improvement which

allowed users to enter different expiry batches within a single order for more accurate tracking. The Stock Deduction module faced challenges with data accuracy, as expired or unavailable supplies still appeared during deduction. This was resolved by real-time data fetching and automatic filtering of expired supplies. Deduction logic was also enhanced to follow the First Expiry, First Out (FEFO) method, ensuring proper stock rotation. During development of Settings and Backup, automatic backups failed to trigger as scheduled. To address this, the team integrated Supabase Edge Functions for reliable, time-based backups and used encryption packages to secure backup data. Minor issues such as password policy mismatches and form validation were also corrected.

Despite these challenges, the system was successfully completed, integrating all modules into a secure and reliable mobile platform that streamlined the clinic's inventory and purchase management processes.

To deploy the final version of the application, an APK file was generated and published through GitHub Releases. A dedicated releases folder was created inside the project directory to serve as the storage location for each release build of the APK. The APK was built using the Flutter command: `flutter build apk --release --no-tree-shake-icons`. The `--release` flag compiles the app for production use, optimizing it for performance and reducing file size. However, the default command without additional flags resulted in missing icons throughout the application. This was due to Flutter's icon optimization feature called tree shaking, which removes unused icons to reduce app size during the build. In some cases, especially when using custom icon sets or dynamic icon loading, Flutter may mistakenly remove icons that are still needed in the UI. To prevent this, the `--no-tree-shake-icons` flag was added to ensure all icons used in the project were retained in the final APK. After the build process completed successfully, the generated APK file located at `build\app\outputs\flutter-apk\app-release.apk` was copied into the releases folder using the following command: `copy build\app\outputs\flutter-apk\app-release.apk releases\familee-dental-v1.0.0.apk`. Renaming and copying the file to the releases folder made it easier to locate and upload the correct version of the APK to GitHub. Once the APK was placed in the correct folder, it was uploaded to GitHub Releases, along with detailed release notes. This process marked the official deployment of the application, making it ready for distribution and installation on Android devices.

RESULTS AND DISCUSSION

Testing

This section presents the evaluation results of the system based on its functional and non-functional requirements. The testing aimed to verify whether the developed Android-based inventory management system met its design objectives, performed according to user expectations, and complied with recognized software quality standards. To verify the credibility of the system and its functionalities, three (3) IT professionals with a minimum of 2 years of experience were invited to conduct the alpha testing. Before the testing phase began, the developers prepared an alpha test consent form and an alpha test checklist; All three (3) testers were provided with the same files. The alpha test checklist consists of a questionnaire regarding specific objectives, functional, and non-functional requirements to thoroughly evaluate the system. To determine the overall passing rate of the system, the evaluation followed the ISO/IEC 25010:2011 standard, which provides an internationally recognized framework for assessing software quality through measurable attributes such as functionality, reliability, usability, performance efficiency, and security. In accordance with the standard, the system's overall passing rate is determined using this formula:

Legend: 1 = pass, 0 = fail

Overall Conformance Level = Number of Passed Responses/ Total Test Response X 100

Conformance Level

95-100% = High Quality – Fully meets requirements and is ready for the next test.

85 – 94% = Acceptable Quality - Meets most functional and non-functional requirements and minor improvements needed.

Below 85% = Low Quality – Significant issues and requires rework before deployment.

Functionality	Tester 1	Tester 2	Tester 3
Does the system verify correct email/username and password?	1	1	1
Does the “Forgot Password” feature send a 6-digit token to the user’s email?	0	1	1
Can the user reset their password using the received token?	0	1	0
Does the “Remember Me” feature keep the user logged in after closing the app?	1	1	1
Does the dashboard correctly display in-stock, low-stock, and out-of-stock items?	1	1	1
Does it display the correct number of expiring and expired items?	1	1	1
Can the user tap on dashboard sections (e.g., Low Stock) to view item lists	1	1	1
Can the user add new inventory items successfully?	1	1	1
Can the user edit and save changes to item details?	1	0	1
Can the user archive and restore items properly?	1	1	1
Does the system follow FEFO (First Expiry, First Out) when deducting stock?	1	1	1
Can the user create and approve purchase orders successfully?	1	1	1
Can the user partially receive supplies in purchase order?	0	0	0
Does the inventory update after a purchase order is approved?	1	1	1
Can the Staff account view purchase orders but not approve or edit them?	0	1	1

Can the Admin/Owner create and manage presets?	1	1	1
Can the staff use an existing service preset?	1	1	1
Is the staff unable to create new presets?	1	1	1
Does stock deduction correctly update inventory levels?	1	1	1
Does each deduction appear in the Activity Log?	1	1	1
Does the reports display accurate inventory data?	0	1	1
Can the Admin/Owner export reports into a .CSV file?	1	1	1
Can the Owner create Admin and Staff accounts?	1	1	1
Can the Admin manage Staff accounts but not other Admins?	0	1	1
Is the Staff account restricted from accessing management or settings features?	1	1	1
Can the Owner create and restore backups successfully?	1	0	1
Are all data and user roles retained after restoring a backup?	1	1	1
Overall Conformance Level (%)	86%		

Table 2. Functionality Testing

The functionality testing of the android-based inventory management system retains a conformance level of 86%. The testing indicated that most key functionalities are working as expected but posed some major and minor inconsistencies regarding purchase order, login module, accounts, and backups. During the 1st testing, these issues include the partial receive function which updates stocks inaccurately when receiving supplies in the partial section. Additionally, the 6-digit token in the forgot password wasn't received. Additional details were also suggested in the

inventory reports to provide accurate supply data. Another minor issue found was that other admin accounts were visible in the admin role. While in the 2nd testing, the issues found during the 1st testing were resolved. However, another problem was noticed in the partial PO; the receipt details were still visible when the PO was rejected. Creating backups also posed a problem wherein the owner role wasn't able to create one. The 3rd test was mainly on issues regarding the UI. These issues were addressed by the developers to ensure smooth functionality testing in the next testing phase.

Reliability	Tester 1	Tester 2	Tester 3
Does the system load each page or feature without crashing?	0	1	1
Does the system maintain data consistency between inventory, dashboard, and activity log?	1	1	1
Does the app continue functioning properly after performing multiple consecutive actions?	1	1	1
Is the data retained after closing and reopening the app?	1	1	1
Does the app handle interruptions (like switching apps or locking screen) without errors?	1	1	1
Overall Conformance Level (%)	93%		

Table 3. Reliability Testing

The results of the reliability testing indicate a 93% conformance level, demonstrating that the android-based inventory management system performs consistently and dependably when handling inventory data. It successfully maintained data consistency across modules such as inventory, dashboard, and activity logs, preserved information after closing and reopening, and handled interruptions like screen locking without errors. However, the slight deviation from a perfect score implies that there were minor issues, such as occasional crashes or slow responses during page transitions, which were later addressed. Overall, the system's reliability ensures stable performance during daily clinic operations.

Usability	Tester 1	Tester 2	Tester 3
Are all buttons, icons, and menus responsive when tapped?	1	1	1
Are screen labels and messages clear and understandable?	1	1	1
Is the app layout organized and easy to navigate?	1	1	1
Is the text readable and properly aligned on all pages?	1	1	0
Is it easy for users to find what they need (e.g., Low Stock items, Purchase Orders)?	1	1	1
Can users easily distinguish between roles (Owner, Admin, Staff) based on available options?	1	1	1
Are notifications and reminders displayed clearly?	1	1	1
Can the user easily switch between Light and Dark mode?	1	1	1
Overall Conformance Level (%)	96%		

Table 4. Usability Testing

The system attained a 96% conformance level in usability testing, indicating that the testers found the application intuitive, well-organized, and easy to navigate. Testers highlighted clear labeling, responsive buttons, readable text, and logical layouts. Features like sorting, filtering, and light/dark modes improved user experience and accessibility. This high rating reflects the system's alignment with the ISO/IEC 25010 usability standards, suggesting that even first-time users can operate it with minimal guidance. Only minor visual misalignments were noted, but they did not affect usability.

Performance Efficiency	Tester 1	Tester 2	Tester 3
Does the app open within 10 seconds?	1	1	1
Do the dashboard and inventory pages load within 10 seconds?	1	1	1
Does saving or updating an item take less than a few seconds?	1	1	1

Does approving a purchase order process without delay?	1	1	1
Does the app respond smoothly even with many inventory records?	1	1	1
Overall Conformance Level (%)	100%		

Table 5. Performance Efficiency Testing

Performance efficiency testing obtained a perfect 100% score, showing that the system met all expectations for speed, responsiveness, and processing efficiency. It consistently loaded dashboards and inventory pages within 10 seconds, processed updates and approvals instantly, and maintained smooth performance even with multiple records. This means that the application efficiently utilizes device resources while ensuring real-time synchronization with Supabase, meeting the ISO/IEC 25010's sub-characteristics of time behavior, resource utilization, and capacity, which is essential for clinic operations that depend on up-to-date inventory data.

Security	Tester 1	Tester 2	Tester 3
Does the system prevent login with incorrect credentials?	1	1	1
Are passwords hidden while typing?	1	1	1
Does the system prevent unauthorized users from accessing restricted modules?	1	1	1
Are role permissions correctly applied (e.g., Staff cannot edit accounts)?	1	1	1
Does the app automatically log out after using the logout function?	1	1	1
Are sensitive operations (e.g., password reset, backup/restore) restricted to authorized roles only?	1	1	1
Overall Conformance Level (%)	100%		

Table 6. Security Testing

Security testing also achieved 100%, proving that the system successfully prevents unauthorized access and enforces role-based permissions. Passwords are encrypted and hidden during input, restricted actions are limited to authorized roles, and all sensitive functions work securely.

Additionally, the system supports data confidentiality and accountability, since only the owner and admin can access logs of user actions. This conformance to ISO/IEC 25010 security attributes guarantees that data integrity and privacy are well protected.

Beta Testing Results

The results of the beta test came from the owner and staff of Tita Doc Pediatric and FamiLee Dental Clinic. The testing was conducted in person while the developers guided the owner, Jilliane Lee, and her staff in navigating each module. The application was deployed through GitHub for efficient installation. The developers also provided the testers with three (3) files necessary for the testing: User Acceptance Letter, Beta Test Consent Form, and Beta Test Questionnaire. The questionnaire uses the Likert scale to measure user satisfaction, usability, and overall acceptance of the system. Only the owner and staff will have access to the inventory management system, wherein it will be used for the operational processes in the clinic. During the testing, the developers guided the testers in navigating the system, which allowed them to seamlessly learn the processes within the application. To interpret the results, we assigned values on each response;

4 = Strongly Agree

3 = Agree

2 = Disagree

1 = Strongly Disagree

and determine the mean using this formula:

Mean = Sum of all scores / number of respondents

Weighted Mean = sum of all question mean / number of questions

To determine the overall perception of the respondents, the interpretation of the weighted mean is as follows:

3.26 – 4.00 = Strongly Agree – The application will greatly benefit the overall clinic operations.

2.51 – 3.25 = Agree – The application will somehow benefit the overall clinic operations.

1.76 – 2.50 = Disagree – The application will not significantly benefit the overall clinic operations.

1.00 – 1.75 = Strongly Disagree – The application will not benefit the overall clinic operations at all.

Statement	Mean	Interpretation
The menus, buttons, and navigation links are organized and easy to understand.	4	<i>Strongly Agree</i>
The system responds quickly and does not lag during operations.	3.66	<i>Strongly Agree</i>
The interface design is visually appealing and well-organized.	4	<i>Strongly Agree</i>
The system provides clear error messages when input is incorrect.	4	<i>Strongly Agree</i>
The login process is straightforward and secure.	4	<i>Strongly Agree</i>
The credentials verification works properly and accurately detects invalid inputs.	4	<i>Strongly Agree</i>
The “Remember Me” feature functions correctly.	4	<i>Strongly Agree</i>
The “Forgot Password” feature allows me to recover access successfully.	4	<i>Strongly Agree</i>
The dashboard provides a clear overview of inventory status (in stock, low stock, out of stock).	4	<i>Strongly Agree</i>
The graphical representation of stock levels is helpful and accurate.	4	<i>Strongly Agree</i>
The dashboard accurately displays expiring, expired, and fast-moving items.	4	<i>Strongly Agree</i>
The report format is organized and easy to understand.	3.66	<i>Strongly Agree</i>
Downloading reports as a .CSV file works properly.	4	<i>Strongly Agree</i>

The generated reports are helpful for monitoring inventory and making restocking decisions.	3.66	<i>Strongly Agree</i>
Adding new items is simple and straightforward.	4	<i>Strongly Agree</i>
Editing item details (name, brand, expiry date, etc.) is easy and functional.	4	<i>Strongly Agree</i>
Archiving and restoring items works correctly.	4	<i>Strongly Agree</i>
Category management (add, edit, archive) functions as expected.	4	<i>Strongly Agree</i>
Archived supplies are properly displayed, restored, and deleted when needed.	4	<i>Strongly Agree</i>
Expired items are accurately displayed in the expired section.	4	<i>Strongly Agree</i>
Expiry segmentation correctly separates items based on expiration date.	4	<i>Strongly Agree</i>
Creating purchase orders is straightforward.	4	<i>Strongly Agree</i>
Editing details in the purchase order (supplier, quantity, expiry, etc.) works correctly.	3.66	<i>Strongly Agree</i>
Supplies with statuses (Pending, Received, Partially Received) are clear and easy to manage.	3.66	<i>Strongly Agree</i>
Uploading receipts and entering reference numbers functions properly.	4	<i>Strongly Agree</i>
Approving purchase orders correctly updates the stock levels.	4	<i>Strongly Agree</i>
The Purchase Order list displays information clearly and is easy to navigate.	4	<i>Strongly Agree</i>
The quick deduction feature accurately reduces item quantities.	4	<i>Strongly Agree</i>

Deducting stocks help me determine the purpose of disposal.	4	<i>Strongly Agree</i>
Creating and using service presets makes stock deduction faster and more efficient.	1.67	<i>Strongly Disagree</i>
Service preset helps me track supplies to be used when performing procedures.	2.00	<i>Disagree</i>
Deducting stocks from a service preset is easy to understand.	2.00	<i>Disagree</i>
Deduction logs help me track supplies that have been deducted.	3.33	<i>Strongly Agree</i>
Notifications for low stock, expiry, and out-of-stock alerts work properly.	4	<i>Strongly Agree</i>
Notifications for purchase order approvals are received correctly and on time.	4	<i>Strongly Agree</i>
Appearance settings (light mode/dark mode) function as expected.	4	<i>Strongly Agree</i>
Customizing system preferences (notifications, appearance, etc.) is easy and useful.	4	<i>Strongly Agree</i>
Editing profile information (name, username, email, password) is simple and functional.	4	<i>Strongly Agree</i>
The role-based access control (Owner, Admin, Staff) works correctly according to permissions.	4	<i>Strongly Agree</i>
The Staff List displays all user accounts within my access permissions accurately.	4	<i>Strongly Agree</i>
Resetting passwords functions correctly and applies the default password as intended.	4	<i>Strongly Agree</i>
Backup and restore features work correctly.	4	<i>Strongly Agree</i>

Backup files are properly listed and sorted by date.	4	<i>Strongly Agree</i>
The activity logs accurately record all user actions and system changes.	4	<i>Strongly Agree</i>
The search bar efficiently retrieves relevant results.	4	<i>Strongly Agree</i>
Sorting and filtering options organize data effectively.	4	<i>Strongly Agree</i>
The log-out feature works correctly.	4	<i>Strongly Agree</i>
The system meets my expectations for managing clinic inventory and supplies.	4	<i>Strongly Agree</i>
The system prevents supply outage, avoids overstocking, and reduces the chance of expired supply usage.	4	<i>Strongly Agree</i>
The system helps me monitor stocks in real-time	4	<i>Strongly Agree</i>
The system helps improve efficiency compared to manual inventory tracking.	4	<i>Strongly Agree</i>
The system performs reliably across all modules and functions.	3.66	<i>Strongly Agree</i>
I am satisfied with the overall performance and usability of the system.	4	<i>Strongly Agree</i>
Overall Weighted Mean	3.83	<i>Strongly Agree</i>

Table 7. Beta Testing

After testing, the overall weighted mean 3.83 suggests that the system's functionalities will greatly benefit the overall clinic operations and is aligned to their specific needs in managing their inventory in real-time, track expiry dates, and provides an effective purchase order for restocking; This also allows them to track orders with partially received purchase orders from suppliers. The developers also received feedback to remove the service presets in the stock deduction module, which was later applied. The service presets were found to be too tedious to use because it required them to double-check whether the supplies included in the preset were actually deducted from the

physical inventory. Removing it simplified the deduction process and made it more efficient for the clinic.

Description of Prototype

Category	Minimum	Recommended
Operating System	Android 8.0 (Oreo)	Android 10 or higher
RAM	4GB	6GB
Processor	Octa-core CPU (equivalent to snapdragon 660 or higher)	Snapdragon 865 or equivalent
Display	6.0-inch screen	10-inch screen

Login

In accordance with our functional requirements, the user shall be able to log in by using their valid credentials. The credentials vary depending on which account role they are allowed to access. If the user forgets their password, they can reset it using the forgot password process wherein an e-mail will be sent; A 6-digit code will be sent to the user's email which is needed when inputting the new password.

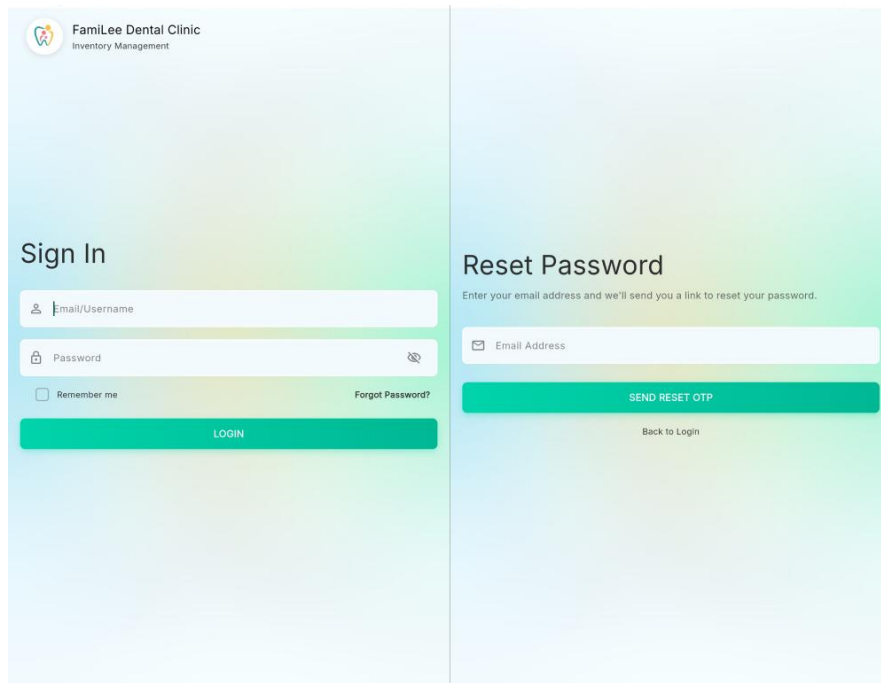


Figure 15.1. Login & Forgot Password

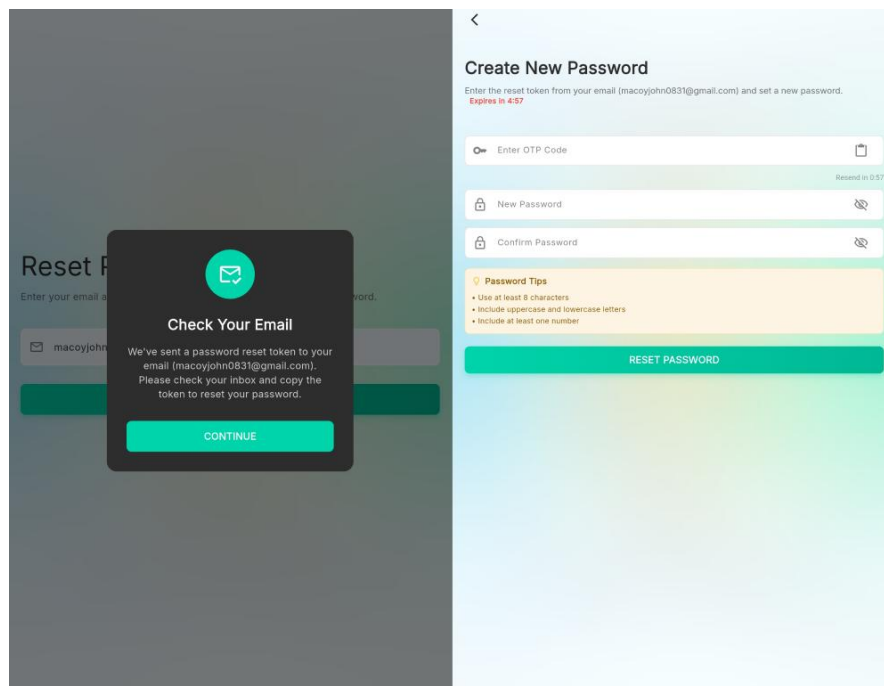


Figure 15.2. Email Confirmation & New Password

Dashboard

This section provides downloadable summaries regarding inventory stocks, expiring and expired, and a summary of the speed of items that are consumed within a certain period. It also displays purchase order summaries that indicates the number of POs with statuses such as open, partial, approval, and closed.

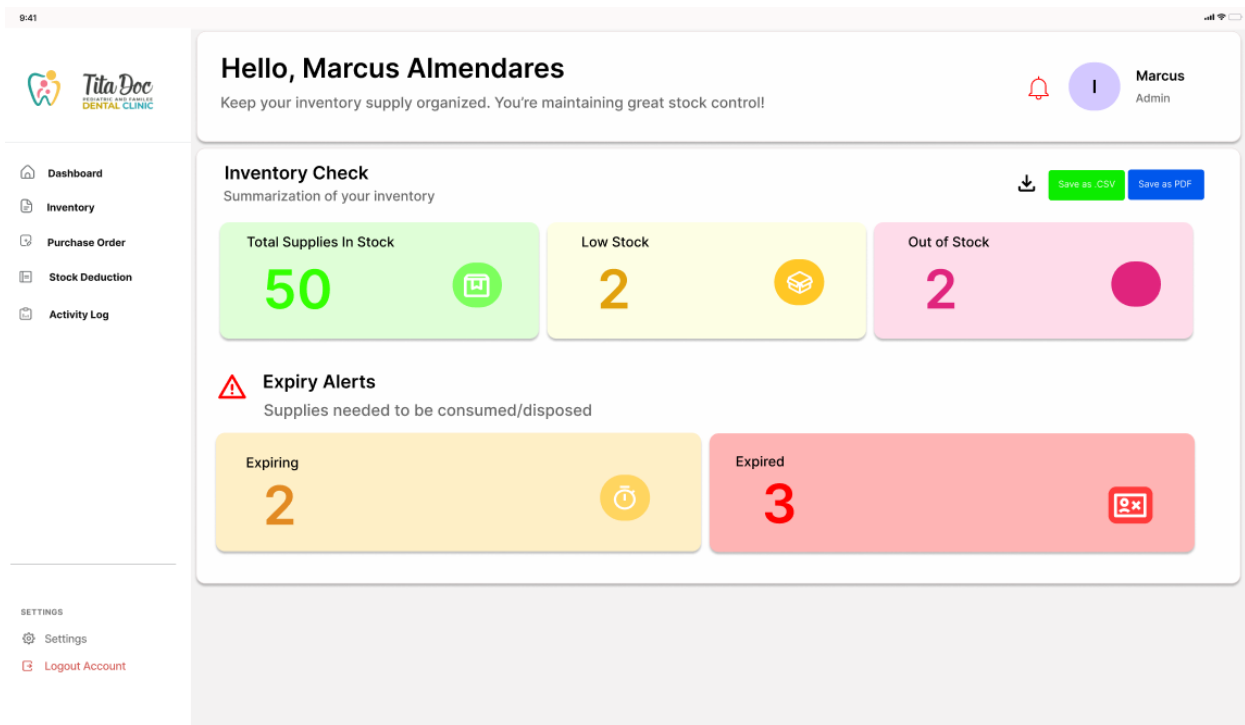


Figure 16. Dashboard

Owner

Inventory

This contains all the supplies and displays their stock statuses. The owner role can access all the functionalities in the inventory module. These include add, expired, and archived supply, add, and edit category. This is to make sure that the owner retains all permissions within the application, especially the inventory module.

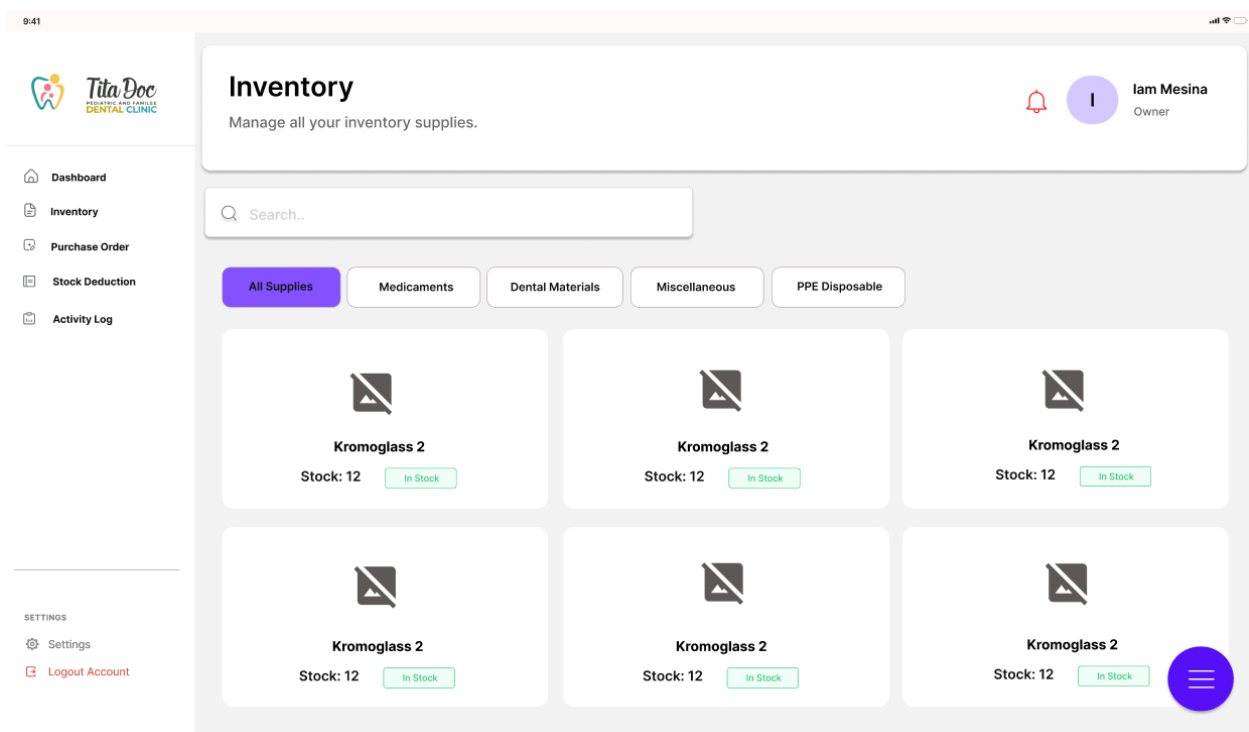


Figure 17.1. Inventory Owner

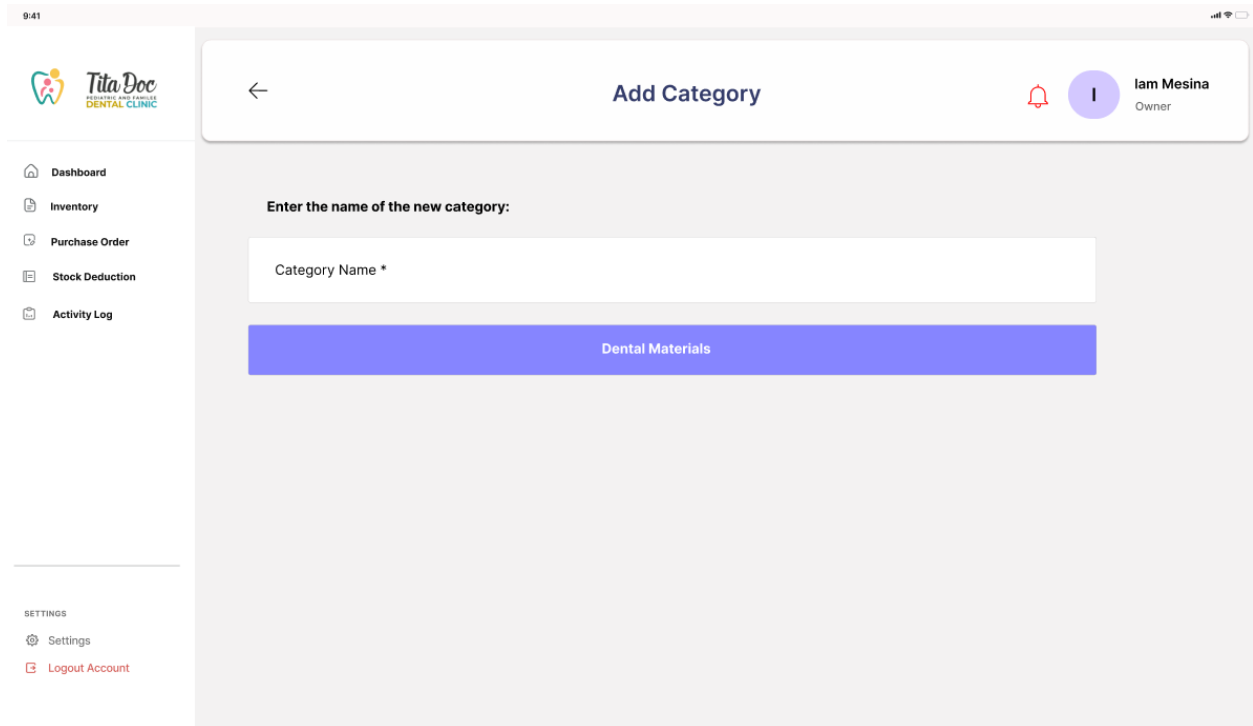


Figure 17.4. Add Category

Purchase Order

This module allows the owner to create purchase orders, and they are also able to flag supplies as “received.” Once all supplies/suppliers are received, the user can input the receipt number, who received it, and an image of the physical receipt given by the supplier. This role can also approve or reject the PO before updating the inventory.

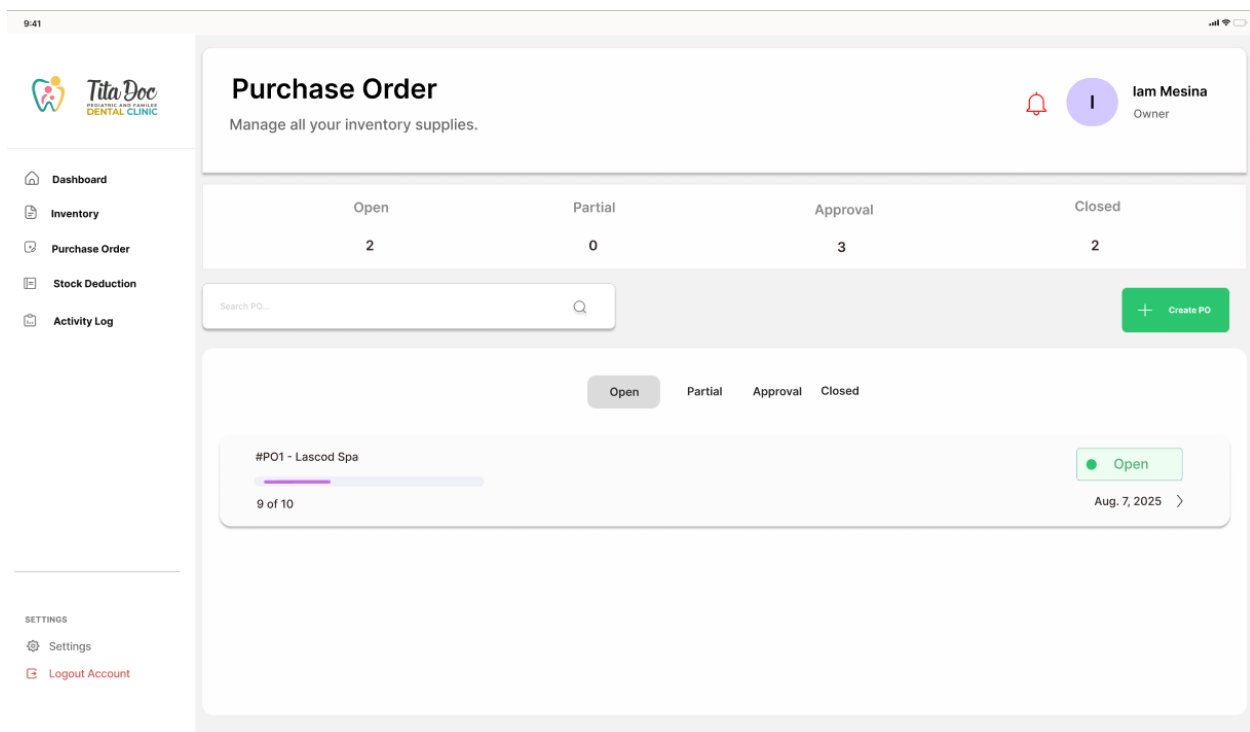


Figure 18.1. Purchase Order Owner

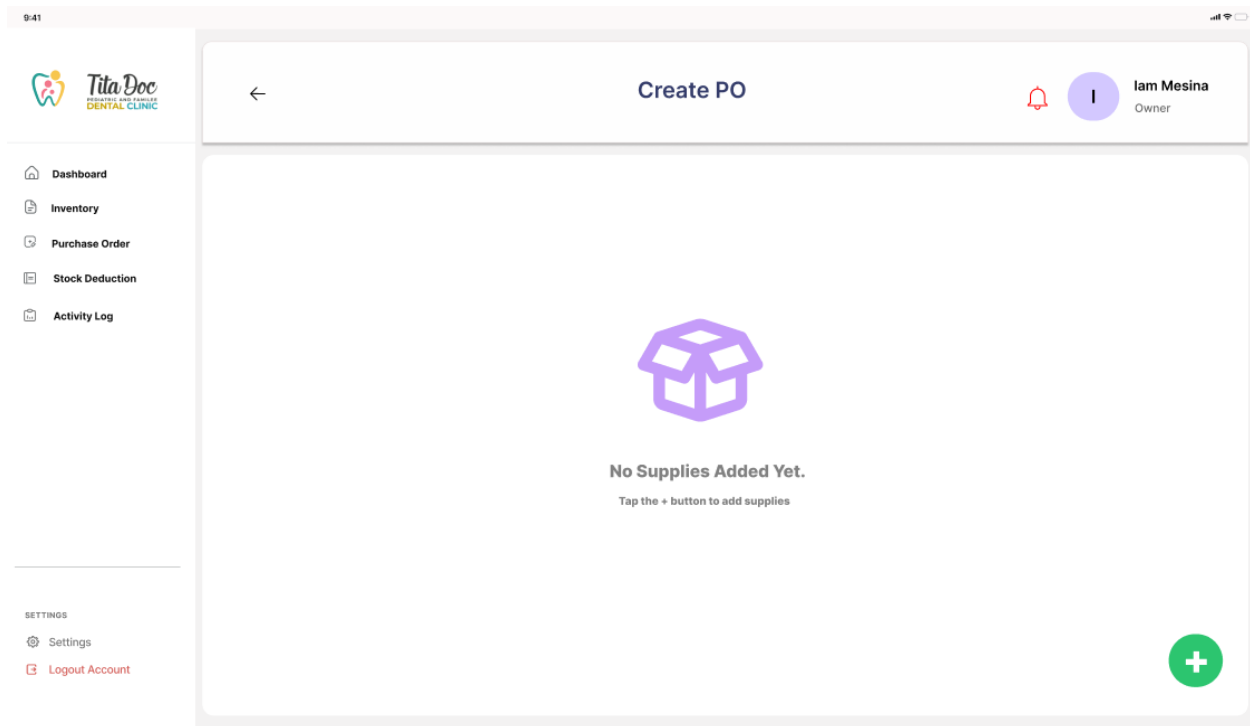


Figure 18.2. Create Purchase Order

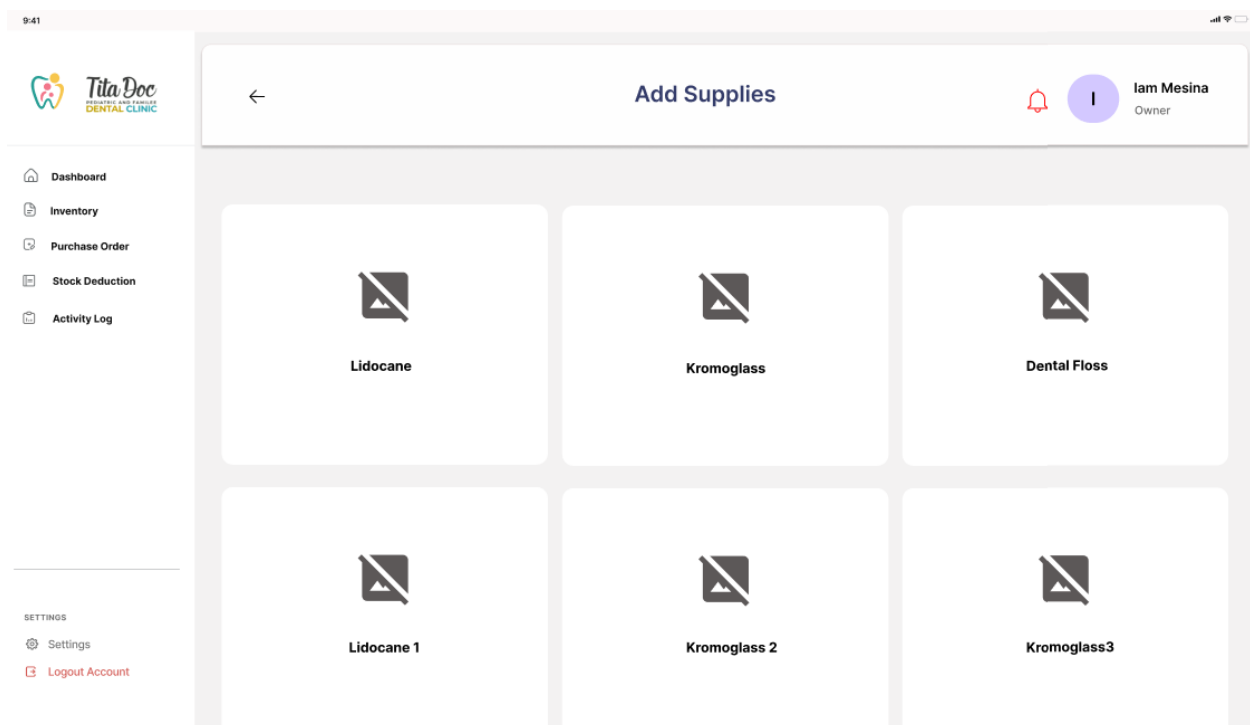


Figure 18.3. Add Supply in PO

Stock Deduction

This module allows the owner to deduct supplies. This is also where the owner will approve and reject submitted stock deductions. The owner can also access the deduction logs section.

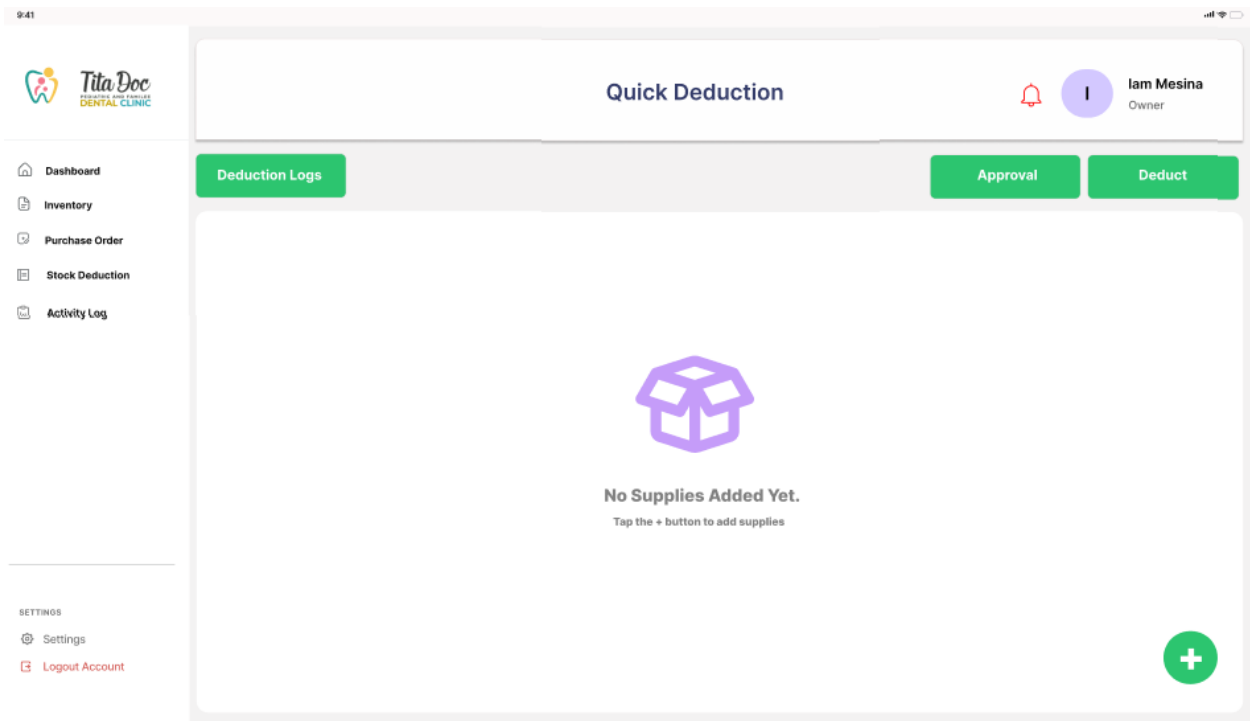


Figure 19. Stock Deduction Owner

Activity Logs

This module shows interactions within the application such as log in, inventory CRUD operations, stock deductions, purchase order creation, and approval. This is only visible to the owner and admin roles.

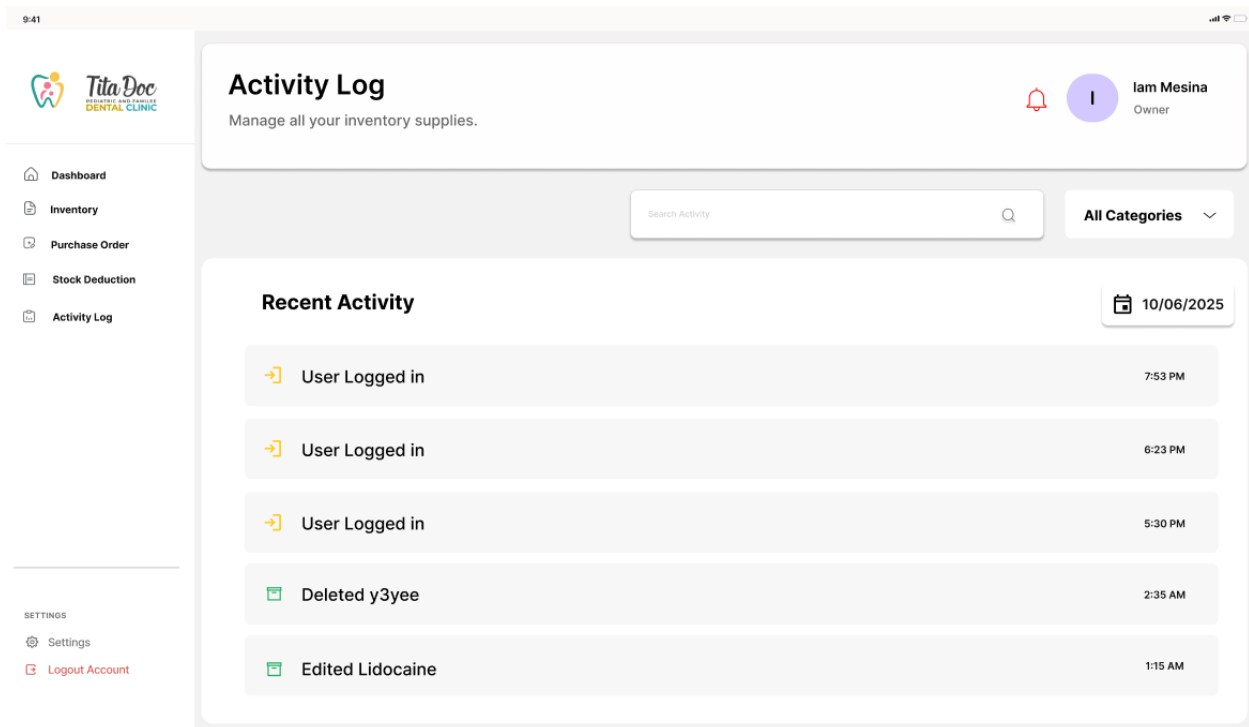


Figure 20. Activity Logs Owner

Settings

This module allows customization of the app's appearance, notifications, and account preferences. The owner can manage user accounts, update profile details, reset passwords, and perform data backup and restore to ensure system security and reliability.

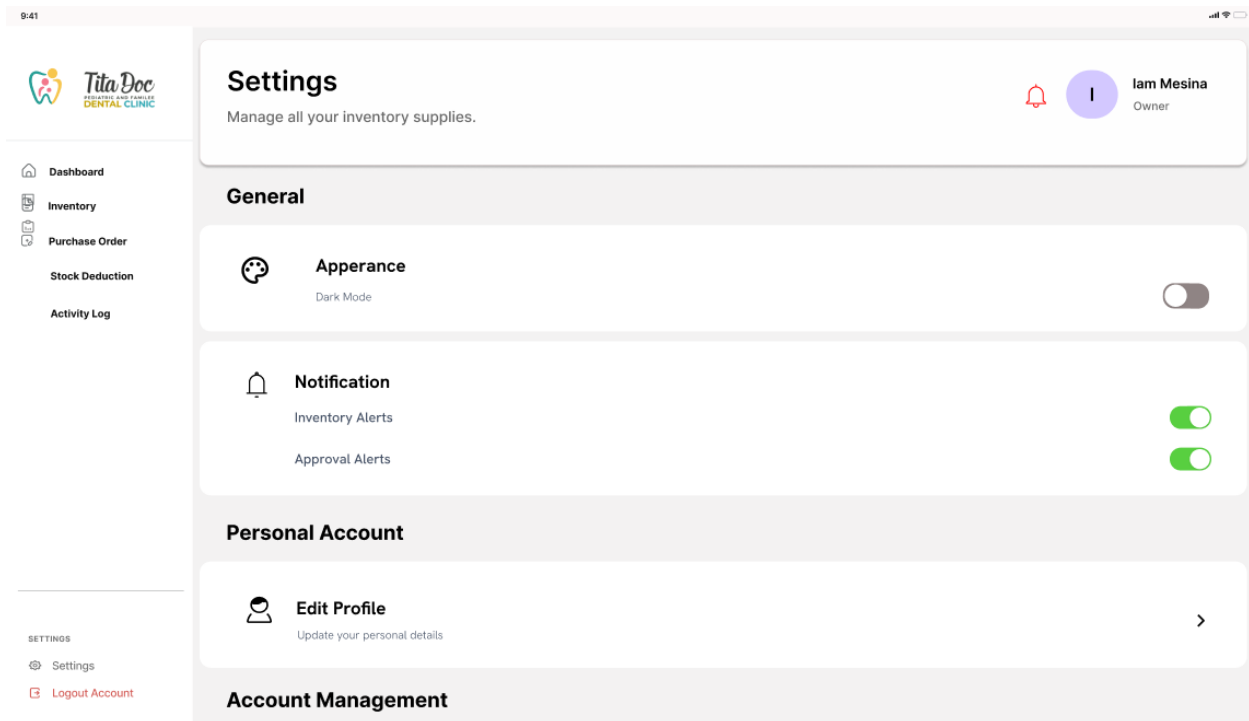


Figure 21.1. Settings Owner

Tita Doc
DENTAL CLINIC

9:41

← Edit Profile

iam Mesina
Owner

User ID
641884DFS-DAD84WE-CAsad7-2DS5Ads152

Name
Iyam Mesina

Username
Owner

Email
lammesina013@gmail.com

Change Password

Save Changes

Dashboard
Inventory
Purchase Order
Stock Deduction
Activity Log

SETTINGS
Settings
Logout Account

Figure 21.2. Edit Profile

Tita Doc
DENTAL CLINIC

9:41

← Employee List

iam Mesina
Owner

Add Employee

M **Marcus Almendares**
marcojohn0831@gmail.com
Admin Active

C **Charles Daroy**
charlesdaroy18@gmail.com
Staff Active

Dashboard
Inventory
Purchase Order
Stock Deduction
Activity Log

SETTINGS
Settings
Logout Account

Figure 21.3. Employee List

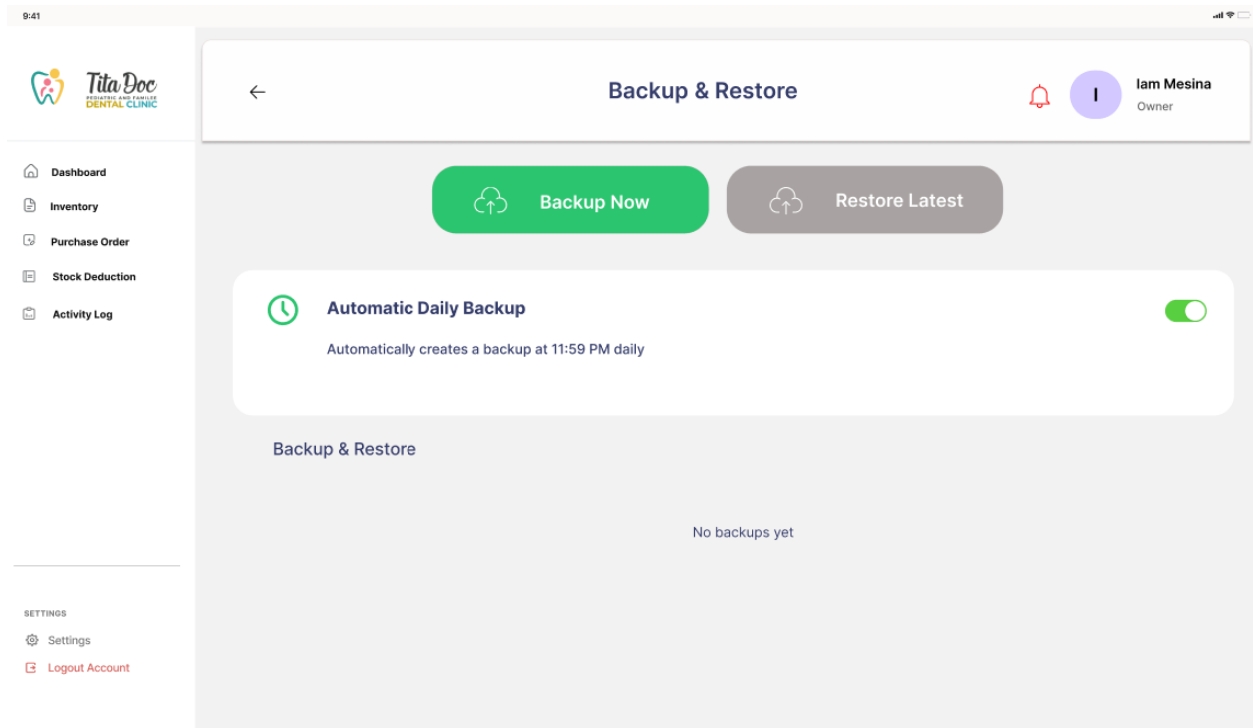


Figure 21.4. Backup & Restore

Admin

Inventory

This contains all the supplies and displays their stock statuses. The admin role can access all the functionalities in the inventory module. These include add, expired, and archived supply, add, and edit category. This is to make sure that the owner retains all permissions within the application, especially the inventory module.

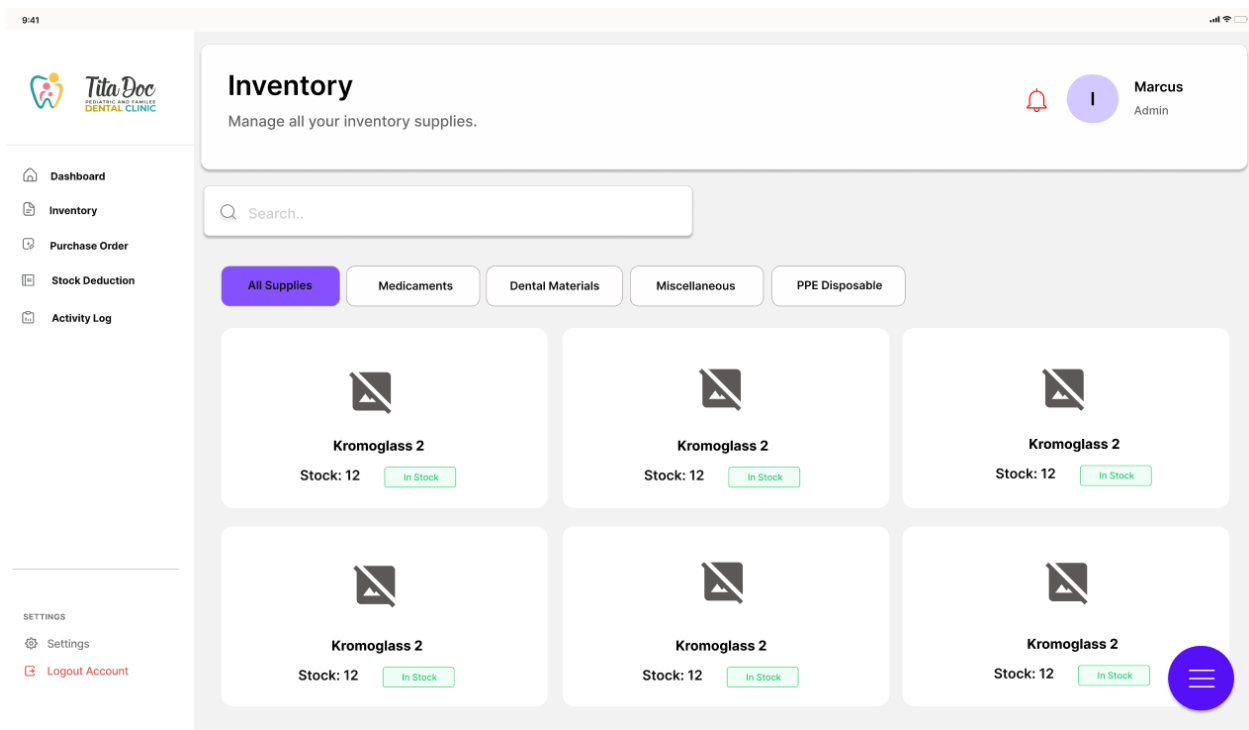


Figure 22.1. Inventory Admin

9:41

←
Add Supply

Upload Image

Item Name *
Type Name *
Supplier Name *

Brand Name *
Cost *
Category *

Quantity
- 5 +

Packaging Unit
Box

Expiry *

Quantity
- 50 +

Packaging Content
Pieces

☐ No Expiry Date?

Cancel Save

SETTINGS
Settings
Logout Account

Figure 22.2. Add Supply

9:41

←
Archived Supplies

Dental Floss
Archived

Filing Repair
Archived

SETTINGS
Settings
Logout Account

Figure 22.3. Archived Supplies

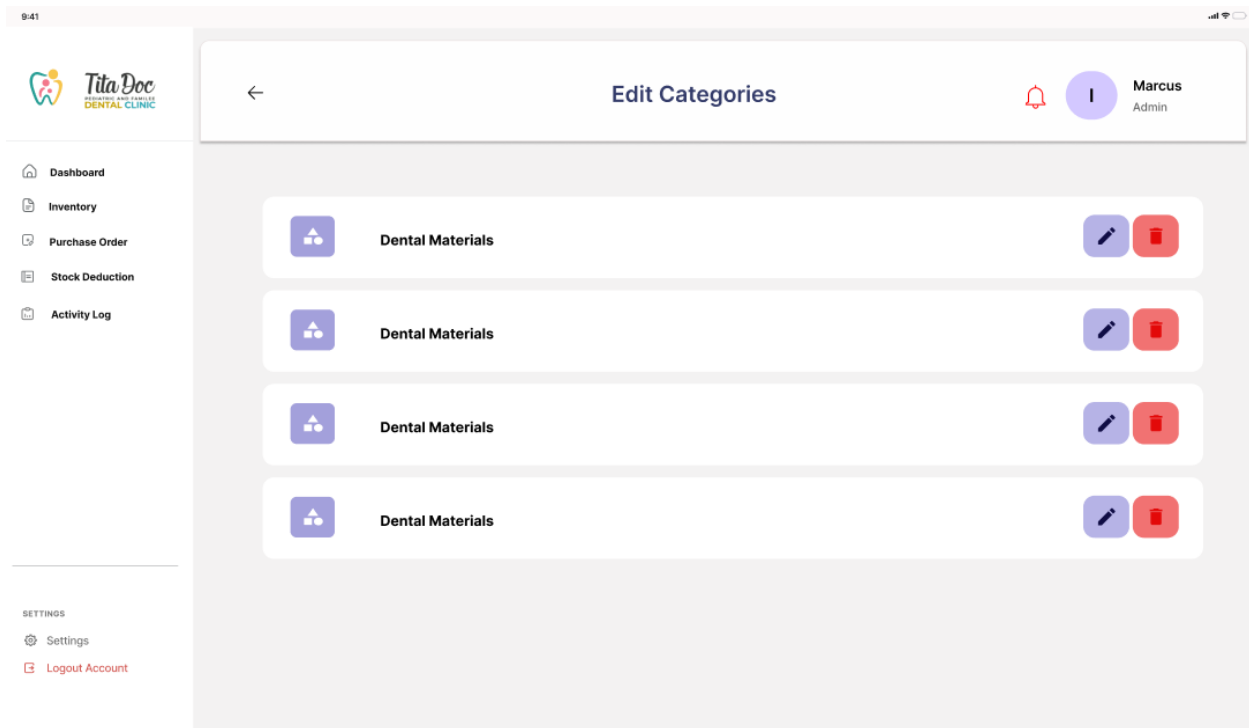


Figure 22.4. Categories

Purchase Order

This module allows the admin to create purchase orders, and they are also able to flag supplies as “received.” Once all supplies/suppliers are received, the user can input the receipt number, who received it, and an image of the physical receipt given by the supplier. This role can also approve or reject the PO before updating the inventory.

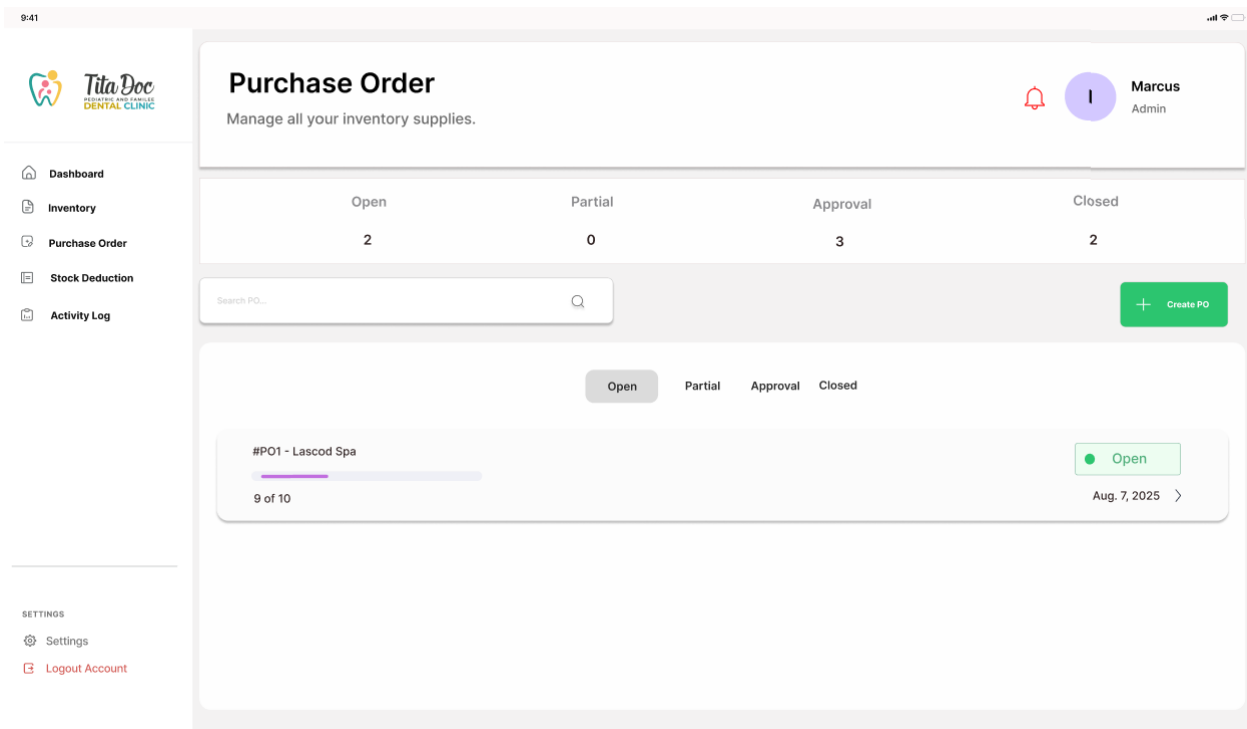


Figure 23.1. Purchase Order Admin

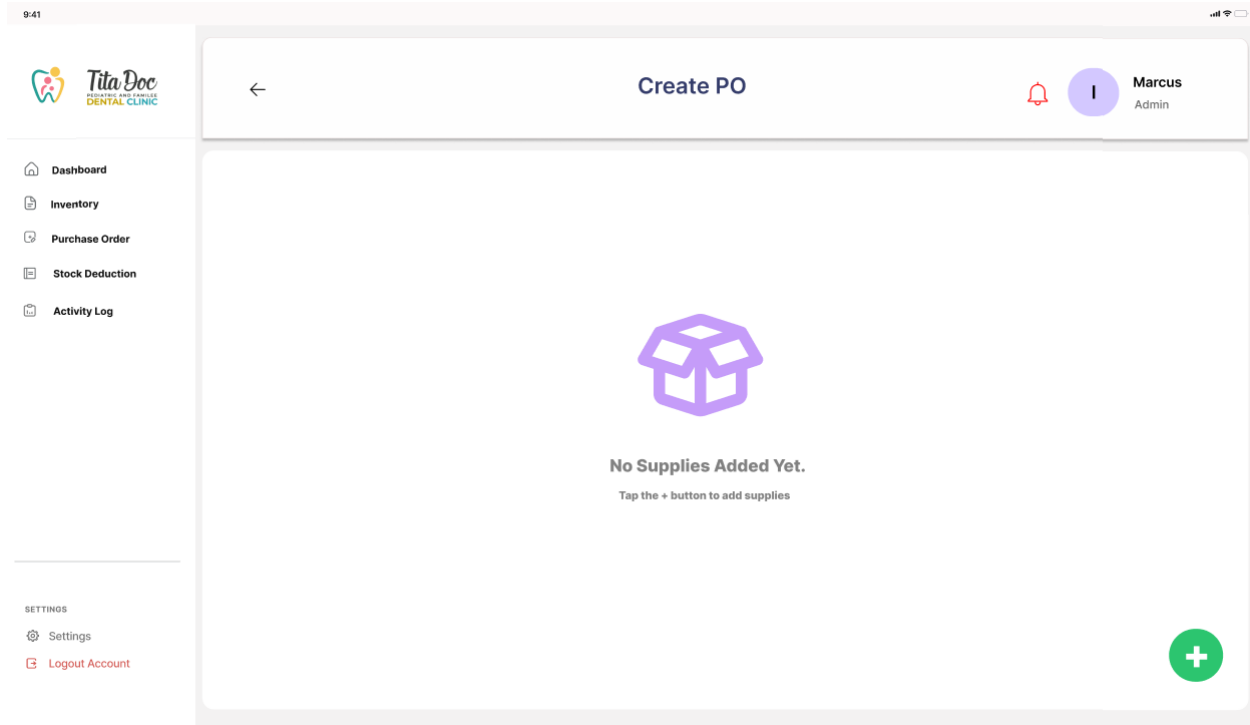


Figure 23.2. Create Purchase Order

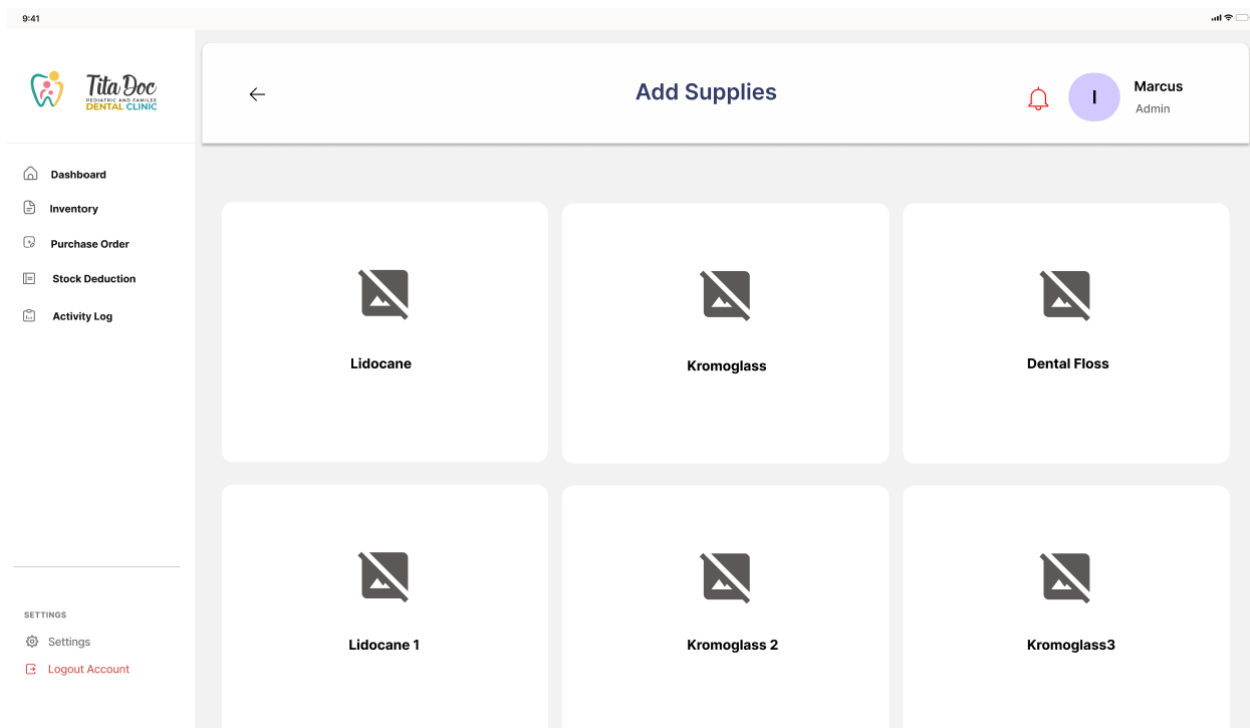


Figure 23.3. Add Supply in PO

Stock Deduction

This module allows the admin to deduct supplies. The admin can also approve, and reject submitted stock deductions.

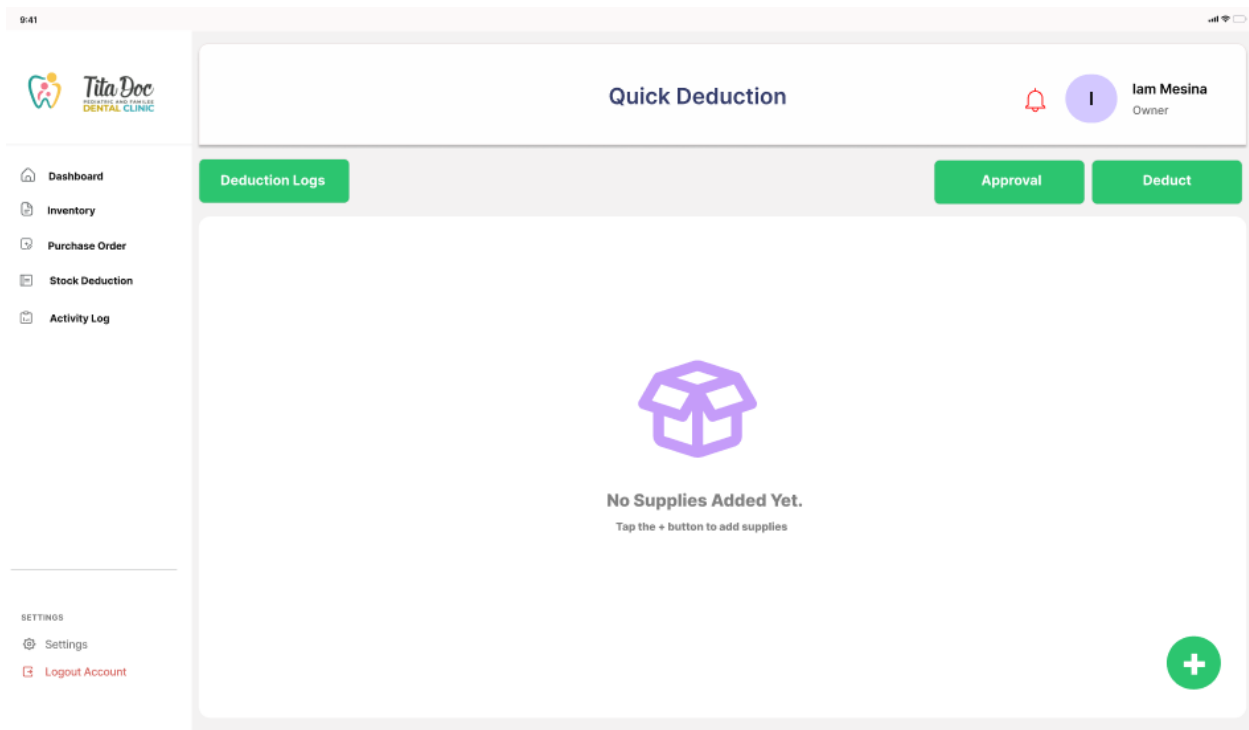


Figure 24. Stock Deduction Admin

Activity Logs

This module shows interactions within the application such as log in, inventory CRUD operations, stock deductions, purchase order creation, and approval. This is only visible to the owner and admin roles.

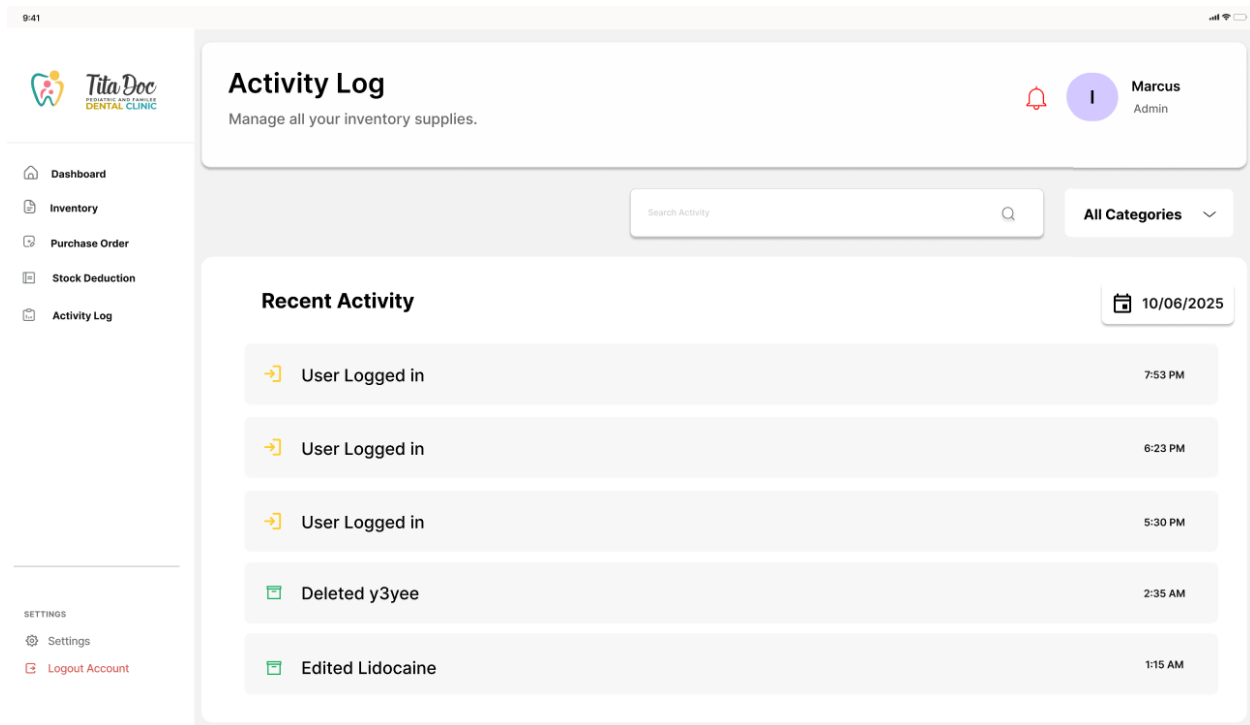


Figure 25. Activity Logs Admin

Settings

This module allows customization of the app's appearance, notifications, and account preferences. The admin can only manage staff accounts, update profile details, and reset passwords to ensure system security and reliability.

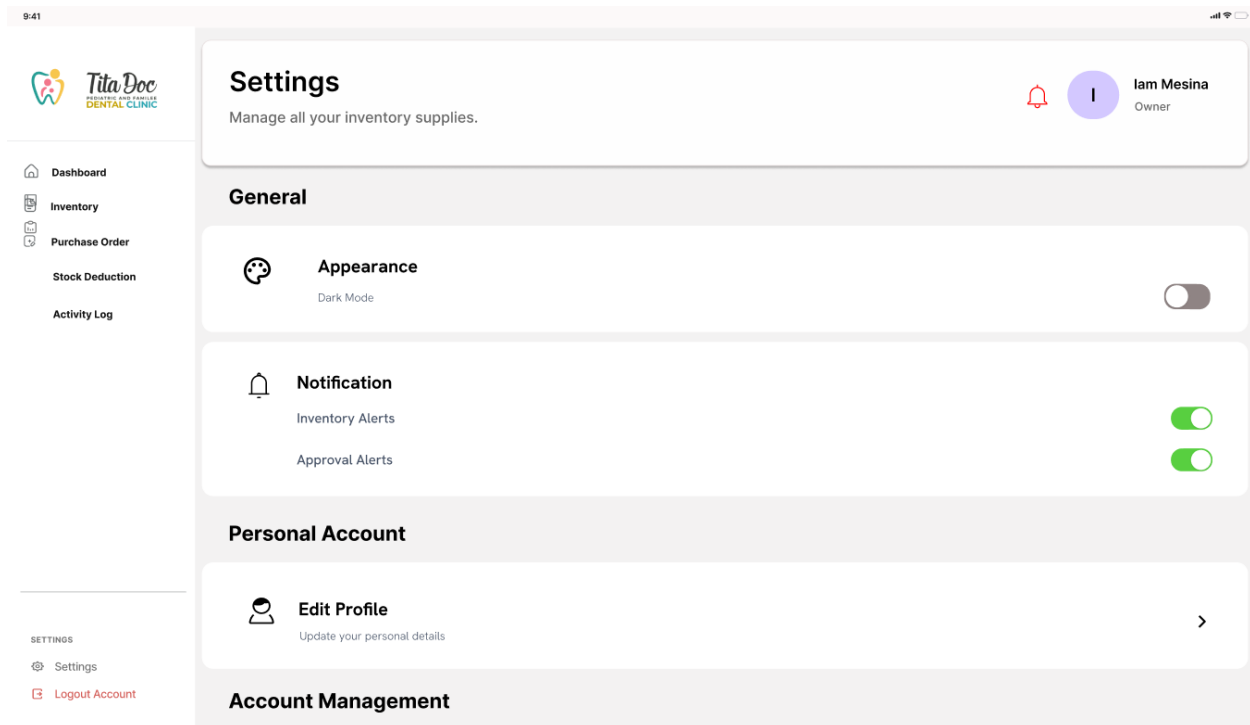


Figure 26.1. Settings Admin

9:41

 Tita Doc
DENTAL CLINIC

Dashboard
Inventory
Purchase Order
Stock Deduction
Activity Log

SETTINGS
Settings
Logout Account

Edit Profile

641884DFS-DAD84WE-CAsad7-2DS5Ads152

Marcus Almendares

Admin

almendares.johnmarcus@gmail.com

Change Password

Save Changes

Figure 26.2. Edit Profile

9:41

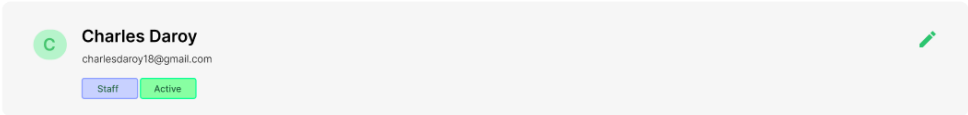
 Tita Doc
DENTAL CLINIC

Dashboard
Inventory
Purchase Order
Stock Deduction
Activity Log

SETTINGS
Settings
Logout Account

Employee List

Add Employee



Charles Daroy
charlesdaroy18@gmail.com
Staff Active

Figure 26.3. Employee List

Staff

Inventory

This module only allows the staff to add supplies and expired supplies. The system also allows the role to edit supply details except for expiry dates.

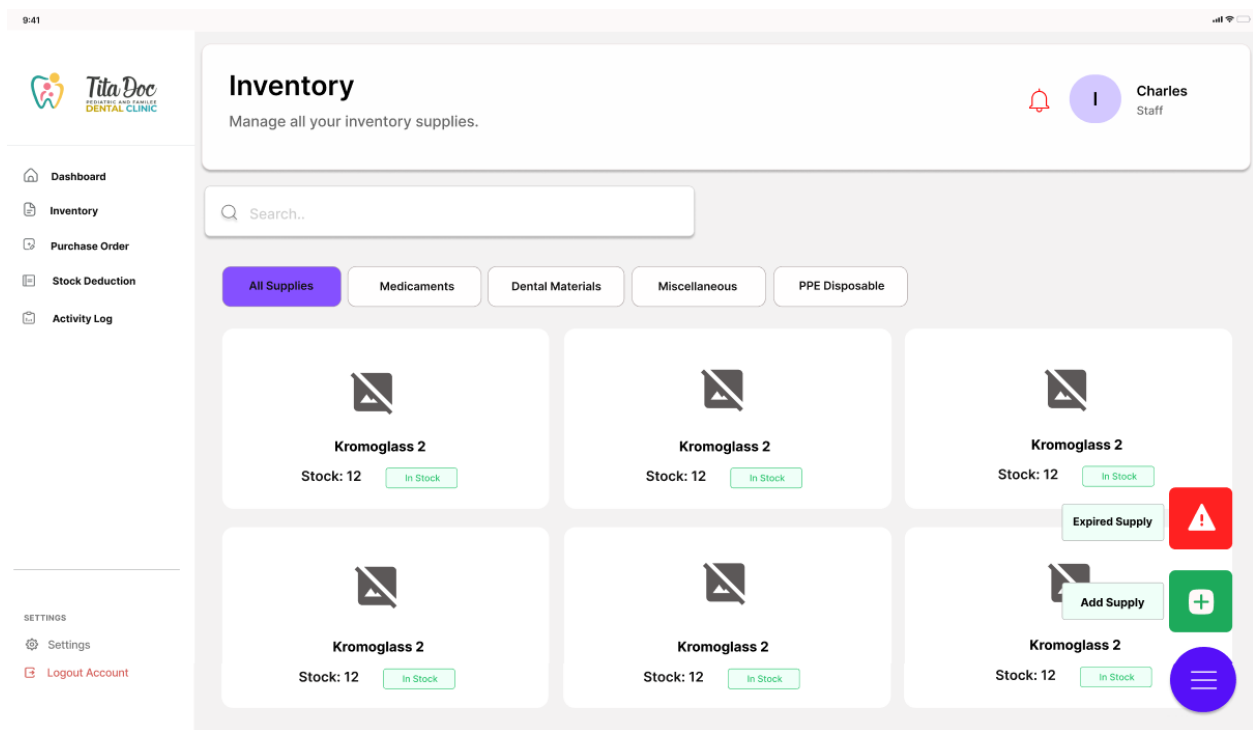


Figure 27.1. Inventory Staff

9:41
Add Item
Charles Staff

Dashboard
Inventory
Purchase Order
Stock Deduction
Activity Log

Item Name *

Type Name *

Supplier Name *

Brand Name *

Cost *

Category *

Quantity

-
5
+

Packaging Unit

Box

Quantity

-
50
+

Packaging Content

Pieces

☐ No Expiry Date?

Cancel

Save

Figure 27.2. Add Supply

Purchase Order

This module allows staff to create purchase orders, and they are also able to flag supplies as “received.” Once all supplies/suppliers are received, the user can input the receipt number, who received it, and an image of the physical receipt given by the supplier.

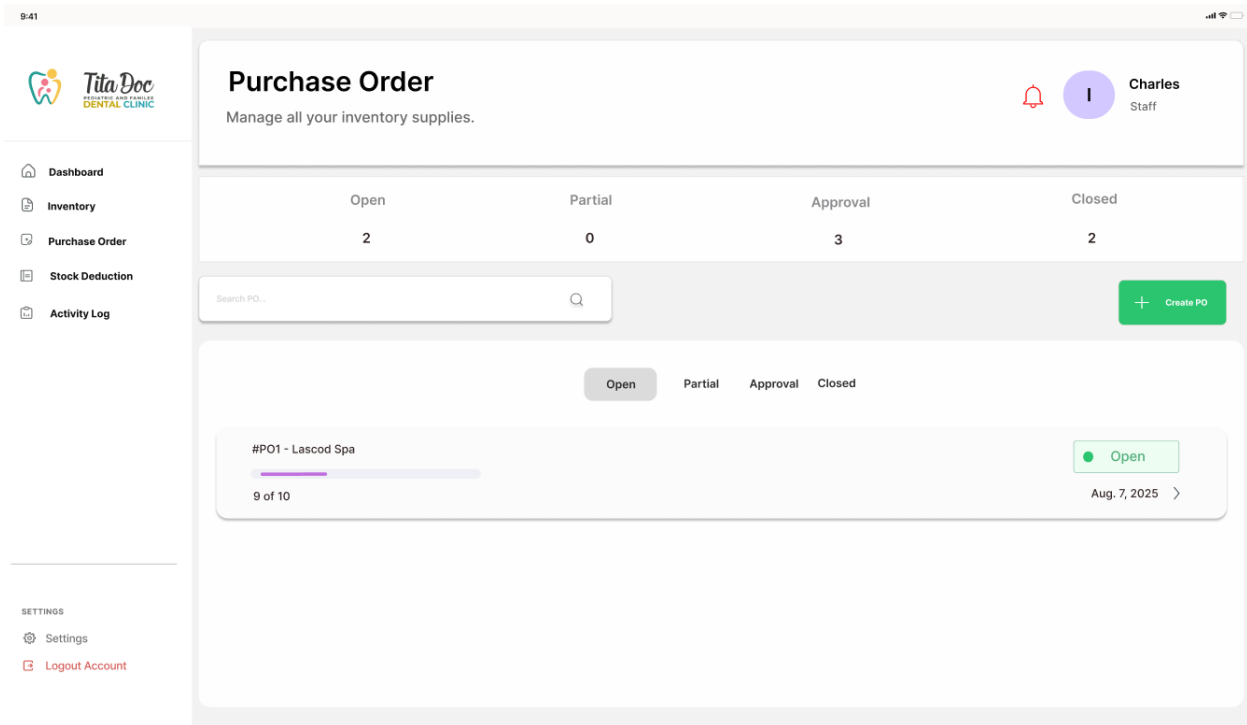


Figure 28.1. Purchase Order Staff

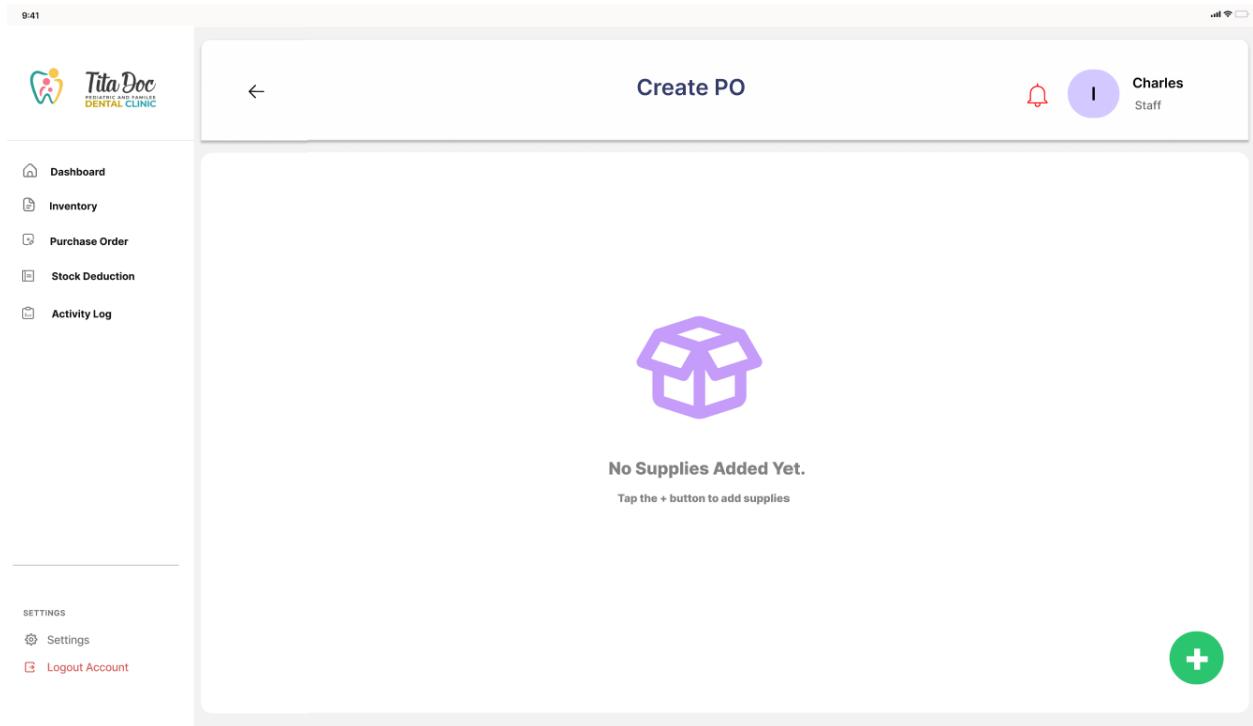


Figure 28.2. Create Purchase Order

Stock Deduction

This module allows the staff to deduct supplies. The staff can only submit stock deductions.

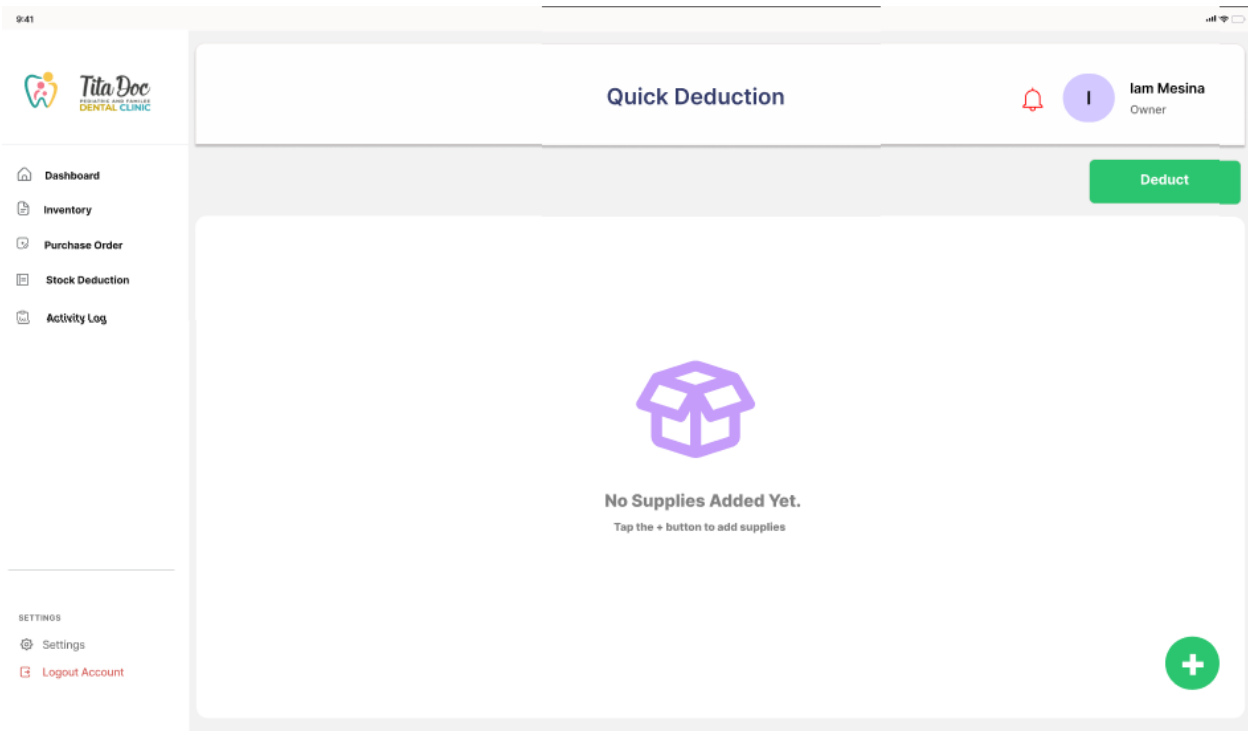


Figure 29. Stock Deduction Staff

Settings

This module allows customization of the app's appearance, notifications, and edit their own profile details such as name, username, email, and password. Unlike the owner and admin, the staff cannot manage other accounts or perform backup and restore operations.

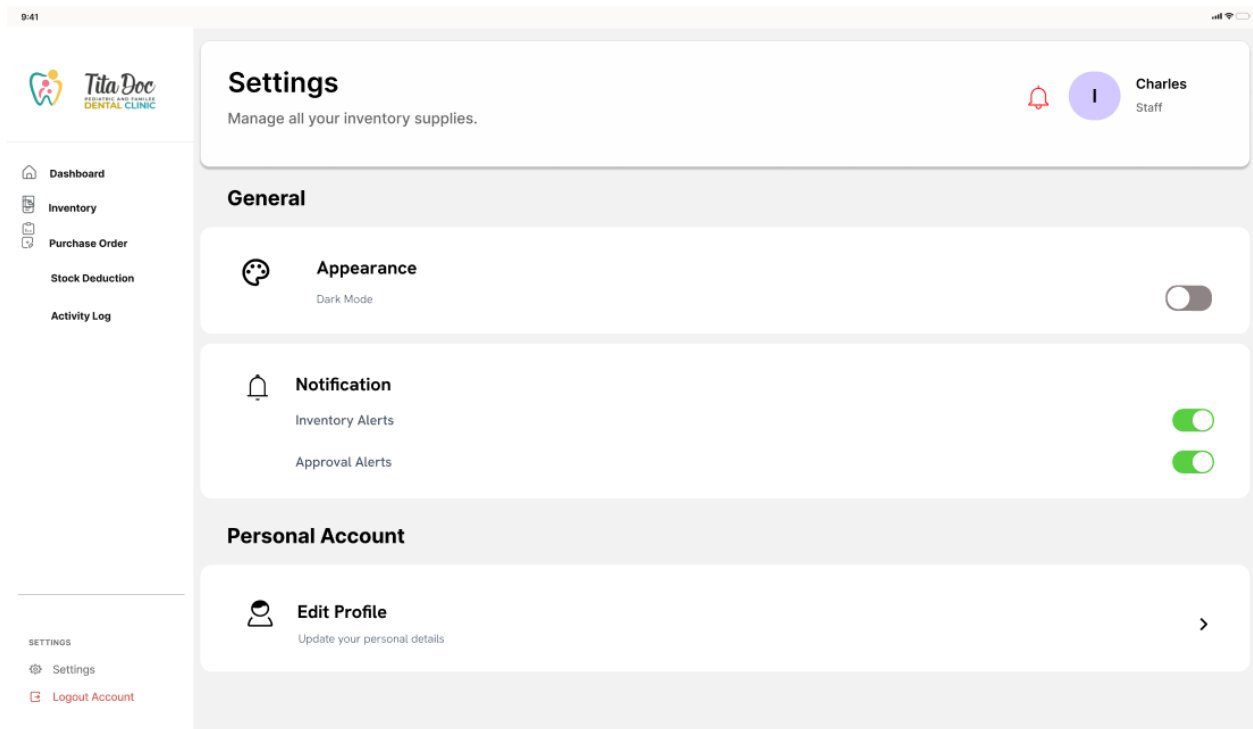


Figure 30.1. Settings Staff

9:41

**Tita Doc**
EDUCATION AND SERVICE
DENTAL CLINIC

Dashboard

Inventory

Purchase Order

Stock Deduction

Activity Log

SETTINGS

Settings

Logout Account

←

Edit Profile



Charles
Staff

User ID

641884DFS-DAD84WE-CAsad7-2DS5Ads152

Name

Charles Daroy

Username

Staff

Email

daroy.charles@gmail.com

Change Password

Save Changes

Figure 30.2. Edit Profile

Implementation Plan

The Android-based Inventory Management System for Tita Doc Pediatric and FamiLee Dental Clinic will be implemented to help Dr. Lee and her staff efficiently monitor and manage their dental supplies. The app will be used on the clinic's tablet but can also run on Android phones. The app requires a minimum of Android 8.0, 4GB RAM, and 64GB of internal storage. An internet connection is needed for real-time data syncing. The app was developed using Dart Programming Language through Flutter framework, while HTML for email template is used for reset password. It uses Supabase for authentication, database management, real-time updates, serverless backend functions, and storage. Specifically, the project integrates the Supabase Auth API for user login and session management, the Database API for supplies, purchase order, activity logs, etc., the Realtime API for live data synchronization, the Edge Functions API for scheduled automated backups, and the Storage API for uploading and backing up files such as images and backup files.

Before deployment, the developers will perform compatibility and load testing to ensure the app works properly on the devices. The APK will be distributed through GitHub offering effective version control, where authorized users can download and install it directly. Security measures such as role-based authentication and backup and restore functions will be implemented to protect clinic data. These features ensure that only authorized users can access or modify records while keeping all information accurate and secure.

CONCLUSION

The Android-based Inventory Management System for Tita Doc Pediatric and FamiLee Dental Clinic's usability, value, and conformance level demonstrated by the alpha and the results of the beta testing are summarized in the conclusion of the system. It is evident that the system has successfully met its general and specific objectives.

The conformance level has resulted in the system's alpha testing has indicated to be of "High Quality" which fully meets the functional and non-functional requirements testing. The functionality testing (86%) confirms that the system fulfilled its intended operations and processes with improvements being needed. The reliability testing (93%) demonstrated that the system performs consistently and dependably under repeated operations, maintaining data accuracy and stability even after prolonged use. The usability testing (96%) showed that users found the system interface intuitive and user-friendly, featuring clear labeling, logical navigation, and efficient layouts that enhance system process performance. The performance efficiency testing (100%) indicated that the system delivers fast response times, quick loading of data, and efficient use of resources, ensuring smooth real-time transactions during clinic operations. Lastly, the security testing (100%) confirmed that the application enforces proper access controls, secure authentication, and data protection measures that prevent unauthorized access and maintain data integrity. This reduces the clinic's reliance on manual recordkeeping, improving task-efficiency, and ensures a smoother user experience.

The system's ability to maintain ideal inventory levels by real-time stock level changes, low stock warnings, and expiry management reduces the possibility of stock outages and improves tracking of expiration dates. The operational efficiency is further improved by how interactive and user-friendly it adds new inventory supplies, creates purchase orders, and deducts stocks using stock deduction with easy steps. The tracking of purchase order status was found intuitive with concise receiving of supplies from suppliers; receipt, recipient, and uploading of receipt image improves the connectivity of the physical receipt and digitalized purchase order. The integration of partial orders further enhances the accuracy of updating supply stocks within the inventory.

With the client's clear agreement with the development of Android-based inventory management system for Tita Doc Pediatric and FamiLee Dental Clinic as evident in the beta testing, the clinic was provided a way to monitor, organize, and update clinic supplies. Provided by a simple way to update and monitor their inventory in real time, the system achieves its objective in creating a centralized inventory for organizing the clinic's supplies. With the integration of expiration tracking in the system, ensure safe and proper usage of supplies. The application clearly notifies approaching expiry dates through alerts. This feature will help the clinic prioritize the usage of supplies following the First Expiry, First Out (FEFO) method, ensuring that closer expiry dates are used. The developers also implemented a reporting feature which organizes and prints summaries of inventory data for documentation and external use. These reports allow users to generate comprehensive reports which contain all inventory and purchase order data. A basic purchase order feature was also developed for restocking; The purpose was further expanded into keeping track of supplier orders, managing partial deliveries, and helping the client to track their mode of payments using the remarks section. This feature provides a structured way for the clinic to manage restocking effectively and maintain accurate inventory records, connecting the physical receipt to the digitalized inventory. The quantities of used supplies connected to their purpose adjust inventory stock levels which keeps the data up to date.

In conclusion, it is evident that the Android-based inventory management system for Tita Doc Pediatric and FamiLee Dental Clinic will be a useful component to the clinic's daily operations. The system addressed the clinic's overall predicament in managing and keeping track of their inventory. Eliminating the need to keep manual records and instead migrate to a digitalized and more organized way of managing their overall inventory.

Recommendations

To further improve the Android-based Inventory Management System for Tita Doc Pediatric and FamiLee Dental Clinic, the following recommendations are proposed. These aim to enhance the system's reliability, efficiency, and adaptability for future development and research. First, it is recommended to enhance any approval system for all staff changes within the application. This will ensure that any modifications made by staff members in the inventory module are reviewed and approved by either owner or admin roles before being updated. This further improves accountability and prevents unauthorized or accidental data changes. Next, a refinement of the purchase order module is advised. Additional features such as supplier feedback tracking and a detailed cost breakdowns (discounts, promo, delivery fees) may improve the clinic's procurement workflow and ensure more accurate order management. Supplier feedback provides a way to track supplier performance, which helps the clinic to determine what suppliers are most reliable. Detailed cost breakdown means showing a more transparent and itemized cost view in the PO (e.g., delivery fee, tax), instead of just the subtotal and total amount. Another recommendation for future development is to add barcode or QR code scanning to make stock updates faster and easier. This feature was not included in the current system because the clinic does not use labeling on their supplies, and implementing scanning would require a workflow where barcodes or QR codes are generated, printed, and attached to items. If the clinic chooses to adopt this workflow in the future, scanning can help reduce manual effort and minimize input errors during stock adjustment.

To increase accessibility, the developers recommend developing a cross-platform version of the system. Extending compatibility to iOS or web-based interfaces will allow the clinic to use the application across various devices, enhancing flexibility and convenience for multiple users. A possible integration with MyMedsPH, the clinic's existing patient record system, is also recommended for future versions of the application. The current system focuses only on inventory management and works separately from where the clinic stores patient information. Connecting the two systems in the future would help centralize clinic operations, make workflows smoother, and reduce the need for staff to switch between different systems. This enhancement, however, will depend on whether MyMedsPH allows their system to connect with other software, since this information is not publicly available. Lastly, integrating a supplier side will make reordering from

suppliers efficient, eliminating the need to manually message suppliers using messenger. Once supplies are low in stock, the user can easily access/message the suppliers which provides a centralized way of managing inventory.

Implementing these recommendations will help further enhance the Android-based Inventory Management System for Tita Doc Pediatric and FamiLee Dental Clinic's reliability, improve operational efficiency, and provide the clinic with a more comprehensive and centralized inventory management tool.

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APPENDICES

APPENDIX A. TRANSCRIPT OF INTERVIEW

Interview 1

Interviewee: Dr. Jilliane Lee, Sister(assistant)

Interviewer: Marcus Almendares, Marvin Ancog, Charles Daroy, Michael Mesina

Date of Interview: 02/14/25

Location of Interview: Google Meet

List of Acronyms: JL=Jilliane Lee, SA=Sister(assistant), MA=Marcus Almendares, MV=Marvin Ancog, CD=Charles Daroy, MM=Michael Mesina

[Begin Transcript 00:00:00]

MA: Uhm, before we start, ah, introduce ko lang po yung team members ko uhm, si, we have here Marvin Ancog, ah in charge sa front end, or yung sa designing po nun application, or system. Tapos si Michael William Mesina is ah, sa back end, or siya po yung mismong developer nung sa pag coding, tapos, si Charles Daroy, sa documentation po, and ako po yung ano, project manager, and, so I'll let Marvin start the interview questions na po.

MV: Ah so, may mga few questions po kami, it's about po naman sa business information. So, first question po is, how do you communicate with your clients? Naka phone po ba email social media or what?

JL: Mostly, social media. Ah, on occasion, may mga phones, nag entertain naman kami ng incoming phone calls, tapos, pero majority ano,

SA: seventy

JL: Siguro mga ninety percent, di lang almost ninety percent social media, mm mm.

[00:01:09]

MV: Okay okay. Okay po ah next question is, how do you handle po yung mga customer concerns and yung mga support ng request?

JL: I personally answer everything. Majority nung mga queries online, phone calls during clinic hours. Pag nandito yung sister ko, sister ko yung sumasagot, pero, anything else ako sumasagot directly.

[00:01:37]

MV: Okay po thank you po ah, ano naman po ah, can you, can you briefly describe your business and what you offer?

JL: Okay, I am a dentist specialising in pediatric dentistry, so mostly kids. Yun lang talaga yung ano ko yun yung ah market ko so... yun we offer services, dental services mostly to kids with ah special needs, yun yun, yung mga ano namin.

MV: Ah ayun po yung target market niyo?

JL: Mm mm.

MV: Yung kids po?

JL: Mm mm.

[00:02:08]

MV: Ah, thank you po. So ah how, how long your business been operating?

JL: how long? we started twenty-twenty one

MV: Ohh.

JL: So going three years.

MA: Okay may, may I know po kung ano po yung range nung kids? Ay, kung ano po yung age range nung kids?

JL: Age range?

MA: Opo opo.

JL: From zero to... ano ba pinakamatanda natin?

SA: Si Kobe, nineteen.

JL: Si Kobe? nineteen.

MA: Mmm, okay okay.

JL: Pero sometimes... sometimes may parents eh

SA (speaking over): [inaudible] special needs [inaudible] twenty-three, meron

JL: Sometimes may mga parents din kami pero very rare.

MA: Okay okay. Ah, thank you po.

[00:02:50]

MV: Next question is, do you have other branches? If yes, where?

JL: Yung personal clinic ko, wala. I only have this one. Pero meron akong parang co-partner ako sa isang clinic sa pasig.

MV: San po sa pasig?

JL: Sa may Caniogan.

MV: Mmm.

JL: Pero I'm rarely there. [inaudible] pag ano lang, special cases lang.

[00:03:19]

MV: Ah, okay po. So next question is how many employees do you currently have and what are their primary roles?

JL: Employees... kasama ka ba dun? (Referring to SA) [laughs]

SA: [inaudible] [laughs]

JL: But technically, parang tatlo.

MV: Ohh.

JL: Yung kapatid ko kasi isa eh so, well sweldado din naman siya, so technically employee rin naman siya. Yung... bale siya, mostly on, ah,

SA: Front desk.

JL: Clinic manager staff. So front desk. Minsan pag kami, although ano kasi siya

SA: Sales.

JL: Hindi all hindi... hindi all the time eh, yung pag ayos nung mga resibo, sales, ah reminder sa patients

SA (speaking over): [inaudible]

JL: Yun minsan communication and patients also, yung dalawa, chairside assist. So sila yung mismong katabi ko kapag naghahandle ng patients. In terms of linis ligpit everything silang dalawa rin.

[00:04:18]

MV: Ah, okay po. So sa ahmm questions naman po is about sa mga challenges na, na parang nafeface niyo dun sa clinic niyo. So first question is, what are the biggest challenges your business is facing, business is facing in terms of daily operations and management.

JL: Ah ako, personally ang problema ko ano, keeping track of my inventory.

MV: Mmm mmm.

JL: Kasi ano lang eh, kasi ako rin yung in charge almost inventory eh, so ano lang siya, ah, hindi ako masyadong nakakapag document properly, so yung inventory sa ano namin sa mga

SA: Gamit.

JL: Sa mga gamit, ah, chinecheck ko lang siya.

MV: Manually?

JL: Visual

MV: Manually na po?

JL: [inaudible] tinitignan ko, oo. Mano mano lang na ganun.

MA: Sinusulat lang po ganun?

JL: Di wala nga eh, sinisingit lang talaga eh

SA (speaking over): Wala [laughs], sinusulat lang siya if ever [inaudible] kailangan.

JL (speaking over): Kailangan na bilhin. Yun lang yun lang yung pinaka nagiging inventory namin.

MA: Okay okay.

JL: Tapos... ano pa ba?

SA: Schedule.

JL: Di scheduling ako rin kasi, kasi medyo magulo yung scheduling ko, ang hirap siyang, mahirap siyang idelegate sa iba, kasi nag aano rin ako eh, I go to other clinics also for, for consultations yung mga ano, may mga on call jobs din ako. So ang hirap idelegate, kasi minsan i have to communicate with the other clinics regards to my ano, schedule. So mostly yun. Ano pa ba? Hmm... bills? Minsan bills nakaka

SA (speaking over): Oo nag kakagulatan? Though, kasi [inaudible].

JL (speaking over): Nakakalimutan ko, di although. Oo minsan nakakalimutan ko, nagkaroon ng time na parang nakalimutan kong magbayad ng tubig. Na kapag susunod na buwan naging disconnection na siya ganun.

MV: Ah. Bills about dun sa mga ano po, sa mga needs nyo dun sa may clinic nyo?

JL: Utilities.

SA: Oo, utilities.

JL: Utilities.

MV: Utilities po?

JL: Mmm mmm.

[00:06:15]

MV: Ah, okay po. Pero usually po, ano po ba yung mga ininventory nyo? Madami po ba?

JL: Madami! Kasi, we have consumables, consumables meaning gloves, head cap, masks, ah, yung, mga baso...

SA: Syringe tape.

JL: Yung saliva injector, mga ganun.

SA: [inaudible]

JL: Iba, iba pa yung

SA: Mismong material.

JL: Dental materials. See we have a lot of dental dental materials also kunwari yung pang pasta, pang bunot, anesthesia, tapos bawat isang klase nun eh kunyari anesthesia, we have...

SA (speaking over): [inaudible] tatlong klase.

JL: Three types in, na nasa inventory namin, tas iba ibang ah, kunyare needle yung mga ganun, so medyo ano rin siya, medyo marami, yung kailangan naming i-keep track.

MV: Okay po.

JL: [inaudible] [laughs]

[00:07:08]

MV: Ah, next question is, are there any task that take up too much time or repetitive?

JL: Hmm... ano ba? Yung ano, yung keeping track sa resibo na ano nung mga binili.

SA: Mmm, oo.

JL: Yun yung minsan yung ano, kasi kunyare, may binili ako ngayon tas pinadeliver, pinapa pinapatabi ko sa kanila yung resibo. Tas minsan pag kunyari “oh yung ganito pa ah nabayaran na ba natin?” hahanapin na namin, hahanapin na naman namin siya isa isa. So yun yung isa sa mga ano namin na minsan, ah, yung tracking also in terms of payment dun sa mga suppliers namin kasi min-, hindi lahat sila ah, outright na nagbibigay ng tseke, kasi most of the time check payments eh, kaliwaan kami, kahit installment yun naka tseke lahat.

SA: Paiba ibang [inaudible]

JL: May mga may mga instances na hindi agad nababayaran ng check or hindi agad nabibigay yung payment, so minsan, sa sobrang ano din... parang may time nagkaroon tayo ng problema dun sa isang ano eh, sabi nila hindi pa daw kami nagbayad, eh alam ko nabayaran na namin, basta yung ganung mga ah, issues minsan.

MV: Mmm, So...

JL (speaking over): Suppliers. Yun yung common na nagiging ano.

[00:08:29]

MV: Oh, so yun, ayun po siguro yung ano mga... eto rin kasi yung tanong ma'am eh, parang nasabi niyo na rin po yung sagot, "Do you counter errors in data entry, financial records or customer transaction?" so ganun po yung problem na naencounter?

JL: Ahhh... wala kaming problema sa data entry kasi wala kaming...

SA: Papers [inaudible].

JL: Yung yung inventory na ano talaga, na... record. Wala kaming ganun eh. Tinry ko kaya lang...

SA: Hindi na [inaudible].

JL (speaking over): Yun lang initial inventory. Tas pagka may nilabas na, hindi na namin natatrack ulit.

[00:09:08]

MV: Hmmm... okay po. So ang next question is, are there challenges in managing employees such as attendance, payroll or task assignment?

JL: hmmm since maliit lang kasi kami, hindi ganun, hindi masyado pa. Kasi tatlo lang naman sila, yung regular na ano... eto iba naman ang usapan kasi namin nito eh (Referring to SA), iba din dun sa dalawang employee ko talaga. Although... siguro ang magiging next na ano ko is once I start ah dun sa mga ano deductibles nila, yung mga government mandated na... ano nila, ano tawag dun (Asking SA)? Government mandated...

SA: Benefits.

JL: Benefits. Yun. Once na nagstart na yun, kasi syempre maiiba na yung payroll mo. Previously kasi ah, they have a fix salary na wala pang deduction, kasi they're still in their ano, yung... trial ano pa nila. Oo. So ngayon, paregular na sila, so I have to include na yung deductions ng, ng mga government mandated benefits nila ah, I'm planning to, to have din na, ano ang tawag dun (Asks SA)

SA: [inaudible]

JL: Health insurance... yung ganun so, yung kung papano ko icompute, ididivide? Papano ko babayaran? San ko babayaran? Yung mga ganun, kasi wala naman akong ano talaga eh, clinic manager na maghandle nun. So everything falls under my responsibility.

[00:10:43]

MV: Mmm... Medyo mahirap rin po pala. Heh[chuckle]. So next question is about naman po sa may technology.

JL: Mmm.

MV: So first question is, do you any, do you use any software or tools to help with your business? If yes, which one?

JL: Ya, meron akong... ahmm

SA (speaking over): Dental [inaudible]

JL (speaking over): Ano tawag dun? Parang dental record system. Sa online siya so all my patient records are there, payments nila. Yun naman ano more less on track yun. Kasi ano na eh since November of last year, we started fully digital na, na charts namin so ah, wala na kaming paper charts ngayon, so lahat naka encode na dun. Ah... in terms of technology ano pa ba? Oh x-rays. Digital na rin yung x-rays namin, so everything lahat nasa computer na.

MV: Naka store lang po sa computer?

JL: Hmmm... Yung ano namin, yung chart, cloud. Web base kasi siya so nakacloud lang yung ano namin yung data. Pero yung sa x-ray, yun sa computer nakastore yun.

[00:11:52]

MV: Ah. Okay po. Next question is, what are the main problems with your com-current, ay. What are the main problems with your current system, if any?

JL: Ah kay ma dun sa ano...

SA (speaking over): Update update na hindi naman kelangan.

JL: [laughs] kasi internal na nila yun eh. Ano naman, so far... Generally happy naman ako dun sa ano nila dun sa system nila. Nagiging problema lang namin, pag nagkakaroon sila ng updates yung mga previous features nila na okay na sana

MV (speaking over): nababa-[inaudible].

JL (speaking over): Nababago.

MV (speaking over): Nag a- ahh.

JL: Alter na, totally, na parang... ah, kunyari yung dati yung, yung pag encode namin ng certain procedures malinaw, maayos. Nung inayot (inayos), nung binago nila yung update nila, nawa, may mga certain ano may certain entries kami na nawala. Eh buti na lang, yun yung mga previous records namin na may paper pa. So, we're able to track pa, na ano pa namin, na review pa namin ulit, na double check pa namin uli, inincode namin ulit dun sa... dun sa records nila before.

MV: Ah, okay. So ano, pwede pa po ba tanong ko if what ano yun, what web po yun?

JL: Ano mymeds...

MV: Mymeds.com po?

JL: Mymedsph.

MV: Ah. May ano may monthly ano po?

JL: Mmm mmm, may subscription siya. Ah, parang, annual kasi yung inaano ko. Mas mas mura kasi pag annual eh. Parang mga eight five siya annually. For the system itself. Although ano naman, sa kanila mabilis naman yung ano nila trouble shooting din, kasi parang chachat mo lang sila sa Facebook. Pag may problema ka natutulungan ka naman agad.

MV: Hmmm, okay po.

JL: Kasi since local din sila.

MV: Ay phil ano lang din po?

JL: Oo, Philippine ano yun, Philippine company yun so which is easier kasi nakaconform talaga siya sa, sa ano ng practice dito sa Pilipinas. Tsaka yun nga ang, ang trouble shooting nila dito lang din mga Filipino lang din.

[00:14:03]

MV: Hmm, okay po. Ah next question is, how do you keep track of finance financial transaction, do you face any issues?

JL: Financial transactions. Dun sa ano dun sa record keeping ano namin meron nakalagay dun. So we can encode there yung binayad ni pasyente. Actually, we can encode also yung mga expenses. Yun lang hindi talaga lang ako, hindi ko na lang natatrack. Kasi may ano rin sila dun parang may summary din sila, kung gaano kalaki yung income na pumasok, gaano kalaki yung, yung expenses for the month. Kung tutuusin, andun na siya, hindi ko lang na mamaximize kasi hindi ko na iincode.

[00:14:47]

MV: Mmm. So next question is do you still use manual processes for some tasks? If yes, which one?

JL: Manual in terms of what?

MV: Manual process [inaudible]

JL (speaking over): Hindi na.

MV: Hindi na po?

JL: Hindi na. Ngayon hindi na kasi I, I, I, ay hindi meron akong part na... paper parin. Pero kasi ano yun eh, consent. So I'd rather keep it in black and white. Ano lang din for legal purposes para at least kung macorrupt man yung cloud nung ano eh

MV (speaking over): [inaudible]

JL (speaking over): sigurado meron akong [inaudible] na ano na pirma ng pasye pasyente. Yun lang naman pero anything else, records, charts, ah, procedures na ginawa sa kanila, yung mga nabill na procedures nila, all are in the cloud.

MV: Ahh. Cloud po nung kaninang binanggit niyo na site?

JL: [inaudible] si mymeds.

[00:15:45]

MV: Ah, okay po. So next question is, would, ah would automating certain task help, if if so, which one? So parang, ayun na po siguro yung ano noh? Yung mymed? Yung [inaudible]...

JL (speaking over): Oo.

MV (speaking over): [inaudible] automating task.

JL: Oo.

[00:16:01]

MV: Ah, next question na po tayo. So do you, do you or your employees have prior experience using business software or management systems?

JL: Wala. Yun na yung pinaka ano namin, pero yung in terms of ah, anong software ba? Wala na akong idea eh, yan lang ginagamit ko eh, yan lang ginagamit ko eh [chuckles]. Yung mga payroll payroll. [inaudible]

SA: [inaudible]

JL: Ganun din yun eh, parehas din.

MV: Mmm.

JL: Yun lang wala na maliban sa yung online ano namin record keeping na system yun lang.

[00:16:43]

MV: So next question naman po is about sa data. Ah Marcus ikaw muna.

MA: Ayun nga po, yung unang question po is, how do you store important business data, customer record, sales history, employee details? Yun po.

JL: Ah yung patient records, yung mga previous namin first two and a half years may paper pa yun, so we still keep the paper, kasi legally we're not allowed to dispose it yet, and yun, digitally, meron na kaming digital ano dun sa record keeping system namin. Ah employee records, naka envelope. [laughs] Ano lang sila sa envelope.

MA: Okay. Ah yung mentioned earlier po na kapag nag update yung software na ginagamit niyo, minsan nawawala yung data?

JL: Oo.

MA: May feature po ba na pwedeng mag back up ng data like in case na mawala mababalik po siya dun sa software?

JL: Meron ata siyang pwedeng back up. Kaya lang yung revert to original, parang yun ang wala.

MA: Ah. Parang yung mababalik lang if, yung [inaudible]

JL (speaking over): yung [inaudible] data na inencode mo.

MA (speaking over): Uh huh.

JL (speaking over): Pero kasi yun yung, yung nawawala kasi yung ano eh specifics nung procedures.

MA: Mmm.

JL: Kasi yun, ano siya. So kunyari we have procedures, we have tooth numbers, yung procedures na iniencode don, we manually enter also based on our practice. So meron siyang feature na parang ililista dun, ililista ko lahat ng services na inooffer ko, and then yun yung lalabas dun sa record, parang nag, na aautomate siya, na pag yun ang inencode ko, automatic ito yung singil. So nung nag ano sila, nung nag update sila, hindi na kasi nila inalaw yung double entry per line. Eh before nagdodouble entry ako. So nung nag double entry hindi, nagkahiwa-hiwalay siya tas yung mga tooth number nawala.

MA Mmm.

JL: Pero ano pa rin naman tama yung procedure, mali na lang yung, yung tooth number yung nawala. Which is also crucial.

[00:18:59]

MA: Mmm, Okay. Ah next question ko is, do you need restricted access for certain employees like admin, manager or some-

JL (speaking over): Mmm mmm, mmm mmm. Ano naman may feature din yun na ganun.

MA (speaking over): Uh huh.

JL: So for staff for example kasi, ako yung admin ah, ako open, open lahat. Lahat ng pwedeng changes, operations, lahat ng data available for me but, may account dun yung mga assistant ko, so that they can access also pero limited.

SA (speaking over): Tyaka bawal yung [inaudible]

MA (speaking over): Mmm.

JL (speaking over): Bawal...

SA (speaking over): Presyo...

JL (speaking over): Oo, bawal yung...

SA (speaking over): Mag tatanggal ng procedure...

JL (speaking over): Mag delete delete...

SA (speaking over): Oo.

JL (speaking over): Yun, bawal sila. Bawal mag alter ng... charges

SA (speaking over): [inaudible]

JL (speaking over): Details, yung mga ganun.

SA: Overall.

MA: Ah may specific ano permissions lang po talaga?

JL: Mmm mmm.

MA: Okay

JL: Ano yun, which is something that I can control also. Kasi I can limit eh.

MA: Okay okay.

JL: [inaudible] ko lang.

[00:19:55]

MA: Yung next po is, are there any security concerns about your data such as unauthorized access or data loss?

JL: Di ko alam. Oh wala naman si hindi ko alam kung may mag hahack ba ng dental record

MA: [chuckles]

JL: System. [chuckles]. Di ko sure pero... Wala naman siguro.

MA: Okay.

JL: Kasi wala naman...

SA (speaking over): And so far, wala din.

JL: O, wala naman din silang mapapala diyan eh.

[00:20:18]

MA: Okay. Yung next set of questions po is regarding sa potential system preferences or requirements. Yung unang question ko is, what kind of system do you think could help improve your business?

JL: Ako, for me, ako, personal lang I think something na pwede rin sa phone. Kasi yun yung problema namin dito eh. Hindi siya, wala siyang application.

SA: Web base lang siya.

JL: Web base lang siya. So if I'm going to, to view it on my phone, naka browser parin ako. Which for me, may edad na ako to ma[inaudible], umiikli pasensya ko [chuckles]. Kapag ka magbabrowse ka pa, ieenter mo pa yung, ieenter mo pa yung details mo dun sa browser, magloload pa siya, tapos since phone ang liit niya. Hindi rin siya maximize dun sa space na available sa phone. If, if ano nga, kasi actually minention ko rin sakanila yan eh.

SA (speaking over): [inaudible]

JL: If meron sana silang parang...

SA (speaking over): Version for phone.

JL (speaking over): Mobile version.

SA (speaking over): Oo.

MA (speaking over): Mmm.

JL (speaking over): Kasi for example, expenses. Diba minsan, kunyari bumili kami ng dalawang litrong, ay dalawang galong tubig, papaano ko pa ieencode yun? Mag oopen pa ako ng, ng laptop ko. Ah I think [inaudible] kung kunyari yung...

SA: [shows phone to JL] Ganyan parin siya.

JL: Kung yung... Yan ang liit.

SA: [shows phone to Interviewers]

JL: Buti nasayo [inaudible] yan. Hindi siya kita.

SA: Wait lang.

MA: Liwanag po [chuckles].

SA: [inaudible]

JL: Hinaan mo. Liitan mo, aanuhan mo ata yung brightness.

SA: Ayan, maliit parin siya.

JL: Yung saakin nga putol eh.

MA: [inaudible] yung text, okay.

JL: Sakin putol. So kailangan ko pang iganun yung ano ko, ivevertical, ipapahalang ko pa siya para makita ko lahat. Which, parang para sakin ang tedious pa.

MA: Okay.

JL: So I think kung kunyare ah paglista ng mga expenses for the day. Kasi sometimes, kunyare ultimo meryenda I have to encode, kasi ano yun eh clinic expense yung meryenda na yun for example. It's something that I have to put input sana right away, kaya lang...

SA (speaking over): Kakalimutan lagi.

JL: Since it's not accessible, hindi siya readily available. Nakakalimutan na siyang ilista yung mga ganun. Tas even yung kunyari, inventory tracking. Ah kunyari nagmamadali during the procedure itself, nagpakuha ako ng isang gamit, hindi na para ma, hindi na mabalik ulit dun sa original inventory, para lang malagay ay ito nabawas na natin. Yung mga ganun.

MA: Okay. [clears throat] So parang ano po. Ay wait, may gumagamit po ba sa inyo ng IOS or pure Android po kayo?

JL: Ah, may isa akong phone na IOS pero majority Android, Windows.

MA: So parang...

JL (speaking over): Yung ano din... ay hindi, pwede pala to sa ios. Ay, oo.

MA: Ang nasa isip po kasi namin, kapag website ulit, pwede rin siyang maaccess sa ios. Pero, kapag magfofocus po kami sa application, baka magfocus na kami sa android, since mas matagal po kasi pag inintegrate pa namin sa ios. So baka ang mangyayari, more on application po kami and may web base din kami. So... Ayun nga po para maresolve yung ah problem niyo sa terms of readability and yung tediousness ng ah, pag open ng website lalo na sa pagcheck ng inventory.

JL: Mmm mmm.

MA: Mas okay na po yung...

JL: [inaudible]

MA: Mag open nalang ng application? Okay, tas dun na lang makikita, then pwede na rin magmanage dun.

JL: Mmm mmm.

MA: Parang ganun?

JL: Oo. Kasi kunwari uhmm... For example, meron kasi kaming tablet dedicated for, for use ng patients sa records nandon so kung kunyare ah managing ng, ng inventory, at least if you just open

the app dun sa tablet, o kunyare, oh ayan na nandiyan na siya pipindot pindot ko na lang ng ganun. Faster. Easier.

MA: Mostly sa tablet po ba kayo nag aano, nag ttrack ng inventory?

JL: Hmm, majority yeah.

[00:24:23]

MA: Okay. So if a system were to be developed, what key features would you need po?

JL: Ako yun, ease of use.

MA: Okay.

SA: Tsaka yung...input siguro agad nung x-ray?

JL (speaking over): Tsaka yung accessibility niya talaga. Alin?

SA: Yung hindi na manually ilalagay. Yung pwede na siya iconnect.

JL: Ay hindi kasi... di ano yun... mahirap yun kasi two different applications.

SA (speaking over): System to system?

JL: Oo system to system yun eh. Hindi kaya yun. Ako for me yun, una, madaling gamitin, ah hindi na siya masyadong... complicated. Kasi for example sa amin sa dentistry, marami ring mga dentista na medyo hindi...

SA (speaking over): [inaudible]

JL: Tech savvy. di ba so...

MA: Opo.

JL: Kailangan yung easy to understand. Tsaka yun nga, kasi since very dynamic din yung trabaho namin, kasi kami dito kulang hindi naman kami, hindi madami yung tao namin eh, so in terms of inventory yun madali ding maintindihan ng assistants.

MA: Mmm mmm.

JL: Kasi syempre hindi lahat ng assistants nakapag aral ng hanggang college. So yung iba... Very minimal din ang knowledge in terms of technology, internet, yung mga applications, ganun. Kung madaling maaano sa kanila yung kunyari o ah... yung in and out ng mga inventory, bawas ka lang isa, tapos yung total inventory mo dun kunyari from four magiging tatlo na lang, tas may warming system siguro, oh last inventory na isang piraso na lang, isang box na lang. Naka warning na siya, naka ano na siguro [inaudible] mga to buy list or yung mga... Low in ano na low in supply na mga gamit. Pwede natin iano, very helpful.

[00:26:09]

MA: Okay po. Ah in terms of customers po per day? Sa tingin niyo po, ilan po yung estimate niyo na pumapasok, like kung ilan yung naaassist niyo ganun?

JL: Average... Siguro average namin per day mga anim na pasyente. Average. Yeah, kasi may days na isa lang, dalawa lang. Pero may days naman naka twelve, thirteen patients kami parang pinakamarami namin nineteen?

SA: [agrees] [inaudible] pamilya [chuckles].

JL: Oo, isang buong ano yun isang buong...

MA: [inaudible]

JL: Ano lang yun ah, [inaudible] to six o'clock.

MA: Uy [chuckles].

JL: Oo.

SA: [inaudible] dentist na [inaudible]

JL: So yun, in terms of ano naman kasi, mabilis lang yung transition namin. Kung cleaning cleaning lang, yun mabilis lang yun fifteen minutes, twenty minutes tapos na siya. So, pero kasi, yun nga yung yung yung ano nun ah, ang sistema na namin nun so habang kunyari ako nakikipag usap pa ako sa patients, assistant magchachange na ng lahat ng kailangang ichange. Someone yun yung sister ko nandito nun, siya yung in charge sa reception, siya yung naniningil, siya yung nakikipag yung mga final reminder sa parents, para mas mabilis yung ano namin turn over ng pasyente.

MA: Okay. Ah may walk-ins po ba kayong inientertain or pure on scheduling po kayo?

JL: Very rare yung walk-ins. Mostly ano talaga, we encourage, even sa page nakalagay yun na strictly by appointment. Kasi una I'm not available every day. So inaavoid din namin yung maghintay ng matagal. Kasi specially we're dealing with children, we're dealing with people with special needs. Very short attention span. Pag nainip yan, mag tantrums na yan. So, we avoid yung masyadong matagal na waiting time also.

[00:27:58]

MA: Okay, I understand. Would you like ano po ba, would you like automated reports and analytics to help track business performance?

JL: Mmm... That can be helpful [coughs]. Although ano naman, yung dun sa system namin ngayon, parang every, everytime I want to see it meron pwede ko siyang machack macheck. Matatrack ko naman, pero hindi automatic yung update niya kunyari ah, monthly wala wala siyang monthly report na ganun. Pero you can track.

MA: Okay.

JL: Pwede naman.

MA: Ano po [inaudible]

JL (speaking over): Actually, hindi naman masyadong particular.

[00:28:36]

MA: Okay. Last question po is, how important is user friendliness in a system for you and your employee?

JL: Very important. Kasi...

MA (speaking over): Mmm.

JL: Kasi kung hindi... hindi nila naiintindihan yung system na ginagamit, kung hindi nila masyadong ah, gets kung papano gamitin. Mas tataas yung chances of error. Especially data encoding. Ah for example when we are trying to shift yung records namin from paper to digital ah, medyo nagkaroon ng mga ano din yun eh mga error din yun, kasi nagkakaroon ng laps kunyari ah, entry ng time, date. Yung minsan nagkakadoble doble. Yung mga ganun. So very important na madali siyang intindihin. Na kahit hindi mo masyadong itraining yung tao, ganto ganto ganto, ituro mo lang ng ganun...

MA: Mmm.

JL: Magegets na nila agad.

MA: Okay. Ummm sang follow up question lang po ah, okay lang po ba na ano magfocus lang kami sa android tas sa ios wag po muna?

JL: Sure, no problem. Kasi ano naman din eh, ang tablet namin android.

MA: Okay okay.

JL: Mas madali rin siya.

MA: Okay. ah so far yun lang po yung mga questions.

[End Transcript 00:29:53]

Interview 2

Interviewee: Dr. Jilliane Lee

Interviewer: Marcus Almendares, Marvin Ancog, Charles Daroy, Michael Mesina

Date of Interview: 04/29/25

Location of Interview: Google Meet

List of Acronyms: JL=Jilliane Lee, MA=Marcus Almendares, MV=Marvin Ancog, CD=Charles Daroy, MM=Michael Mesina

[Begin Transcript 00:00:00]

MA: Okay so may question po kami regarding sa ano, process niyo po sa clinic-

JL: Mmm.

MA: Yung parang work process.

JL: Okay.

MA: Ano siya, kunwari po, sa ano... sa normal day, pwede po malaman kung pano po yung routine niyo? Like, para po malaman po namin yung process niyo sa clinic.

JL: Anong routine?

MA: Like ano po-

JL: [inaudible] ng patient or starting from opening ganon?

MA: Ah starting from opening po. Like-

JL: Starting from opening... Uh... Assistants ko yung mga nag o-open. So pag open nila, yun bukas lahat, ilaw, aircon, uh nag lilinis sila nang konti before we start, and then yun, may tablet ako dito, chinecheck nila schedule so naka ano naman kami... naka google calendar na yung ginagamit ko para always naka sync yung ano namin, yung schedule for the day.

MA: Okay.

JL: Tapos more and less naman may, minsam may notes ako dun kung ano gagawin sa patient. Pag dumating na si patient, ppull up lang yung chart, nandun din sa tablet, tapos pwede nang iano yun, pwede nang... kasi since digital charting na rin ako, nandun naman na lahat ng kailangan

niya. Then yun we do procedures. Uh, depende, case to case basis kung ano kailangan gawin ni pai- sa patient.

[00:01:27]

MA: Okay, yung sa ano po, yung sa appointments, diba po meron listahan kayo?

JL: Mmm.

MA: Ano po- yung mga nakalista po dun kung ano yung mga gagamitin na materials?

JL: Wala. Wala nakalagay sa listahan ah-

MA: Okay.

JL: Pero uhmm, more or less ano naman na, alam na nung mga assistants ko kung ano yung mga gagamitin. For example, medicaments ba iyan, kung ano yung mga materials na kailangan depending on the procedures na gagawin-

MA: Okay.

JL: So yun naka... may briefing na sila pero wala kaming definitive list na physical, meron lang talaga ano, tinuturo lang namin sakanila before, nung nag start sila.

MA: Ay parang ano lang, tinatatak niya lang sa utak.

JL: Oo, parang ganun nalang siya.

MA: Okay so-

JL: Minsan, kasi yun nga, since there are certain procedures na may kailangan ng mas specific na instrument for example, kukunin nalang sa drawer or kung ay kailangan pang specific na material, kukunin nalang kung saan siya nakalagay.

[00:02:31]

MA: Okay. Yung sa materials niyo po, yung, ano ano po yung categories niyo po dun?

JL: Ah, materials marami yan, consumables, disposables, meron kaming ah... may ano kasi rin yan eh, parang may category per procedure, may mga ganun din siya. May instruments, may

instruments pang pasta, may basic instruments din, may instruments pang bunot, actually, complicated siya. Kasi madami.

MA: Mmm.

JL: Sobrang dami ng materials. Tas ibat ibang... per material may ibat ibang sizes pa siya. Kunyari uhmm... ano ba? Meron kaming tinatawag na burr, yung ginagamit for drill, so yun ibat ibang design, ibat ibang sizes, ibat ibang grits. So medyo... kung kunyari iaano sa inventory yun medyo madaming details yung kelangan.

[00:03:31]

MA: Okay. Yung sa services niyo po, ano po yung mga prices niyo?

JL: Prices?

MA: Opo.

JL: Uhmm... Meron naman kaming base price for basic procedures, for example consultation may base price siya. Uhmm... Cleaning and fluoride may base price rin siya pero, it can vary depending on ah... Kunyari mas complicated yung procedure, may add on siya. So hindi siya always fixed.

MA: Ah okay. So, kapag ano po ba, isa po ba sa mga factor yung price ng materials sa price ng isang service?

JL: Mmm. Actually, with my prac- ah... depende, depende. For example, crowns. Kasi may crowns ako na ginagawa eh. Ah, yun dependent yun sa material kasi, for example, may dalawang klaseng crowns: stainless steel, tsaka yung zirconia. Pag stainless steel gagamitin ko, ibang price siya. Pag zirconia gagamitin ko, ibang price. Pero, for example, restoration sa pasta. Regardless kung ano material ginagamit ko, ano naman parang medyo nilawakan ko yung range, para it would cover both. Kasi there are certain situations na hindi pwedeng ah... kasi parang hindi mo bibigyan yung patient. Ang option would be dapat nasa amin ang final option kasi depending on the case ng pasyente.

MA: Okay.

JL: Yun lang yung- pero yung in terms of the crowns, kasi yun, mahirap siya i-centralize yung presyo nun, kasi sobrang laki ng ano niya eh, ng material cost niya.

MA: Mmm mmm. So ano po, per operation nag depende pa rin po yung price nung sa service?

JL: Ah... more or less naman, nag sistick kami dun sa base price. Bihira naman akong...

MA: Parang di nagbabago?

JL: Rare occasion naman yung nagcchange siya nang onti or tataas nang onti. Yun nga depending on severity dun sa, ah... Actually, yung lalim nga nung procedure. Yung kunyari pasta sa lalim hindi na rin masyado eh. More of sa difficulty ng case ng patient. Kasi kumbaga nag vavary na lang yun, yung pinaka professional fee eh. Kasi, we have to include that kasama nung material cost.

MA: Okay, ayun thank you po.

[End Transcript 00:06:06]

Interview 3

Interviewee: Dr. Jilliane Lee

Interviewer: Marcus Almendares, Charles Daroy, Michael Mesina

Date of Interview: 05/28/25

Location of Interview: Google Meet

List of Acronyms: JL=Jilliane Lee, MA=Marcus Almendares, CD=Charles Daroy, MM=Michael Mesina

[Begin Transcript 00:00:00]

MA: Yung starting from opening hours po, can you walk us through starting from how you prepare for patients up to closing hours?

JL: Hmm, okay. So upon opening, so yung mga assistants ko muna mag aano nun, they prepare everything. Linis muna ng buong clinic and then they prepare the materials for use for the particular day. So per patient may nakaset na ano yun, ah nakaset na, na instruments. Usually, ang nireready muna nun, basic instruments. Ah basic instruments as well as yung mga consumables na gagamitin nung mga tao. For example, for me and the assistants, headcap, face mask, gloves. So per patient... ahmm, actually all through out the day pinapalitan lang naman namin yung gloves. Headcap and facemask hindi na. So usually, one per day lang yun, on the average. Ah papalit lang kunyari napunit or nacontaminate yung face mask, yun lang palit. And then per patient may nakaset na yun. So, ah sa table, for example nakaprepay yung tray, per tray nandun yung basic instruments, ah nakaprepay din baso, ah suction example, yun yung mga basic na nakalagay. And then depending on the procedure, dun lang magdadagdag ng mga necessary materials, and then dun lang iprepay also necessary materials, so it varies from day-to-day operation. Depende kung ano yung procedures na gagawin. But, basically yun basic instruments are always there, yung mga basic consumables, yung baso, yung ah meron kaming barrier na tinatawag, so per day papalitan lahat yun. Ahmm, so meron din in between patients meron din kaming disinfectant na ginagamit to ensure na malinis everything for patient. Ah in terms of inventory naman ng ibang materials, so yun, as I've mentioned, depende kung anong gagawin for the day. So kunyari pasta. Pasta varies also kung which material ang gagamitin. Ah kung ano yung kailangan nila, yun lang nilalabas [inaudible]. So everything, ah lahat ng pwedeng gamitin, nakastock sa [inaudible]. So it varies na lang dun sa procedure tsaka sa material na gagamitin for that particular procedure.

[00:02:26]

MA: Okay. Ah yung, namention niyo po diba nung on busy days merong nineteen patients na cater niyo? Pano niyo po napagsabay sabay yun?

JL: One after the other, One after the other. Hindi, hindi kaya sabay sabay kasi mag isa lang akong dentist eh. So what we do, pagka ganun na maraming pasyente, pagka ganun, usually apat kami, ako, dalawang assistant, tas yung sister ko. Yung sister ko yung nagiging front desk officer. So after procedure, ahmm inaano ko na lahat ng mga kailangang gawin. Ay ng mga nagawa. Binibigyan ko siya ng summary. Tapos nakikita naman dun sa system namin yung costing. So siya

na yung maninigil. So ako focus pasyente lang talaga. Ang turn around dun as in immediately after procedure, ligpit lahat, linis lahat palit bago. Kaya kaya siyang maaccomodate mabilis ng marami.

MA: Mmmm.

JL: And usually pag ganun karami, ano yun mga magkakapatid, kaya kayang sunod sunod. Minsan we cater to mga three, three siblings at a time. Minsan sumasabay parents kaya umaabot ng ganun karami.

[00:03:40]

MA: Mmmm. So half... af... uh... after the procedure po. How do you clean up and handle used materials?

JL: So clean up, ano kami. So yung tray ko kasi, what I do, yung tray ko may plastic siya. So yung plastic na yun, lahat ng waste per patient nakalagay dun. Tapos, so lahat ng ano yun, nalawayan, dugo, nandun yun siya. Tapos meron kaming ahmm, [inaudible] ano dedicated trash bin for wastes, biological wastes. Ayan so yun meron kaming hauler na nag cocollect ng mga infectious wastes. So hindi siya kasama dun sa regular wastes namin. So naka ano yan siya dun sa likod ng clinic. Meron kaming canister dun. Usually, monthly ang kuha dun eh. So what we do nalang para maprevent yung spillage, per patient naka seal yung mga ano nila, nakabuhol yung mga plastic nila, and then [inaudible] collect dun sa ano, sa likod. Tapos may separate ano rin kami, separate space bins for [inaudible] for example, or medicine. May separate yun na kinocollect din nung hauler every month.

MA: Okay. Yung ano rin po, kapag gumamit po kayo ng mga essential materials sa procedure, pano, pano po yun nakekeep track? Sinusulat niyo po ba or?

JL: Actually yun hindi namin halos nakikeep track. Kaya medyo mahirap. Kaya medyo naano kami sa inventory. Ang nakikita lang namin yung. Eto example ah, yung glass ionomer cement, yung capsules nun, meron akong specific area dun sa cart ko na may mga nakalagay. So dun lang, nirerefill lang namin yun. Tapos from there, kunyari nauubos na, tsaka lang namin ilalabas yung box ulit and then we refill. Tapos ang ginagawa na lang namin for example, yung mga consumable namin na maraming boxes. Nakalagay lang siya kunyari one o kunyari gloves. Ang ginagawa ko

ay order one dozen at a time. So per batch nilalagay namin kagaya one over twelve. Sabi ko dun sa assistant ko “oh kukuha niyan kailangan ah sunod sunod” para we track already pang ilang box na yung nagamit namin. So sabi ko “pag mga nasa number ten na tayo, number eleven, you let me know right away para maka, maka order na”. Yung mga ganun medyo mas madali kasi nga nananumberan. Kaya lang yung iba, kasi for example we have certain items tulad ng chlorhexidine rinse. So yun, we have to check lang every now and then ah yung bote ba niya mabigat bigat pa, mukhang marami pa ba or more or less tinatantsa rin namin. Ummm, so we rely mostly on what we see, kung ano nakikita namin doon sa, kasi yung cart namin, yun yung mga exisiting na ginagamit. So pag medyo nagiging low na rin yung, yung gamit dun, chinecheck na rin namin yung backup niya, tsaka yung stocks.

MA: Okay. Si charles po ba ano, may tanong ba?

[00:06:56]

CD: Hello po doktora ahmm.

JL: Hi.

CD: Ahmm, clarifi, clarification lang po about sa stock monitoring niyo po or sa inventory monitoring niyo po.

JL: Mm mm.

CD: Pag, diba po na mention niyo po na visually checking lang po like kung napansin niyo nalang po na onti na tsaka po kayo mag rerestock. Pano niyo po natatrack yung mga kailangan niyo po ireorder?

JL: Yan tinitignan, ano lang tinitignan na lang namin. Tas every now and then nitong May, diba nagkaroon kami ng convention. Before, parang a few weeks before the convention, sinabihan ko na yung assistant ko “oh mag inventory kayo, tignan niyo kung ano na yung current na laman ng mga stocks natin”. Eh, tinignan na niya doon more or less, so we have an idea kung anong meron, para alam ko rin kung ano yung mga bibilhin ko. Yung ano naman, may mga gamit kasing mga fast moving. For example, gloves. Yan mabilis yan eh, parang I think every month halos umoorder ako. Yun ah yun nga, sa lalagyanan namin naka number naman siya ano lang one over twelve, two

over twelve, ganyan. Yung mga medyo slow moving items, yun we just keep track dun sa mga boxes din. Kasi mostly nakabox naman siya na may mga number din naman. So we keep track doon nalang din. So more or less, hindi na, [inaudible] sisilipin nalang namin. Kasi yung cart na ginagamit namin, so yun yung mga ng materials na kailangan namin. So also, ahmm, meron din kaming certain materials for example na isa isa lang. Yun nga yung isa isa mas madali naming makikita yun. So pag medyo malapit na maubos, nangangalahati na, we take note. Sinusulat, may cabinet kami na sinusulatan nung assistant ko kung ano yung mga kulang or low na. So next purchase, yun na yung mga bibilhin namin.

[00:09:05]

CD: Okay po. So... moving on nalang po tayo sa mga about sa expiry or mga waste management niyo po so.

JL: Mm mm.

CD: Paano niyo po nachecheck yung mga, paano po ba yung proseso niyo ng pagcheck po ng mga expired materials or yung mag eexpire na po? Ganun.

JL: Ahmm, more or less may, kasi dun sa, ang ginagawa ko kasing pag stack ng mga boxes nila, nakaharap yung expiration date. so from upon purchase, tinitignan na namin yun. Tas inaano na namin so inaarrange namin according to the pinakamatagal na nag eexpire. So kung ano yung mas maagang mag eexpire yun yung unang gagamitin. And then kung kunyari, expired na we dispose. Proper disposal inaano nga namin dun sa, ah, meron kasi kaming mga infectious waste so sinasama namin siya dun. Hindi namin siya sinasama sa regular waste na tinatapon every day.

[00:10:06]

CD: Okay po. So speaking of disposing po. Pag may mga naprepare po kayong materials or items na gagamitin for a procedure, ano, tapos hindi po nagamit, ano po yung ginagawa niyo po? Binabalik niyo po ba sa inventory or dinidipose niyo po yun?

JL: Ah depende. Usually ano naman, kung, ah, for example, ahmm, capsules. Ang capsules naman, pwede namin siyang ibalik. Kasi hindi pa naman siya used. Pero kung kunyari ah, we have bonding agent. Si bonding agent kasi, pinapatak yun eh. So once na napatak na siya, nagamit o hindi, tapon.

Ganun siya. So depende kung anong material. Usually naman, bihira kami nagdidispense ng hindi nagagamit. Kasi we try to ano pa rin eh, syempre ahmm, tipid tipid din ng konti. Hindi kaagad dispense ng dispense unless it's really necessary for the procedure.

CD: Okay po.

JL: Pero ano yun, pag basta questionable contaminated tapon.

[00:11:14]

CD: Okay po. So ahmm. About sa ano naman po, during procedure po ahmm, kunwari po may kailangan po kayong extra materials, pano niyo po natatrack kung ano ano po yung mga yun?

JL: Hmm. Ano yun, as for need basis lang talaga. So ano naman, yun nga, ah in terms of inventory kasi, wala kaming kunyari minus one minus one, wala kaming ganun eh, kasi hindi na rin, ah medyo mahi, sabi ko nga sa inyo, sobrang dami. Hindi ko pa sinama yung ano, nung nung tinanong ni Marcus sakin yung inventory, hindi ko pa sinama yung crowns. Kasi pag sinama ko yung crowns, malilito talaga kayo. Kasi per crown, per tooth, we have six sizes. So sa baby teeth, we have twenty teeth times six. So ganun karami yung crowns, eh yung crowns ko, hindi lang isang klase. We have [inaudible]. So lahat yun may inventory. So medyo ah, tracking wise, medyo mahirap sa ganun. Ah yung mga ahmm, may iba kasi kaming materials na hindi lang naman siya for one patient. For example, composite. Composite materials naka syringe yun siya. So we just get a piece. Kapinggot lang. Kung ano lang yung kailangan for the day or for that particular procedure. And then, yung ano na yun, nirerecap lang. And then keep ulit dun sa lalagyanan niya. Pero yung mga individualized talaga. Yun, immediately after use, discard.

[00:12:49]

CD: Okay po. And uhm, tinatrack niyo rin po ba yung mga manufacturing date nung mga supplies niyo po?

JL: Hindi na. Hindi ko na natatrack pa. Ang tinitignan ko lang yung expiration.

CD: Opo, okay po.

JL: Kasi mas crucial yung expiration than the manufacturing. Although ano rin naman kasi, ah we have certain materials kasi na shelf life niya maikli lang. So for example anesthesia. Anesthesia usually one year lang yan talaga eh. Yung ibang mga consumables, roughly around two years to three years. So tsaka most of the time, bihira naman din kung may binebentahan ng sobrang tagal na yung manufacturing date, kasi hindi ko rin tatanggapin, kasi nga shelf life nila mas iikli from the manufacturing date.

CD: So bale po inuuna niyo po gamitin yung malapit na po mag expire?

JL: Mm mm.

CD: Okay po.

JL: Ganun. Hindi, hindi kasi, minsan hindi kasi nag aapply yung first in first out. Kasi depende sa kung ano yung dineliver samin. For example, anesthesia. There are certain situations na yung pangalawang set ng anesthesia na binigay, mas matagal expiration. So uunahin natin kung ano yung mas mabilis o mas malapit na expiry.

CD: Okay po. So, so far naman po, wala na po akong question.

[00:14:12]

MA: Ayun po regard ay. Regarding sa pag order ng or pagreorder ng mga materials, ano po yun? Through online, by call, or by message?

JL: Usually, ano na message. Memessage ko lang yung ah, for, for specific products may certain ano kasi sila eh, mga ahente. So usually, may mga ahente yan. So meron na akong mga contact lagi. Kunyari ah, crowns, Ito yung bibilhan ko. Sa anesthesia, sila yung bibilhan ko. Sa yung mga basic requirements o yung basic ah materials na ginagamit lagi, meron akong contact na dental supply. Sa lahat ng mga consumables ng gloves, headcaps, dun ako umoorder. Imemessage lang yun. Actually, ngayon mas mabilis na. Imemessage mo lang sila online, ischedule lang nila yung delivery tas yun dadalhin ka. Unlike yung iba na...

MA: Iba iba po yung suppliers niyo?

JL: Mm mm. Iba iba. Kasi for example ah, yung mga generic na mga consumables, yun pwede sa dental supply. kasi sila mas, kinecarry nila lahat eh. From gloves, mask etc... dun ako kumukuha. Ah specific products. For example, si glass ionomer. Meron akong contact na sila talaga yung distributor. So dun tayo, dun ako pumupunta sa source mismo. Kasi most of the time, mas mabibigyan nila ako ng discount. Mas nabibigyan nila ako ng mas mahabang payment terms depende dun sa amount ng binili ko.

[00:15:49]

MA: Kapag ano po, yung supply item out of stock dun sa supplier, san po kayo bumibili or nag iiba po kayo?

JL: Sa dental supply na. Kasi sa dental supply, usually sila mas maraming [inaudible].

MA: Ah okay.

JL: Although ano naman, [inaudible] siguro sa heads up. So yun may mga instances lang kailangan namin ng alternatives. Si... yung mga dental supplies, mas marami silang stocks. Ang ano lang syempre mas mataas ang konti. Kasi nag aadd sila ng ano eh, nag aadd sila ng konting profit dun sa products.

[00:16:22]

MA: Mmm. Si... usually po sino yung naghahandle ng mga orders?

JL: Ako rin.

MA: Ah kayo rin.

JL: Oo. Kasi, I have personal relationships dun sa mga ano eh, dun sa mga suppliers. So more or less, minsan naawitan ko sila na instead of usual na bibigay. Kunyari two months lang yung cheke, napapahaba ko minsan ng mga three months, four months, ganun.

MA (speaking over): May list of ah... ay sige po.

JL: Yun, minsan kasi sa samin ah, we have to develop a certain relationship also sa supplier. Para, kunyari yan, meron silang promo, meron silang mga bagong products, ah, I have supplier na they

give me samples also for me to try out, mga ganun. So they give me special deals na hindi nila binibili sa iba, parang ganun. May, may nalalambingan ko sila kunyari ng extended, yun nga extended na, na payment terms. Mga ganon. Kasi maliit lang naman yung clinic ko. It's a single, single proprietor clinic so hindi ko na dinelegate yung ano doon. So yung, pang ano lang, lista lista, yun ang dinedelegate ko dun sa mga assistant ko, pero ordering ako na.

[00:17:41]

MA: Okay. Meron po ba kayong list of items na kapag ioorder niyo, ganun?

JL: Alin? Panong list of items?

MA: Parang ano, like you always order tapos nag chchange po ba siya overtime?

JL: Yeah, it varies din eh, kasi depende. Ah kunyari. Although yung mga ang regular talaga gloves, yan yung mga disposable. Kasi mabilis yan eh. Per patient parang kumakain kami ng anim na gloves agad. Three pairs at a time per patient. So ganun kabilis yung turn around ng gloves namin. Ah, it varies from time to time. Tsaka minsan ano rin, kinocollect ko muna. Para isang orderan na lang. Isang... Ah isang bayaran na lang din. Collective na yung payment nila. Para hindi rin mahirap magtrack ng ano, ng payments.

[00:18:39]

MA: Ahmm yung mga receipts po ng orders, san siya, san niyo po siya kinikeep?

JL: Tinatago ko lang dun sa drawer ko. Tas minsan may envelope ako, pero yun nga, hindi na rin, hindi ko na rin siya naoorganize ng maayos.

MA: Hmmm.

JL: Sa tracking wise, nasa isang lalagyan lang sila lahat. Sinasabihan ko lang din yan “oh itago niyo lahat ng resibo, kasi just in case meron tayong kailangan ibacktrack, nasa isang lalagyan lang siya, alam natin kung saan ilalagay”. Pero yung para ilista lista pa hindi na namin nalilista.

MA: Sige meron ka pa bang ano ano Charles tanong?

[00:19:20]

CD: Ahmm may follow up question lang po pala ako so...

JL: Sure.

CD: Pag kunwari po may nagpa appoint po sa inyo, tapos po na prepare niyo na po yung mga items or products po na gagamitin niyo po sa procedure tas bigla pong nagcancel, siguro po, ahmm ano, ano po gagawin niyo po sa mga items na yun?

JL: Mmm, usually ano naman, hindi naman namin nilalabas agad. So ano lang balik lang. Basta hindi nagamit...

CD (speaking over): Ah sa...

JL: Kahit nagcancel sila okay lang. Hindi naman siya ano, walang wastage, kasi hindi naman nagagamit. Kasi for example, instruments. We don't open up the pouches unless nandiyan na yung patient mismo. So kung kunyari hindi sila dumating, babalik lang namin sa drawer.

CD: Mmmm.

JL: Ganun lang siya so wastage hindi naman na siya, wala, walang wastage sa ganun. So we can always put back the, the stocks.

CD: Okay po. Thank you po.

[End Transcript 00:20:20]

Interview 4

Interviewee: Dr. Jilliane Lee

Interviewer: Marcus Almendares, Marvin Ancog, Charles Daroy, Michael Mesina

Date of Interview: 10/31/25

Location of Interview: Tita Doc Pediatric and FamiLee Dental Clinic

List of Acronyms: JL=Jilliane Lee, CA=Clinic Assistant, MA=Marcus Almendares, MV=Marvin Ancog, CD=Charles Daroy, MM=Michael Mesina

[Begin Transcript 00:00:00]

MA: Okay so let's start po sa tracking po nung mga items sa app tulad nung bib. Ayan, eto po yung packaging unit or content. Per pack may 50 pieces, pero yung tracking pa rin po niyan is per pack.

JL: Oo. Alanganin kasi kung per piece eh. Kasi hindi naman namin binibilang yung ano eh, yung bib eh. For example samin, since kids yung patient namin, minsan we cut the bib into half, tas meron kaming separate na lalagyanan dun sa mga naputol na. So usually, ang pinaka tracking namin, pagbukas ng cabinet, kung ilan pa yung pack. Tas ang ano ko naman sa mga assistant ko, pagka kunyari nabuksan na yung last pack or isang pack nalang natitira of whatever we have, inaano na yun, naka... may sinusulatan kami diyan na yun na yung way namin to check. "Oh ito malapit na maubos", tas yun, pagka ano, dun na ako mag oorder. More or less, sinasabay namin yung pagbili para isang delivery nalang. Mga ganun. So yung sa bib talaga, for us, ako mismo, personally ang hirap itrack niyan ng isa isa. So kung sa ano ang tinatrack niyan per piece, edi paano na yan diba? Kasi medyo... ang isang pack, ilan ba yung laman ng isang pack?

CA: 100 pieces po.

JL: 100? Oh, 100 yung isang pack eh. So actually, medyo slow naman yan siya. Tas hinahati pa namin sa gitna yun. Yung patients namin maliliit diba? So ganyan lang naman kailangan icover eh. We only use one whole of the bib pagka adult yung patient. So pano yung mga hinahati namin diba kung dyan is isang piraso. So medyo... yung samin ah, dito sa sa sa sa practice namin. Medyo maano yung tracking na paisa isa lang. For us ah, it's better na we do it by pack. Hindi mo naman din kasi mabibili ng per piraso eh. Diba? So why bother doing an inventory per piece? If hindi mo naman siya mabibili per piece. Kasi usually, we start purchasing talaga... hindi kami maghihintay na sampung piraso na lang, yung mga ganyan ah, kasi we buy it in pack. Kasi, for example, cotton ball. Alangan namang bibilangin pa namin isa isa yung cotton ball [chuckles]?

MV: Kahit nga po sa may spray eh.

JL: Oo. Diba yung spray? Hindi rin eh. We only track it by the bottle. Sa anesthesia siguro pwede yan per blister pack. Kasi ang packaging ng anesthesia... per box, may limang blister pack. Siguro yun, pwede pa per blister pack which is a pack of ten. For pouches hindi rin. Kasi kunyari, alangan namang iano ko, pagkasampu na piraso tsaka ako bibili. Usually, we start ordering materials pag

isang set nalang, isang pack. For example, mga ganyan, povidone, mga 1 to 2 bottles na natira order na kami uli. Eto sa dentin conditioner, usually bumibili lang ako dalawa, pag nabuksan na yung pangalawa, dun lang ako mag order ulit. Kasi medyo ano yan eh, medyo mahal siya tsaka hindi naman siya kelangan ng sobrang dami. We use this sa per pasta ano lang isa, one drop one drop lang.

[00:08:24]

MA: Okay. May natrack po ba kayong mga bottles per mL ganun?

JL: Bottles per mL [chuckles]? Punta tayo dito para makita niyo. Etong dentin conditioner... tignan mo expired na pala ito, di ko na ano, ngayon ko lang nalaman nung nakuha na. Tignan mo, binili ko ito 2023 pa, hindi parin na ah... kasi nga mabagal siya. Per patient na pasta, one drop. That's all we need. If imemeasure natin talaga... diba ang 1mL is equal to 15 drops? Pero hindi naman kasi super accurate ito eh. Minsan nag fflow siya ng mabilis eh. So mahirap siya itrack talaga. For example, bonding agent. Pano mo itrack yung bonding agent? Per bonding agent is just like this. Drops lang. Kaya mahirap for us na itrack or imeasure yung mga ganung bagay. That's why ang ginagawa namin is always by the bottle. Itong glass ionomer cement, mahirap iquantify. Kasi yung gagamitin natin per patient, per cavity can vary from patient to patient. Di mo naman pwede igrado kada patient diba? Kasi it varies kung kunyari malaki yung cavity. Yung consumption mo, pwedeng iba dun sa ngipin, pwedeng iba dun sa isa pang ngipin eh.

[00:15:06]

MA: Ilan po yung mga quantifiable na ano niyo, supplies?

JL: Ang makaquantify talaga natin na per piece? Yung GIC kaya yan per piece.

CA: Gloves po?

JL: Gloves? Usually 3 pairs. Pero there are situations in the middle of procedure... kunyari masyadong madumi na yung gloves ko, palit. Nabutas, palit. So hindi mo rin siya pwedeng sabihin na ganito lang lang yung gagamitin natin for patient. Kasi if the material is compromised already, you have to replace. Gauze rin, mahirap. Alangan naman ililimit ko lang ito 2 per patient, eh pano kung kailangan ko mas marami? Hindi talaga siya pwede per piraso kasi mahirap, di na rin namin matatandaan mga yan na pabago bago kung ilan, mas lalo na at too busy kami sa procedure para ilista mga yan. Ang hirap naman nun noh, kada palit mo magtrack track ka pa. Ang monitoring namin sa ganto per pack. Sa ginagawa niyo, siguro may options depende sa materials. Pipili nalang

ganun. So usually, pagka yan, per box na lang talaga. Tulad niyan, sinabihan nila ako “Doc last box available”. Usually, it would last a few days parin naman eh. Kung one to one edi 50 days pa rin ito. Ano pa ba... wipes. Alangan naman per pull iaano natin [chuckles]. Ang hirap na monitoring na ganun.

[00:20:31]

MA (speaking to other interviewers): Wala naman silang way ng pag track ng ano kada spray.

JL: Oo. Hindi talaga. Hindi praktikal. Kasi alangan naman ilimit yung assistant mo na “oh dapat hanggang tatlong spray lang”. Hindi naman din ganun kasi you have to cover yung entire surface na kunyari sa chair. Isspray namin yan. Hindi naman pwedeng limit to ano, edi isang area lang yun.

MA: Okay po, makes sense naman po.

JL: Eto tignan mo inabutan ng expiry ito eh. Oh, may laman pa. Kaya lang kami nag palit? Expired na. So technically, hindi rin siya nauubos. Eto namang kapalit niya, since 2027 pa naman expiry, hindi ako para magstock ng bago. Kasi kung mag stock ako ng bago, makukuha ko same batch lang din niya. By the time na papalitan ko ito kasi expired na, expired na rin yung pamalit ko. So usually, pagka ganun, hindi kami bibili ng ano talaga, ng extra.

[00:24:02]

MA: Okay po. So, ay may dinagdag po pala kami para kahit papano po connected sa services niyo. Ayan kunyari extraction. Tas yung patient information din...

JL: For me ah, minsan, hindi ko naman masyadong malalagay yung ibang info ng patient dyan eh unless I have a specific assistant to do that for me. Kasi for example dun sa isang clinic na pinupuntahan ko, yun multi chair yun ah. Kahit maraming tao, I didn’t see them na tinatiyaga pa nila, na isa isahin yan per patient. Hindi siya ganun ka praktikal for us. Kasi ako charting nga lang ang dami kong backlog ng charting mga isang buwan na ata yung backlog ko eh. I try to to to chart and to make notes per patient per day. Kaya lang, there are certain situations. Kunyari nag 10-15 patients ako a day, every 30 minutes ang turn around ko ng patient. I don't have much time para pa mag ano ng details. Minsan nakakalimutan na nga eh.

MV: Include po kasi namin ito kasi parang ano daw po ata, sa may staff niyo daw po. Kasi pano daw po pag nagbawas lang bigla item, tapos parang walang marerecord na san ginamit.

JL: Ah... oo.

MM: Kung sino daw po yung ano...

JL: In charge? Pwede siguro. Talagang magkakaroon ka ng dedicated staff para gawin yan. If you want to really include this one magiging ano yan... may dedicated person ka to do that usually, assistant. For single chair practices, for solo practices medyo ano na sya... additional workload pa for us. Kunyari gloves, alangan namang “ay sa ganitong, kay ganitong pasyente ba naka ilang gloves ako?”. It will not change rin naman yung ichacharge mo kay patient. So more or less ano rin naman eh. Kunyari sa sa sa gloves we have yung yung, yung interval ng pag order. So for us that's a gauge also. Pagka shorter duration na than before eh medyo magdududa na ako bakit sampung araw pa lang oorder na naman tayo ulit, di naman ganun kadami patient mga ganun. Kasi kaya ka nga mag gaganito eh, para you want to keep track of the the materials that you have. Para mas convenient. Kaya lang kung magdadagdag pa ng additional task of tracking per day, medyo added work yun for us. Kahit yung sa mga nakikita ko na bigger practices, pagka ano, even the assistants are busy. Kasi sila rin ang maglilinis, sila magliligpit. Kumbaga, natapos na kami sa trabaho, sila dun palang magsstart magtrabaho kasi dun sila palang maglilinis. Unless siguro they have a dedicated person para dun sa... which is an additional head pa na babayaran.

[00:33:52]

MA: Edi itong parang manual deduction tama lang po?

JL: Oo. Kasi kunyari pagka ganun, ang gagawin namin, meron kaming small listahan dito, nakailang GIC kami tas tsaka na lang namin itransfer dito. Pero yung para in the middle of the procedure, in between patients, medyo... it can be difficult in certain situations. Siguro kung wala masyadong patient okay lang. Pero pag sunod sunod, mahirap.

MM: Eto po kunyari po, kunyari naubos yung isang box, ganto lang po yung process.

JL: Oo, yan deduct nalang 1. Kunyari naubos na yung isa...

MM: Ayan, eto anim na pack, tas naubos yung isa, yan.

JL: Oo. By the pack.

MM: Okay lang po sainyo ganto?

JL: For me, yeah. Practically... kasi kaya ka nga nag app para mas madali. Yun ang mas ano eh, yun ang mas convenient eh. Pag naubos na yung isang pack oh, minus 1. Gumagawa ka ng app para mapadali ang mga bagay bagay. Hindi para madagdagan yung trabaho mo. Ay, eto pala kelangan niyo makita ito kasi nandito yung mga ano... ayan dyan. Andito yung main na stocks.

Tapos per box... kunyari nag bukas kami ng isang box, nandito siya nilalagay. Ayan kung kunyari nag open na kami ng box, yung mga box na open, tas irerefill lang namin yan. So dyan kami tumitingin kung konti nalang. Kapag naubos lang yung mga nandito, tyaka lang kami kukuha dun para ilagay dito. Pag kunyari naubos isang box na nakalagay dito, minus 1 sa app niyo.

[00:40:27]

MA: Bale nilagyan din po pala namin nung reason or purpose kung bakit idededuct. Like ganun contaminated, deducted...

JL: Expired. Yun ang isa sa mga usual. Madalang lang naman yung may mga missing. Siguro yung mga missing missing, sa malalaki talaga, yung malalaking kumpanya. Mas madali mag nakaw dun eh, mas madali mag puslit. Tsaka kasi, ako mismo rin yung nagchecheck most ng inventory. Unless idedelegate ko sakanila “oh pakicheck yung ganito, pakicheck yung ganyan”. In terms of stock deduction per service, that would be applicable to bigger companies. They have the staff eh. Mas marami silang tao, mas marami si kumbaga... accountability wise, mas maraming pwedeng maging accountable in terms of loss ng materials. For us, mas madali kasi namin mamonitor eh. Mas obvious pag may nawawala. Tulad nung “oh bakit parang ang bilis maubos?” diba. Yung isang clinic na pinupuntahan ko, may listahan lang sila na ilan na yung pumasok, ilan na yung lumabas. Kunyari fluoride nagstock sila 50 fluoride, icocounter-check nalang nila yun dun sa mga services na nabigay. Itong listing na ito per patient, more applicable sa larger practices na may multiple chairs for operation, multiple assistants and different doctors. Pero samang smaller... I don’t think magkakaroon pa ng enough energy to list down.

MM: Kailangan talaga mabilisan eh.

JL: Oo, kailangan convenient talaga. Straight forward ganun. Naubos 1 box, bawas. Kung per piece yan iisipin pa, tatandaan pa namin kung ilan nga ba talaga. Tapos yan, isa isa icheck tas per patient pa tapos... medyo ano na yun.

[00:48:09]

MA: Tsaka hindi rin naman po per service nababawas lahat nung nasa list na yan eh diba?

JL: Mmm mmm. Yeah for example, yung head cap namin, usually that lasts the entire day. Hindi naman kami para magpalit per patient. Even yung mask, unless soiled.

CD: Edi hindi magiging reliable po yung result kung ganun?

JL: Oo. Kasi kunyari yung kapatid ko, pag kapatid ko nandito, minsan naiiwan niya kung saan saan yung bib niya, kukuha nanaman siya bago. Ay yung mask niya pala.

MA: Tapos hindi rin naman po pwede na kada kuha cotton ball in the middle of a procedure ilalagay muna dito sa ano...

JL: Yes, hindi siya ano... it's impractical to do it per service per ano. May mga materials kasi kami na mahirap iquantify per patient.

MA: Sa fast-moving po pala ganto.

JL: Ayan. Tama lang.

MA: Sa daily po ba, wala naman kayong nababawas na supply?

JL: Hindi ganun kalaki. Ilang piraso lang talaga. Tapos yung gagamitin o kelangan sa patient.

MA: Edi hindi po kailangan ng daily na fast-moving?

JL: Hindi masyado eh. Weekly monthly oo? Even for bigger ano, I think weekly? Kasi nga sa isang araw, minsan iba iba ang ano naman, flow ng patient eh. Tulad ngayon, patient namin tatlo.
[00:53:18]

MA: Okay. About po pala sa system na ginagamit niyo ngayon, yung mymed, would you like it po ba na integrated or connected yung app namin dun?

JL: I don't... di lang ako sure ah, pero ang alam ko kasi, kung ganyan diba lalapit kayo niyan sa kanila mismo?

MA: Opo, ganun po mangyayari kasi di naman po available yun sa public...

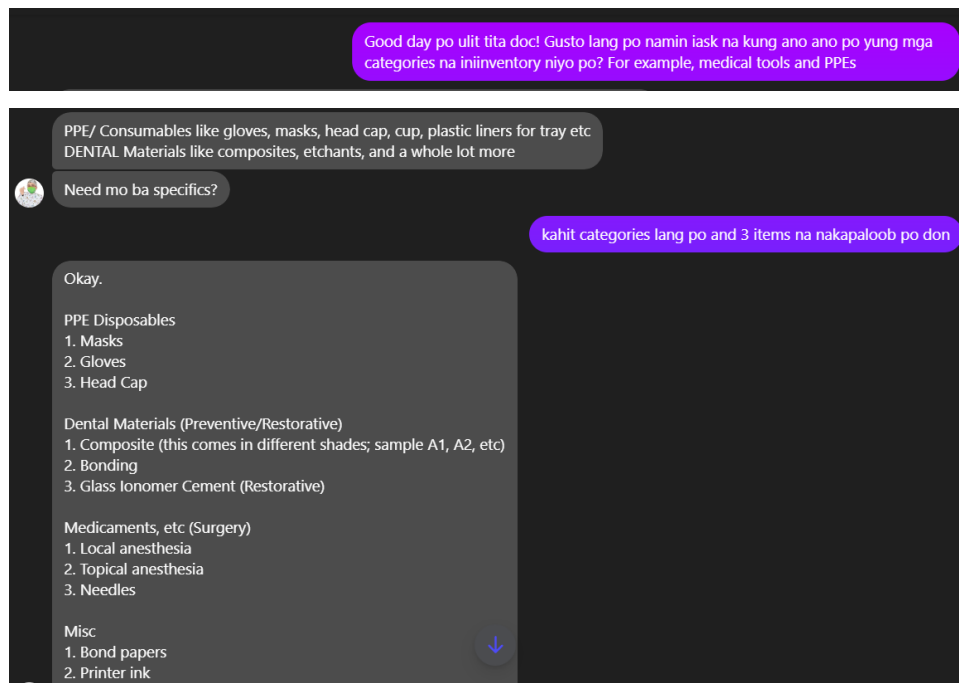
JL: Oo, ayun nga. Mag inquire kayo niyan sa kanila [chuckles]. Ewan ko lang kung ano ah... For me ah, wag niyo nalang iconnect kasi maproseso yan at tyaka di naman yan priority natin. Ang kailangan ko lang naman ngayon is yung pang inventory eh, kasi tulad kanina, diba nakita niyo may mga di ako napansin na may expired na pala. Napansin ko lang nung kinuha ko yun at chineck yung expiry date. Eh kung meron akong inventory tulad nung sainyo, most probably napa... matagal ko na nakita yung expired na yun and kung ano ano yung malapit na mag expire.

MA: Mmm, okay po. Thank you po. I think we should end the interview here now so we can proceed na rin po sa uli sa Beta Testing. Para makapag test na rin po silang assistants niyo po.

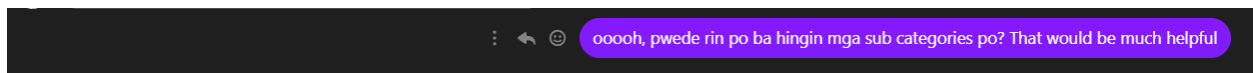
[End Transcript 00:55:40]

Direct Messages with Client

1. Asking for Sample Categories



2. Asking for Subcategories



Okay.

PPE Disposables

1. Masks
 - a. Pink
 - b. Blue
2. Gloves
 - a. Xs
 - b. Small
 - c. Medium
3. Head Cap

Dental Materials (Preventive/Restorative)

1. Composite
 - a. Packable
 - a.1. A1
 - a.2. A2
 - a.3. BW
 - b. Flowable
 - b.1. A1
 - b.2. A2
2. Bonding
3. Glass Ionomer Cement
 - A. Luting
 - B. Restorative
 - B.1 Fuji VII
 - B.2 Fuji IX
 - B.3 Equia Forte

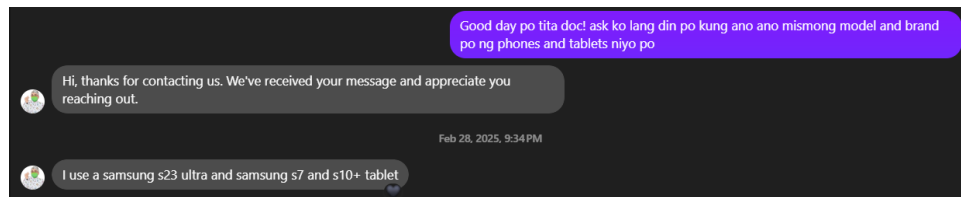
Medicaments, etc (Surgery)

1. Local anesthesia
 - A. Lidocaine
 - B. Articaine
2. Topical anesthesia
3. Needles
 - A. Extra Short
 - B. Short
 - C. Long

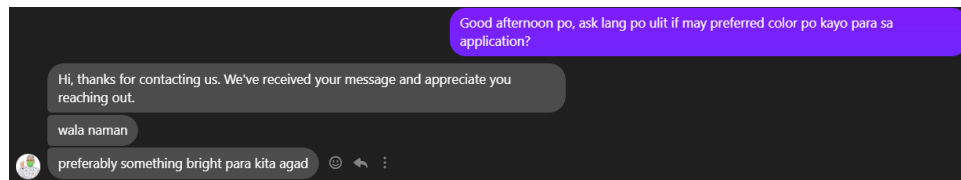
Misc

1. Bond papers
 - A. Short
 - B. Long
2. Printer ink
 - A. Black
 - B. Cyan
 - C. Yellow
 - D. Red
3. Envelops

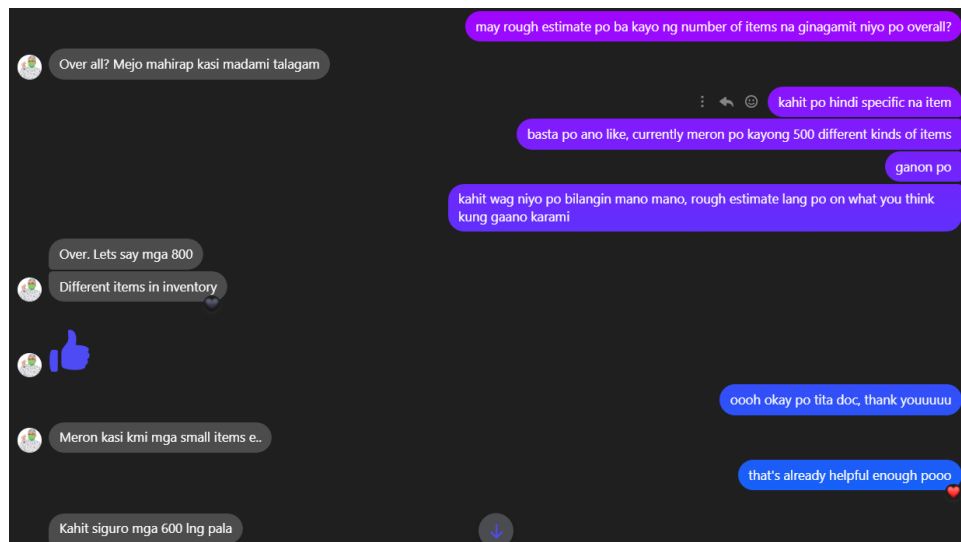
3. Devices used

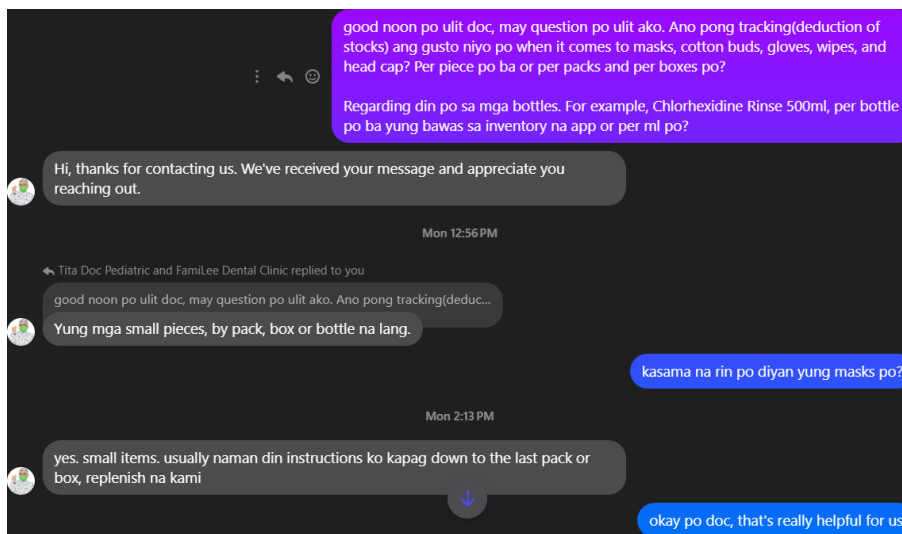
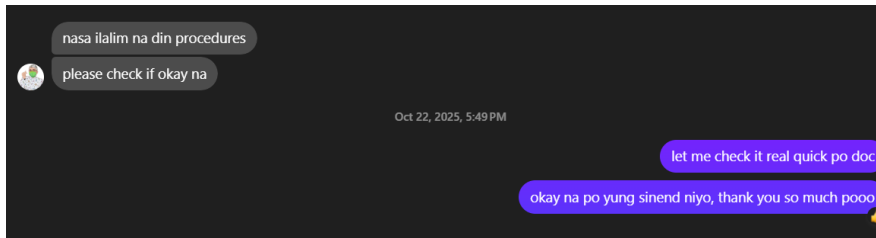
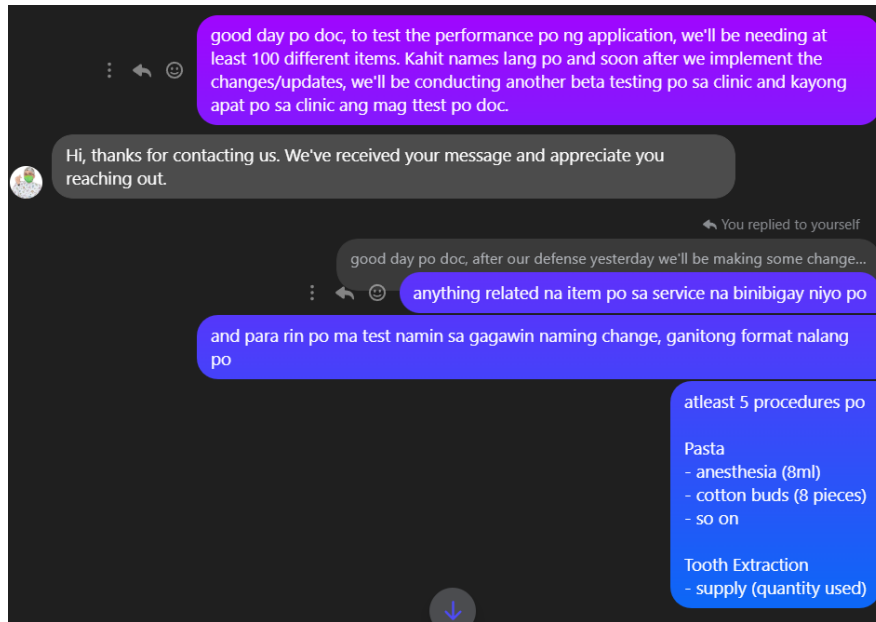


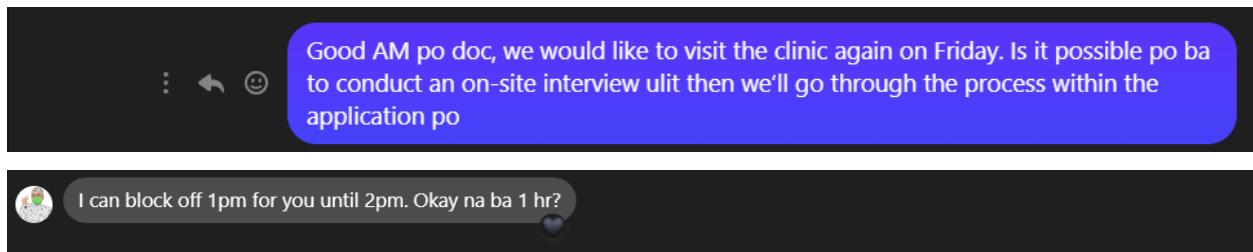
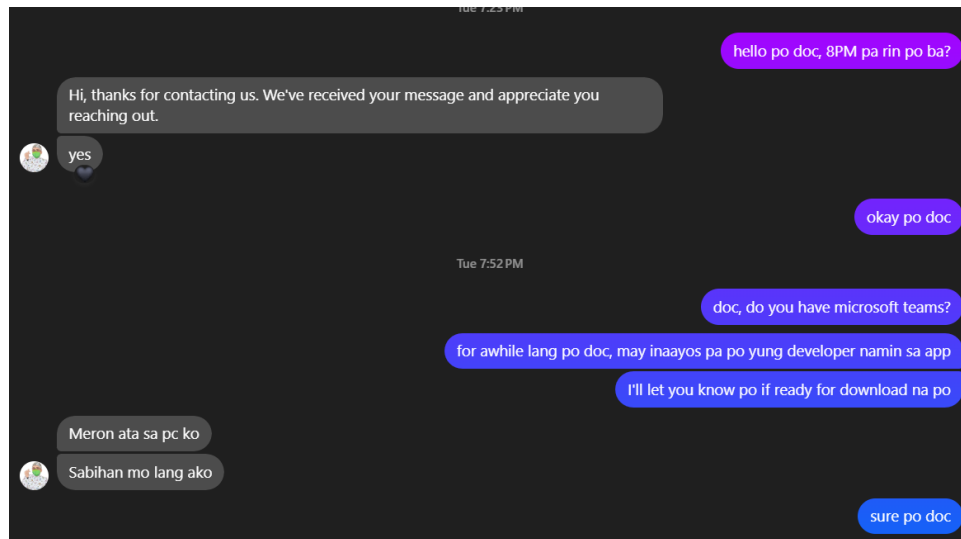
4. App Theme



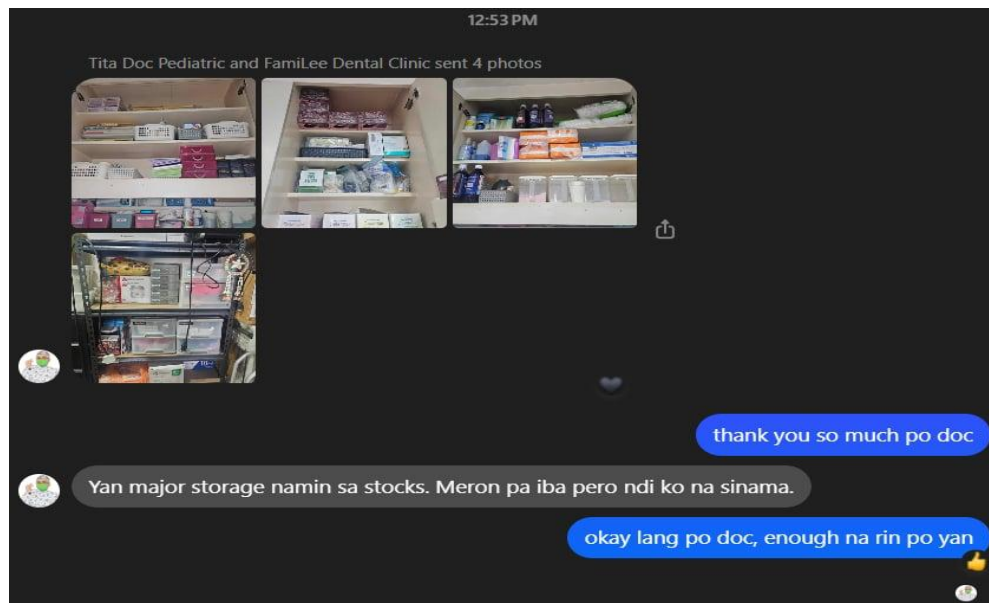
5. Supplies





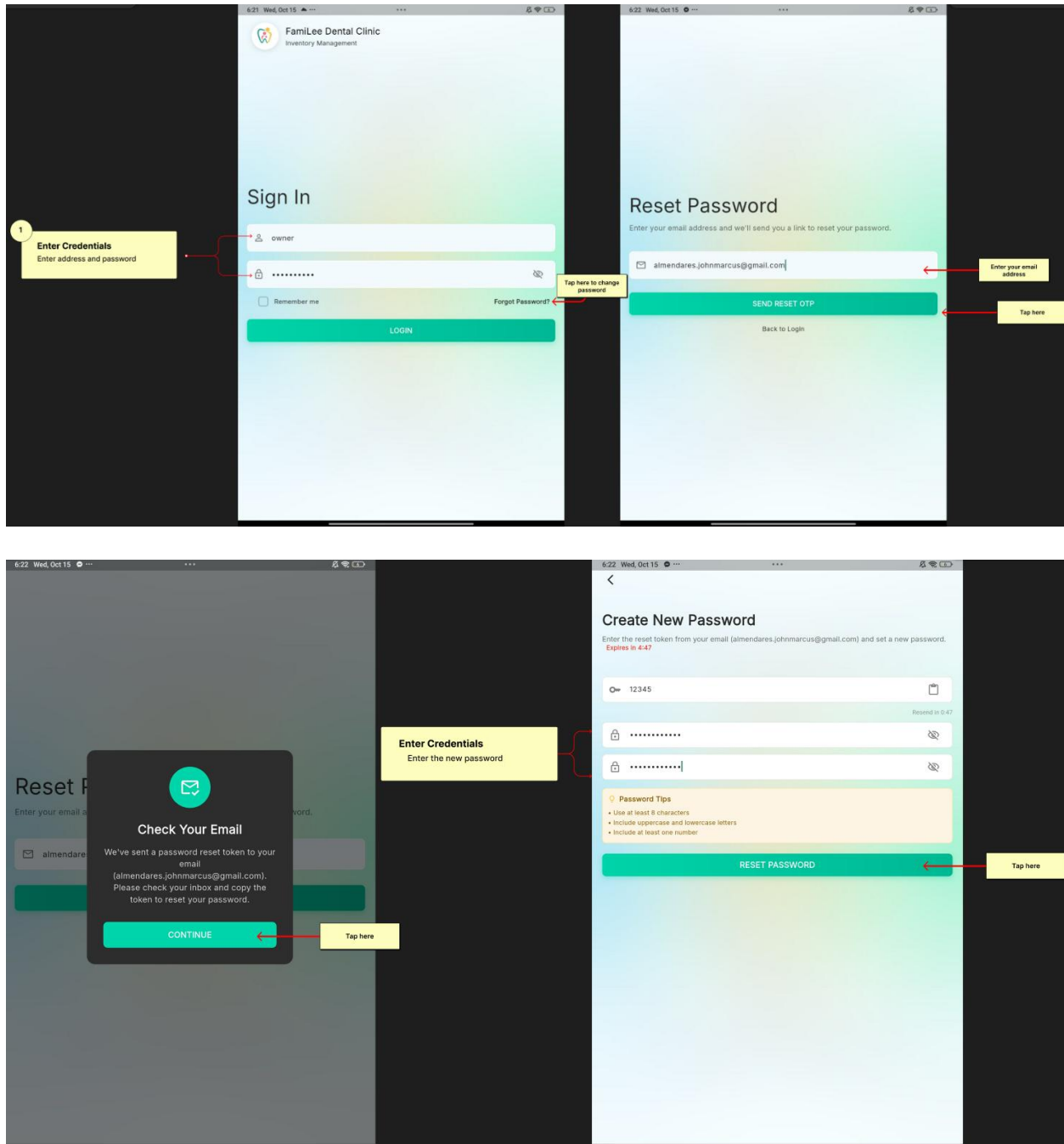


8. Asking for clinic's storage picture



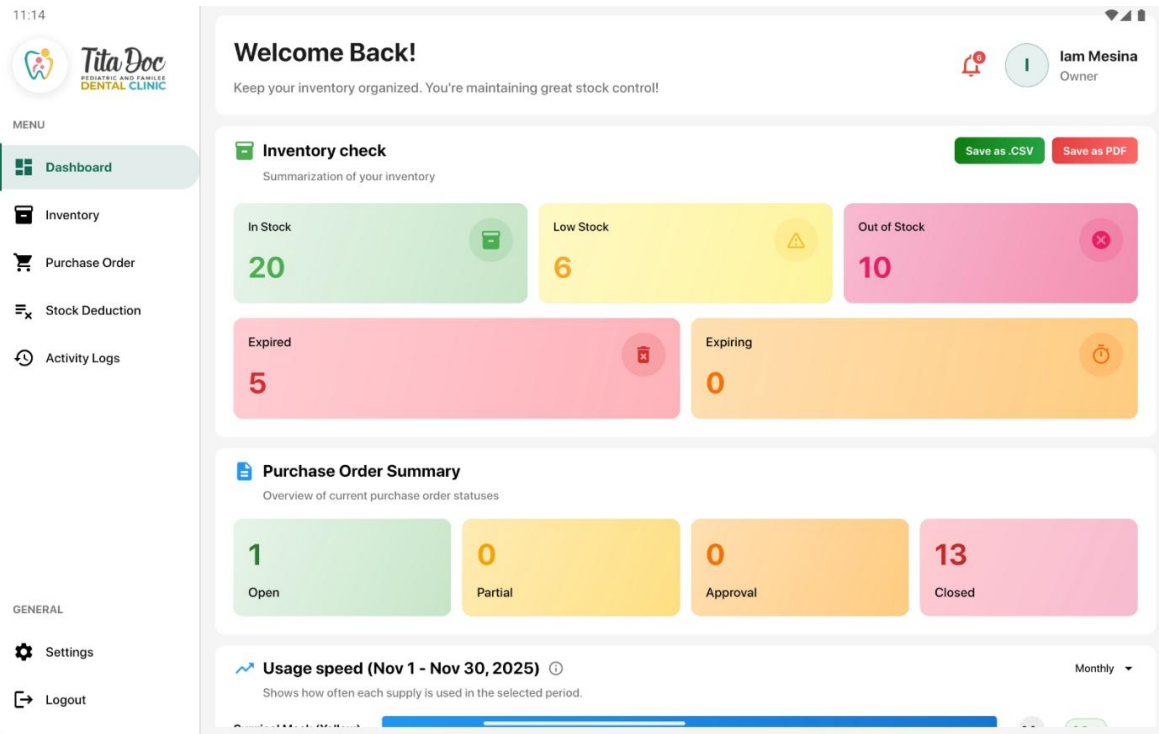
APPENDIX B. USER'S GUIDE

1. How to Login and Forgot Password



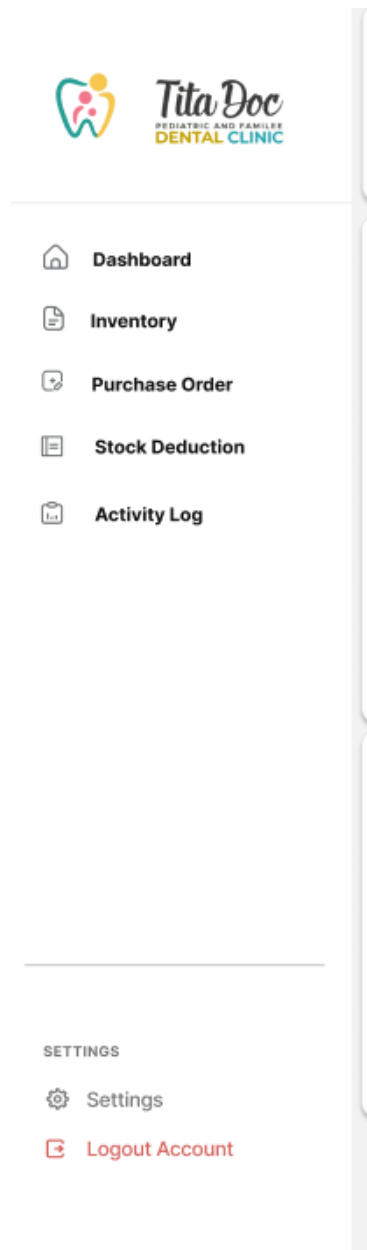
2. Dashboard Navigation

- The dashboard provides summaries from all modules within the app
- To download these reports, click on the “save as PDF” and “save as .CSV” buttons.



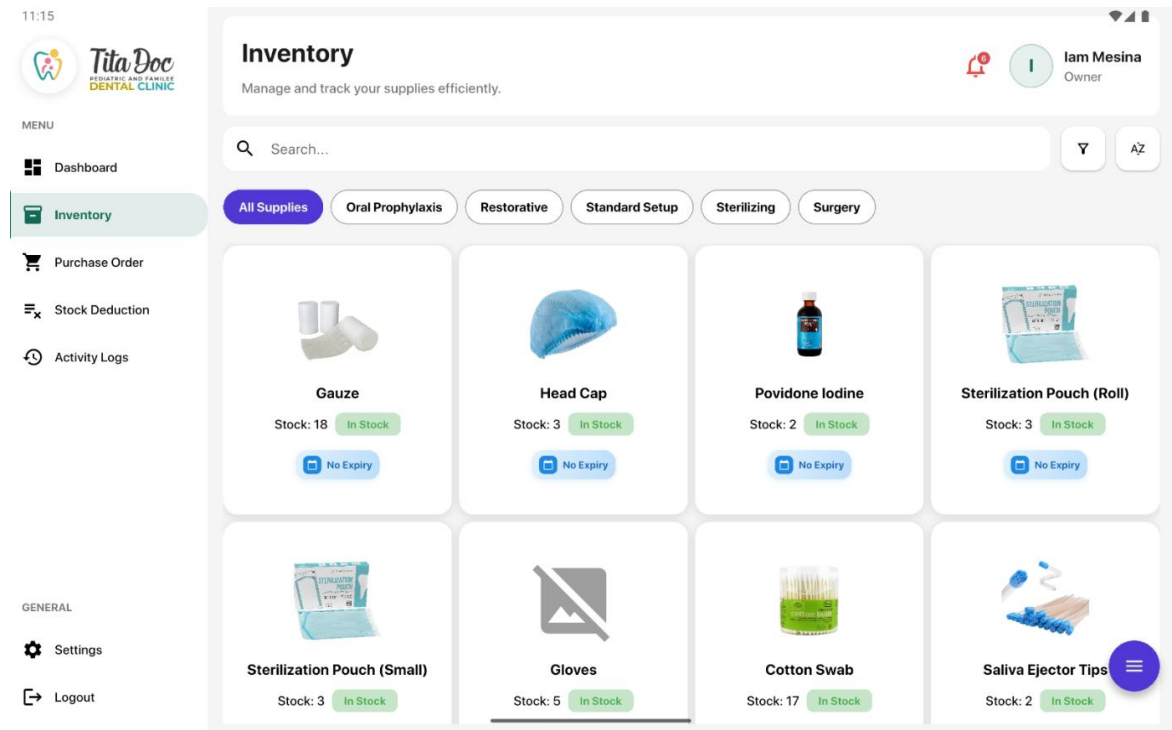
3. SideBar Menu

- The user can use the sidebar to quickly navigate different modules.



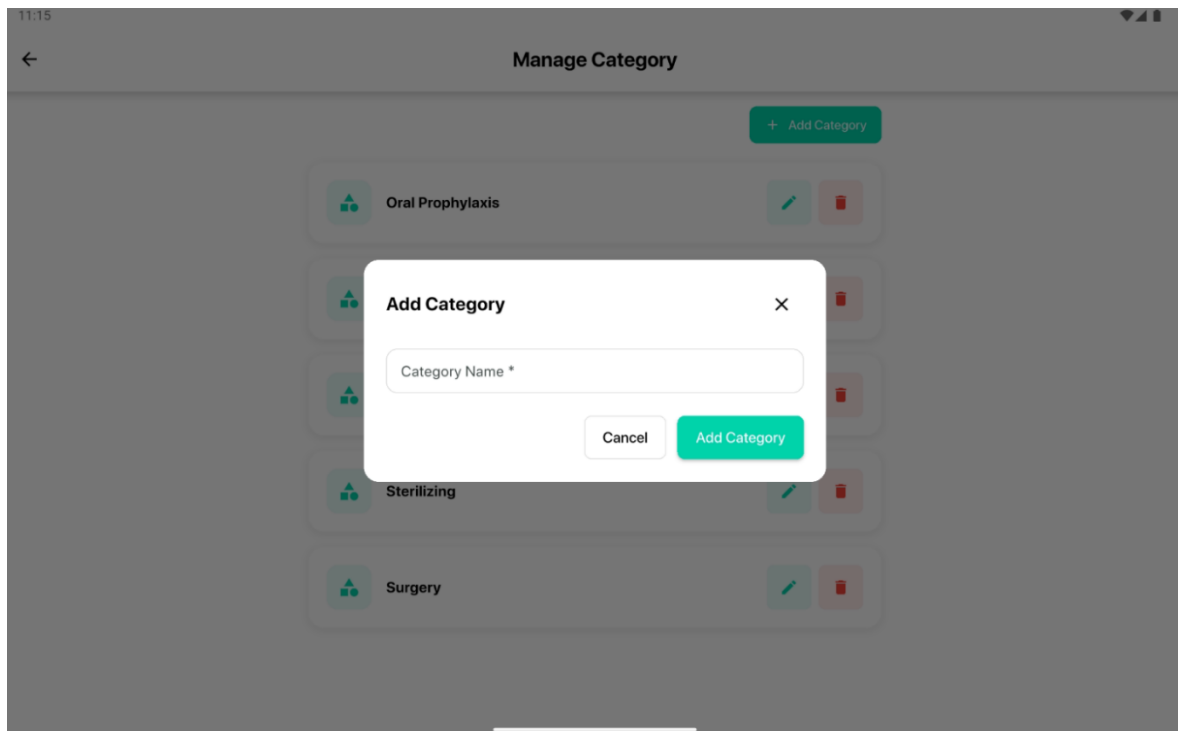
4. Inventory Navigation

- Tap Inventory from the navigation.
- Use search/filter (Category, Low/Out/Expiring) and sort.
- Tap an item to open Supply Details.



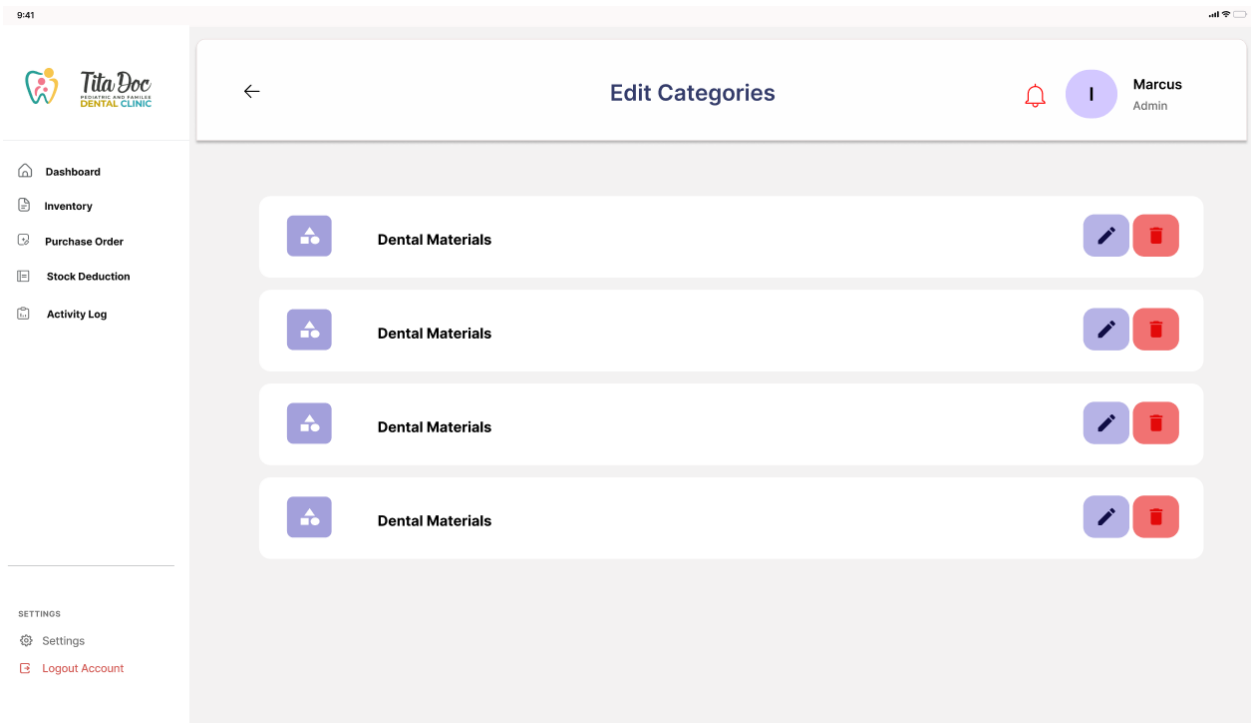
5. How to Create Category

- Simply input category name and add category.



6. How to Manage Categories

- To edit, tap the pencil icon to customize existing category names.
- To delete, tap the trash button.



7. How to Add Supplies

- Tap the “Add Supply” in the Floating Action Button.
- Add the supply details.
- Tap Save

9:41

 **Tita Doc**
DENTAL CLINIC

Dashboard
Inventory
Purchase Order
Stock Deduction
Activity Log

SETTINGS
Settings
Logout Account

← **Add Supply**   **Marcus**
Admin

Search..

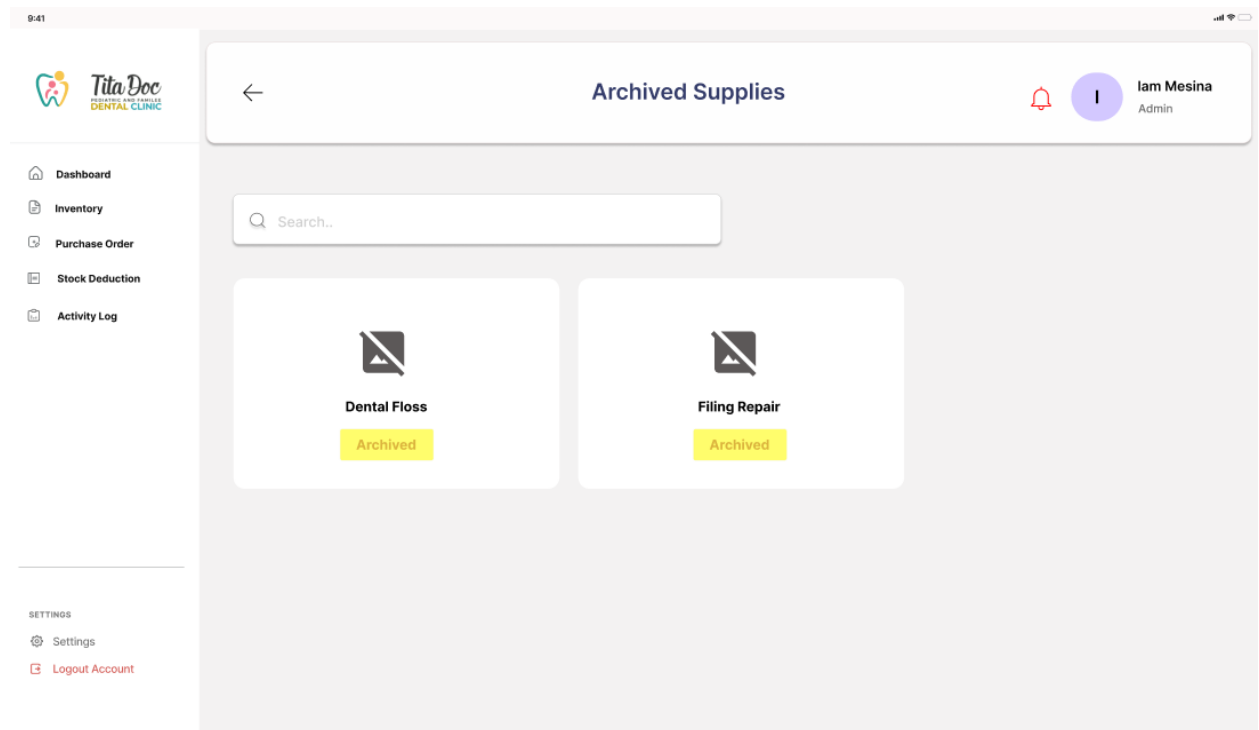

Upload Image

Item Name *	Type Name *	Supplier Name *
Brand Name *	Cost *	Category * 
Quantity - 5 +	Packaging Unit Box 	Expiry * 
Quantity - 50 +	Packaging Content Pieces 	☐ No Expiry Date?

Cancel **Save**

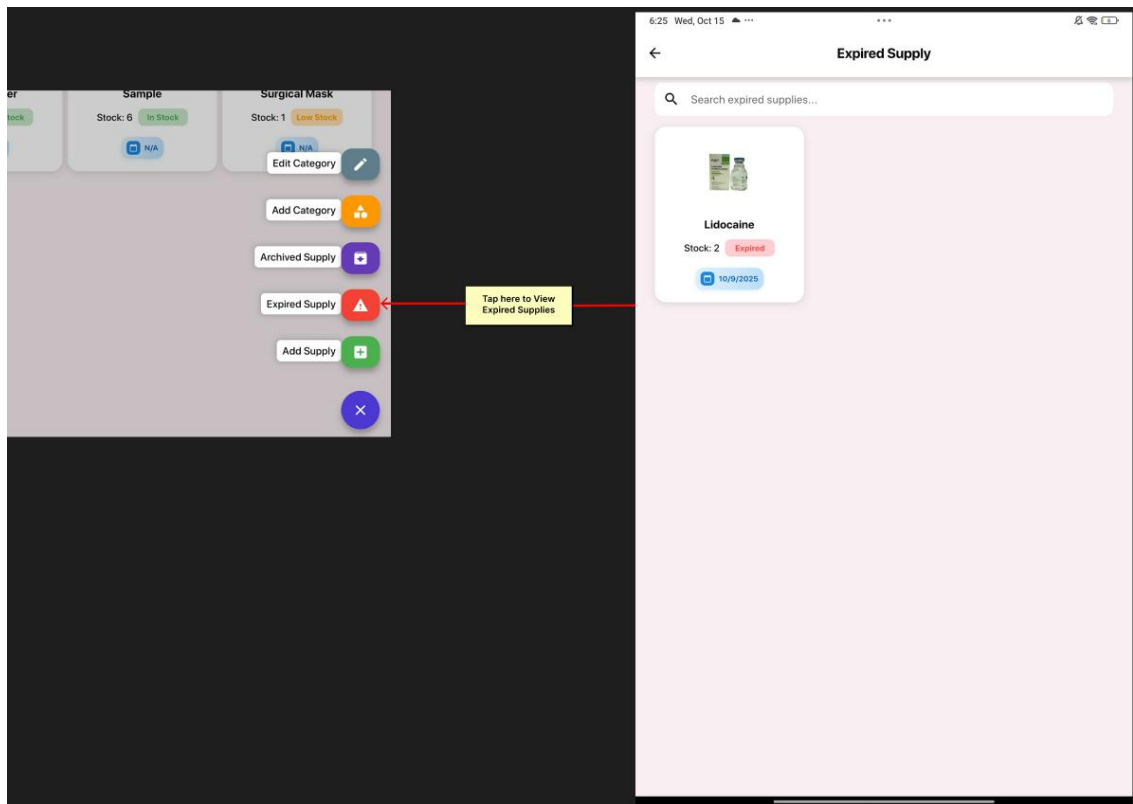
8. How to Manage Archive Supplies

- In the supply details, simply tap the yellow button on the top-right corner of the screen.
- Once archived, it will store that supply in the archived section where you can restore and delete said supply.



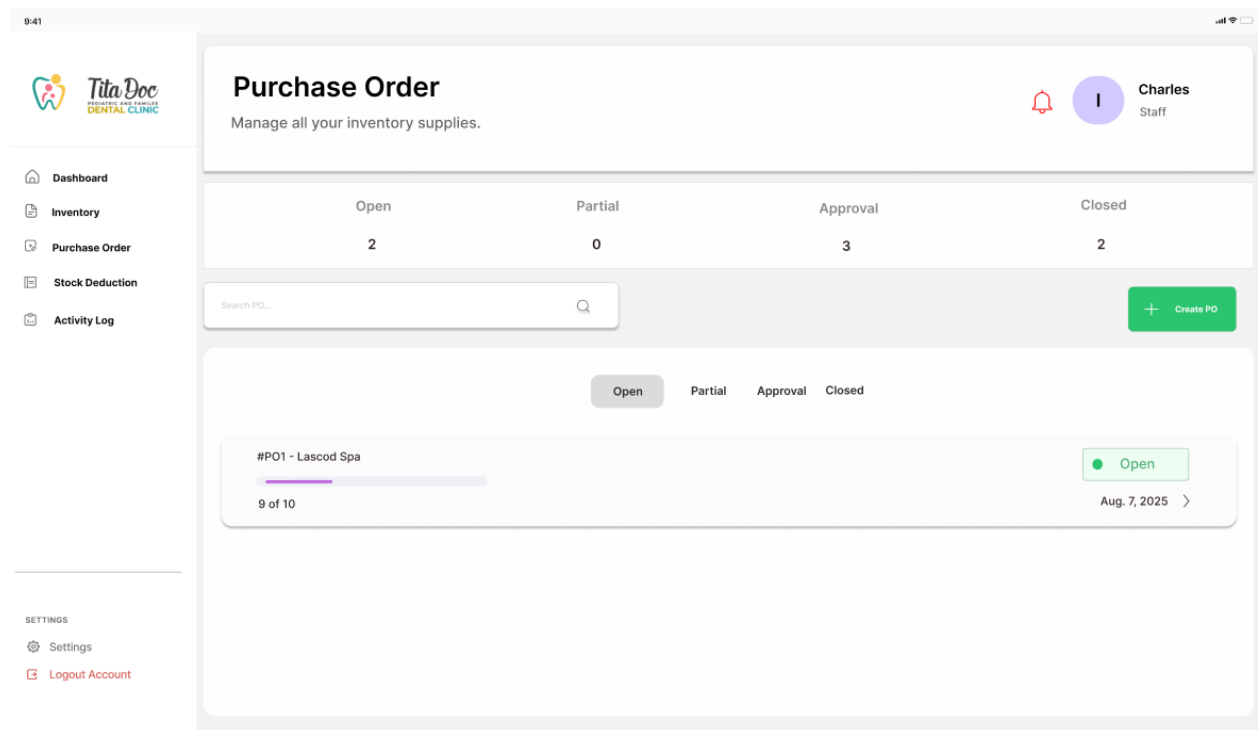
9. Expired Supply

- Navigate to the floating action button then tap expired supplies.
- You can tap a supply card in order to view the details and dispose using the trash button on the top right.



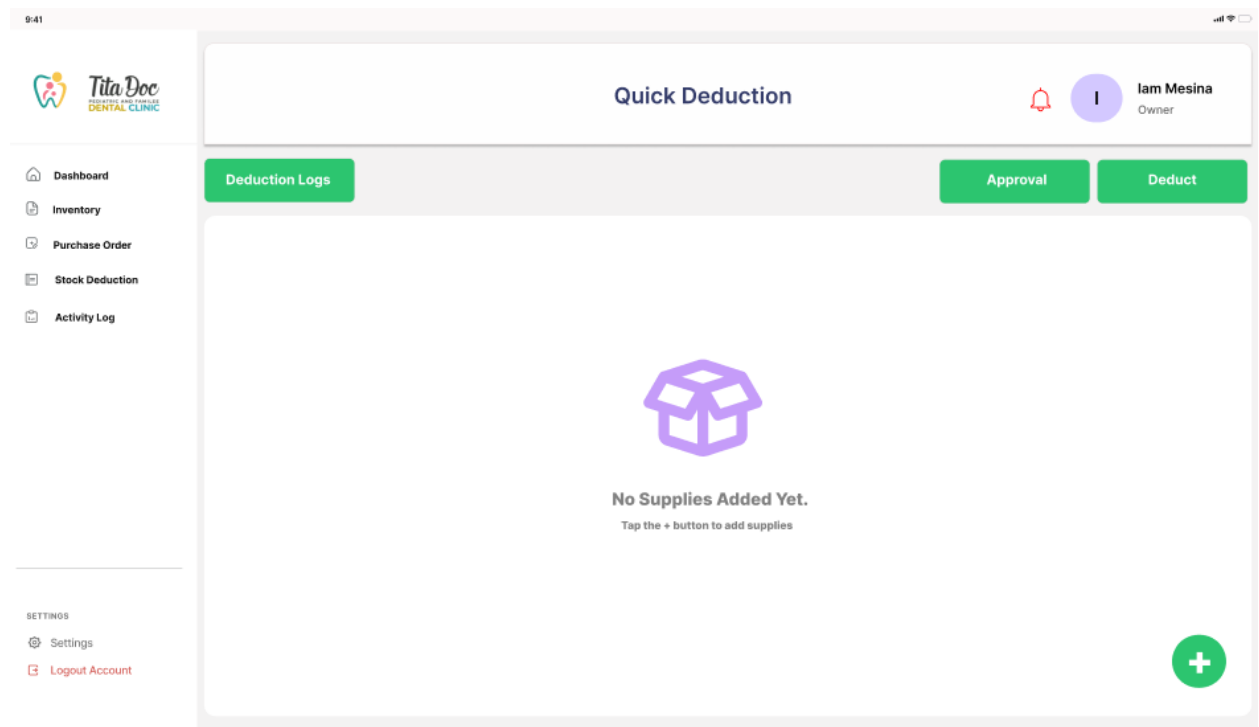
10. Purchase Order

- To create a PO, tap “Create PO” button.
- Add supply then select a variant, enter supply details, and save.
- To receive a PO, open a PO card.
- For an item with pending quantity, tap Partial Receive.
- Modal Quantity Received appears showing: the item, the expiry dates set at PO creation, and numeric fields beside each expiry to enter received quantity per expiry.
- Enter quantities received for the shown expiry dates.
- Then tap confirm.
- If all items are received, tap “Receive” button.
- Enter receipt details, recipient, remarks, and receipt image.
- The admin then approves the PO to update stock levels.



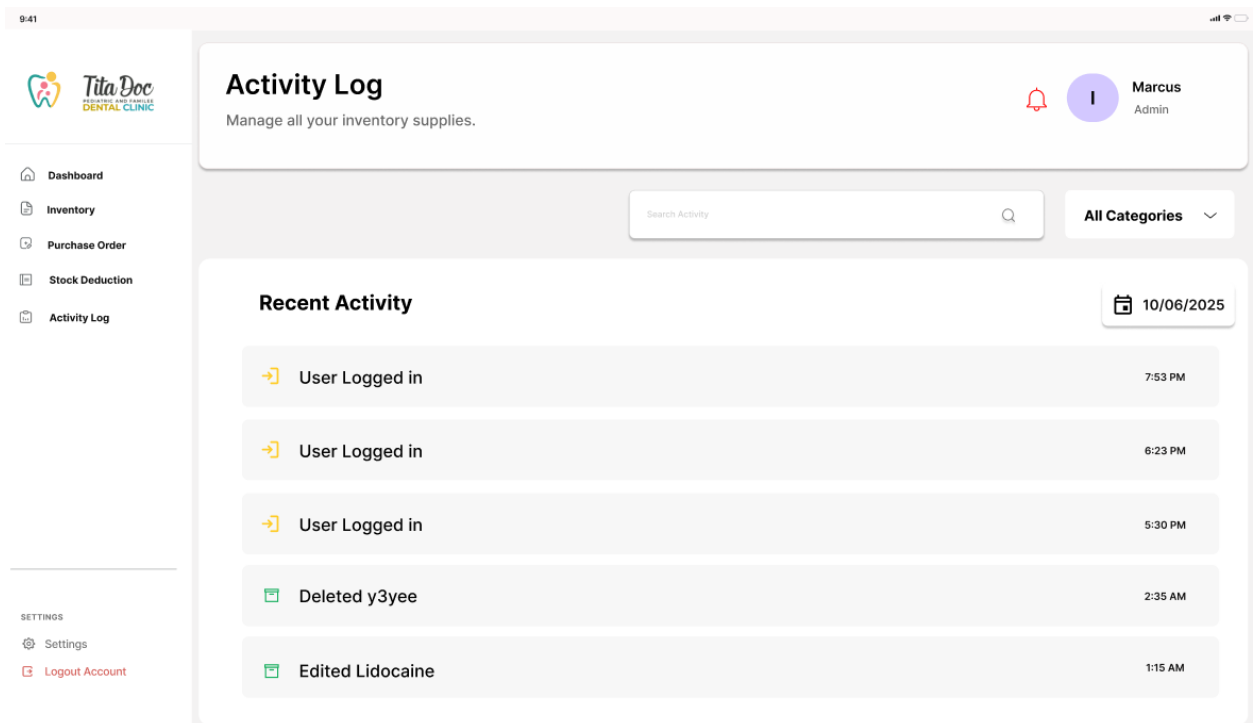
11. How to Deduct Stocks

- To deduct, add a supply using the “+” button.
- Tap a supply card, select a type (if there are any), then specify how many stocks to deduct.
- Select a purpose as to why you’re deducting stocks.
- Then tap deduct.
- Once complete, the admin/owner shall now approve the change.
- You can view all deduction history in the “Deduction Logs” button.



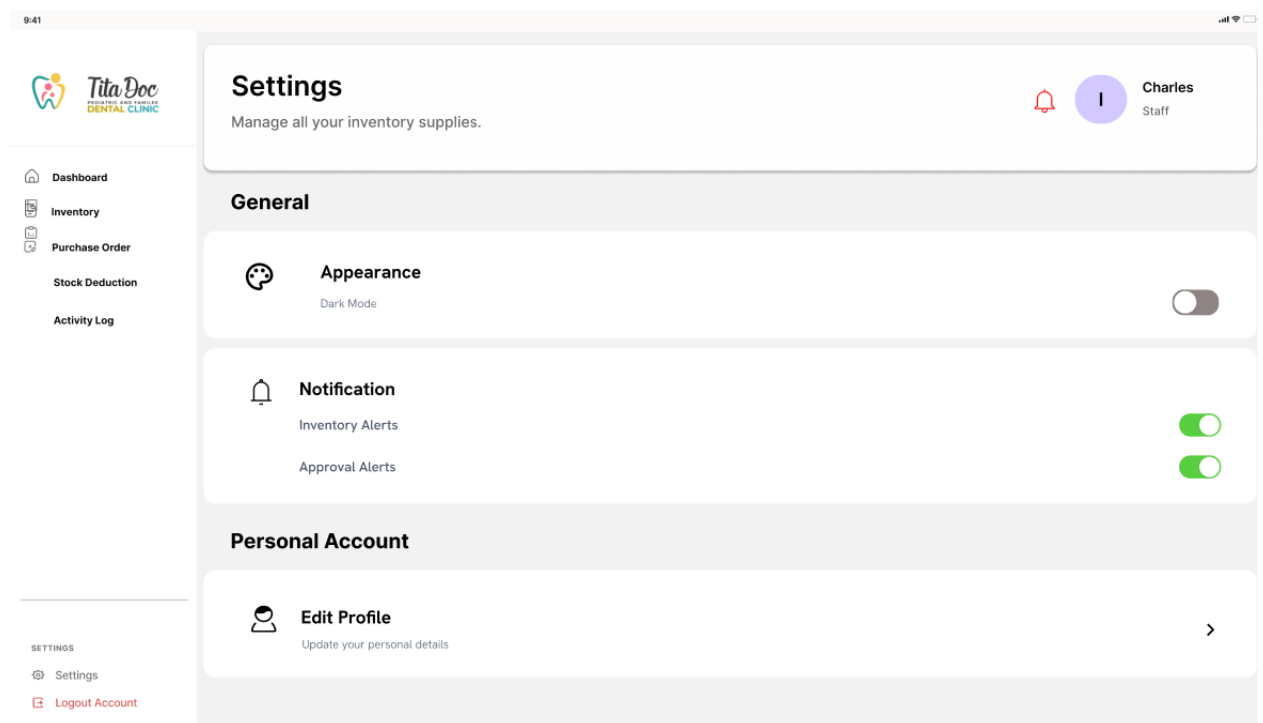
12. Activity Logs Navigation

- To view an activity log, the admin/owner can simply tap an activity card to display activity details.
- You can choose a category to display activities regarding Login, Inventory, Purchase Order, Stock deduction, and settings.
- You can choose dates by tapping the date picker and display all activities within that day.



14. Settings Navigation

- The user can access the general, personal account in this section.
- Only admin and owner accounts can access the employee management.
- Backup and restore is only accessible by the owner.
- In the general settings, you can toggle dark mode and light mode, and notifications such as inventory alerts and approval alerts.
- The user can edit their profile details by editing their names, email, username, and change passwords.
- To add an employee account, navigate to the employee management.
- There, the admin/owner can create an account by tapping the “Add Employee” button.
- Input credentials such as name, email, username, and roles.
- Once done, tap save.
- The admin/owner can also edit these accounts and reset their passwords.
- The backup and restore, the owner can manually backup by tapping the “Manual Backup” and restore using the “Restore latest” buttons.



APPENDIX C. RESOURCE PERSONS

BUSINESS PERMIT



PHOTOS OF TITA DOC PEDIATRIC AND FAMILLEE DENTAL CLINIC INVENTORY











INTERVIEW CONSENT

Capstone Consent Form

Team member names:

Almendaras, John Marcus L.

Ancog, Marvin

Daroy, Charles

Mesina, Michael William

Dear Dr. Lee,

We are conducting an interview as part of our capstone/thesis making for Bachelor of Information Technology at STI College Ortigas-Cainta. The purpose of this interview is to develop a system that will benefit your establishment. We genuinely appreciate your willingness and insights that will contribute to our study.

To ensure accuracy in gathering information, the interview will be audio/video recorded. Gathered data and recordings will be kept confidential and used solely for academic purposes. Your participation is voluntary and you may withdraw at any time without consequences.

Figure 31. Consent Form

Acknowledgement and Consent

I have read the consent form and recognize that my participation in this study is entirely voluntary and that I am free to withdraw at any time during the course of the study without consequence.

I understand that any information resulting from this study will be strictly confidential.

 Client Signature	02/09/25 Date
 Almendares, John Marcus L.	02/08/25 Date
 Ancog, Marvin	02/08/25 Date
 Daroy, Charles	02/08/25 Date
 Mesina, Michael William	02/08/25 Date

Figure 32. Acknowledgement and Consent

ALPHA TEST CONSENT FORMS

Alpha Test Consent Form

Project Title: Android-based Inventory Management System for Tita Doc Pediatric and FamiLee Dental Clinic

Team Members:

Almendares, John Marcus L.

Ancog, Marvin

Daroy, Charles Michael

Mesina, Michael William

School: STI College Ortigas-Cainta

Dear Mr. Ancheta,

We are conducting an alpha test as part of our capstone/thesis making for Bachelor Information Technology at STI College Ortigas-Cainta. The purpose of this testing is to evaluate the functionality, usability, and overall performance of our application before it proceeds to the beta stage or final release. Your participation will help us identify potential issues and improve the system based on actual user feedback. By signing this form, you acknowledge and agree to the following:

1. The information you provide (feedback, error reports, etc.) will be used solely for academic and development purposes.
2. No personally identifiable information will be shared outside the project team without your consent.
3. The system is in an early testing stage; bugs may be evident.
4. With your permission, we would also like to collect a copy of your Curriculum Vitae(CV) for documentation and record-keeping related to our project

Thank you for your time and valuable contribution.

Figure 33. Consent Form Ancheta

Acknowledgement and Consent

I have read the consent form and recognize that my participation in this study is entirely voluntary and that I am free to withdraw at any time during the course of the study without consequence. I understand that any information resulting from this study will be strictly confidential.

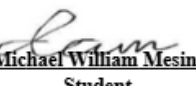
 _____ Kyle Joshua B. Ancheta Tester's Signature	_____ 10/28/2025 Date
 _____ John Marcus L. Almendares Student	_____ 10/28/2025 Date
 _____ Marvin Ancog Student	_____ 10/28/2025 Date
 _____ Charles Michael Daroy Student	_____ 10/28/2025 Date
 _____ Michael William Mesina Student	_____ 10/28/2025 Date

Figure 34. Acknowledgement and Consent Ancheta

Alpha Test Consent Form

Project Title: Android-based Inventory Management System for Tita Doc Pediatric and FamiLee Dental Clinic

Team Members:

Almendares, John Marcus L.

Ancog, Marvin

Daroy, Charles

Mesina, Michael William

School: STI College Ortigas-Cainta

Dear Ms. Navarez,

We are conducting an alpha test as part of our capstone/thesis making for Bachelor Information Technology at STI College Ortigas-Cainta. The purpose of this testing is to evaluate the functionality, usability, and overall performance of our application before it proceeds to the beta stage or final release. Your participation will help us identify potential issues and improve the system based on actual user feedback.

By signing this form, you acknowledge and agree to the following:

1. The information you provide (feedback, error reports, etc.) will be used solely for academic and development purposes.
2. No personally identifiable information will be shared outside the project team without your consent.
3. The system is in an early testing stage; bugs may be evident.
4. With your permission, we would also like to collect a copy of your Curriculum Vitae(CV) for documentation and record-keeping related to our project

Thank you for your time and valuable contribution.

Figure 35. Consent Form Navarez

Acknowledgement and Consent

I have read the consent form and recognize that my participation in this study is entirely voluntary and that I am free to withdraw at any time during the course of the study without consequence. I understand that any information resulting from this study will be strictly confidential.

 _____ Mar Anjolie Navarez Tester's Signature	_____ 10/25/2025 Date
 _____ John Marcus L. Almendares Student	_____ 10/25/2025 Date
 _____ Marvin Ancog Student	_____ 10/25/2025 Date
 _____ Charles Michael Daroy Student	_____ 10/25/2025 Date
 _____ Michael William Mesina Student	_____ 10/25/2025 Date

Figure 36. Acknowledgement and Consent Navarez

Alpha Test Consent Form

Project Title: Android-based Inventory Management System for Tita Doc Pediatric and FamiLee Dental Clinic

Team Members:

Almendaras, John Marcus L.

Ancog, Marvin

Daroy, Charles

Mesina, Michael William

School: STI College Ortigas-Cainta

Dear Mr. Nakpil,

We are conducting an alpha test as part of our capstone/thesis making for Bachelor Information Technology at STI College Ortigas-Cainta. The purpose of this testing is to evaluate the functionality, usability, and overall performance of our application before it proceeds to the beta stage or final release. Your participation will help us identify potential issues and improve the system based on actual user feedback.

By signing this form, you acknowledge and agree to the following:

1. The information you provide (feedback, error reports, etc.) will be used solely for academic and development purposes.
2. No personally identifiable information will be shared outside the project team without your consent.
3. The system is in an early testing stage; bugs may be evident.
4. With your permission, we would also like to collect a copy of your Curriculum Vitae(CV) for documentation and record-keeping related to our project

Thank you for your time and valuable contribution.

Figure 37. Consent Form Nakpil

Acknowledgement and Consent

I have read the consent form and recognize that my participation in this study is entirely voluntary and that I am free to withdraw at any time during the course of the study without consequence. I understand that any information resulting from this study will be strictly confidential.

 Emil Patrick Nakpil Tester's Signature	10/26/2025 Date
 John Marcus L. Almendares Student	10/26/2025 Date
 Marvin Ancog Student	10/26/2025 Date
 Charles Michael Daroy Student	10/26/2025 Date
 Michael William Mesina Student	10/26/2025 Date

Figure 38. Acknowledgement and Consent Nakpil

BETA TEST CONSENT FORM

Beta Test Consent Form

Project Title: Android-based Inventory Management System for Tita Doc Pediatric and FamiLee Dental Clinic

Team Members:

Almendaras, John Marcus L.

Ancog, Marvin

Daroy, Charles

Mesina, Michael William

School: STI College Ortigas-Cainta

Dear Ms. Lee,

We are conducting a beta test as part of our capstone/thesis making for Bachelor Information Technology at STI College Ortigas-Cainta. The purpose of this testing is to evaluate the functionality, usability, and overall performance of our application to cater your business processes. Your participation will help us identify potential issues and improve the system based on actual user feedback. By signing this form, you acknowledge and agree to the following:

1. The information you provide (feedback, error reports, etc.) will be used solely for academic and development purposes.
2. No personally identifiable information will be shared outside the project team without your consent.
3. With your permission, we would also like to collect a copy of your business permit for documentation and record-keeping related to our project
4. The clinic staff will also participate during the testing.

After the test, the application's official deployment at the clinic's environment will not be carried out yet. As such, further developments will be made before the official talks for the deployment.

Thank you for your time and valuable contribution.

Figure 39. Beta Test Consent Form

Acknowledgement and Consent

I have read the consent form and recognize that my participation in this study is entirely voluntary and that I am free to withdraw at any time during the course of the study without consequence. I understand that any information resulting from this study will be strictly confidential.

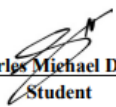
 Jilliane Lee Client Signature	10/31/2025 Date
 John Marcus L. Almendares Student	10/31/2025 Date
 Marvin Ayco Student	10/31/2025 Date
 Charles Michael Daroy Student	10/31/2025 Date
 Michael William Mesina Student	10/31/2025 Date

Figure 40. Acknowledgement and Consent Form

User Acceptance Letter

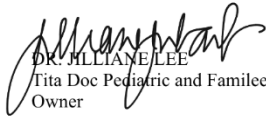
User Acceptance Letter

Tita Doc Pediatric and FamiLee Dental Clinic

989 Banawe St. Brgy. Manresa, Quezon City

October 31, 2025 This is to formally acknowledge that I, JILLIANE LEE, owner of TITA DOC PEDIATRIC AND FAMILLEE DENTAL CLINIC, alongside my staff, KIMBERLY HAJEL ROMERO, and LEIZEL GARCIA, have reviewed and tested the Android-Based Inventory Management System developed by the BSIT students of STI College Ortigas-Cainta.

Following a comprehensive review and testing of the system. I hereby acknowledge that the application conforms to the clinic's overall process. The features that are demonstrated such as real-time inventory tracking, simple purchase order, and stock deduction are expected to improve the clinic's operational efficiency.

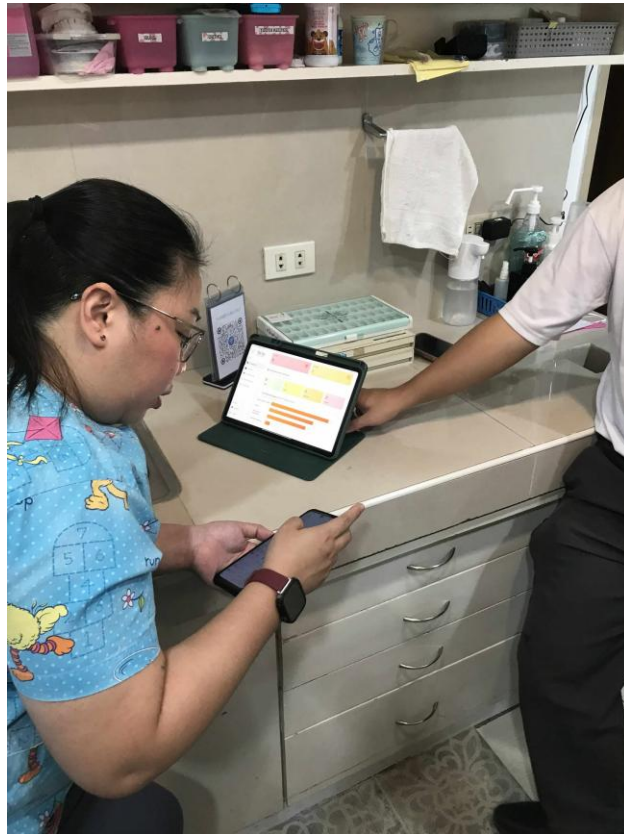

DR. JILLIANE LEE
Tita Doc Pediatric and FamiLee Dental Clinic
Owner


KIMBERLY HAJEL ROMERO
Tita Doc Pediatric and FamiLee Dental Clinic
Staff


LEIZEL GARCIA
Tita Doc Pediatric and FamiLee Dental Clinic
Staff

Figure 41. User Acceptance Letter

BETA TEST PICTURES



ACCOMPLISHMENT FORMS

ACCOMPLISHMENT AND CONSULTATION FORM

INSTRUCTION: List all the activities, improvements, or accomplishments in your Thesis/Capstone Project Documentation and System/Prototype. This form may be reproduced as you go along with your thesis/capstone project. This form should be submitted to your Thesis/Capstone Project Adviser every week.

Thesis/Capstone Project Title: _____
 Week Number: _____

ACTIVITY/ ACCOMPLISHMENT	REMARKS/ COMMENTS/ SUGGESTIONS/ DELIVERABLES and DUE DATE
• Checking of Transcript • Questions for project proposal	- Title proposal (objectives and problems) - The web based app has lesser functions than the app. (still deciding) - Application shall track the fast moving / slow moving stocks.
Prepared by: John Marcus Almendares/02/17/2025 Micheal William Mesina/02/17/2025	Marvin Ancog/02/17/2025 Charles Daroy/02/17/2025
Checked by:  Mr. Christian Sulapas Thesis/Capstone Project Adviser Date Signed: _____	Noted by: Thesis/Capstone Project Coordinator Date Signed: _____

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■■■

ACCOMPLISHMENT AND CONSULTATION FORM | FT-CRD-187-00 | PAGE 1 OF 1

Figure 42.1. Accomplishment and Consultation Form (Week 5)

INSTRUCTION: List all the activities, improvements, or accomplishments in your Thesis/Capstone Project Documentation and System/Prototype. This form may be reproduced as you go along with your thesis/capstone project. This form should be submitted to your Thesis/Capstone Project Adviser every week.

Thesis/Capstone Project Title: _____
 Week Number: 6

ACTIVITY/ ACCOMPLISHMENT	REMARKS/ COMMENTS/ SUGGESTIONS/ DELIVERABLES and DUE DATE
• Consultation for Chapter 1 • Clarification of features and functions	- Project Context must must be 2 pages. - Summary discussion of the the client problem in in proj. context - insert add articles on proj. context and paraphrase. - Reference article used Apa format and 20 articles.
Prepared by: Almandares, John Martinus L. Name of Group Member/Date Masing, Michael William D. Name of Group Member/Date	Ancas, Marvin Name of Group Member/Date Dany, Charles Michael Name of Group Member/Date
Checked by: Name of Thesis/Capstone Project Adviser Thesis/Capstone Project Adviser Date Signed: mm/dd/yyyy	Noted by: Name of Thesis/Capstone Project Coordinator Thesis/Capstone Project Coordinator Date Signed: mm/dd/yyyy

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Figure 42.2 Accomplishment and Consultation Form (Week 6)

ACCOMPLISHMENT AND CONSULTATION FORM

INSTRUCTION: List all the activities, improvements, or accomplishments in your Thesis/Capstone Project Documentation and System/Prototype. This form may be reproduced as you go along with your thesis/capstone project. This form should be submitted to your Thesis/Capstone Project Adviser every week.

Thesis/Capstone Project Title: _____
 Week Number: _____

ACTIVITY/ ACCOMPLISHMENT	REMARKS/ COMMENTS/ SUGGESTIONS/ DELIVERABLES and DUE DATE
- Project Context - Purpose and Description - Scope and Limitations - Partial completion of Chapter 11	(Use the flow) - Use the new document format for thesis * Chapter 1: Intro w/ RRL * Chapter 2: Methodology - Remove User Errors in the limitations
Prepared by: Almendarez, John Marcus L. Name of Group Member/Date Anon, Marvin Name of Group Member/Date	Daroy, Charles Name of Group Member/Date Mesina, Michael William D. Name of Group Member/Date
Checked by: <i>Christina N. Sulayac</i> Name of Thesis/Capstone Project Adviser Thesis/Capstone Project Adviser Date Signed:	Noted by: Name of Thesis/Capstone Project Coordinator Thesis/Capstone Project Coordinator Date Signed:

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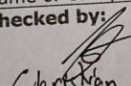
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Figure 42.3 Accomplishment and Consultation Form (Week 8)

ACCOMPLISHMENT AND CONSULTATION FORM

INSTRUCTION: List all the activities, improvements, or accomplishments in your Thesis/Capstone Project Documentation and System/Prototype. This form may be reproduced as you go along with your thesis/capstone project. This form should be submitted to your Thesis/Capstone Project Adviser every week.

Thesis/Capstone Project Title: _____
 Week Number: 9

ACTIVITY/ ACCOMPLISHMENT	REMARKS/ COMMENTS/ SUGGESTIONS/ DELIVERABLES and DUE DATE
- Title - Design Overview - Confirmation of ^{new} Capstone format	
Prepared by: Almenares, John Marcus L. March 20, 2025 Name of Group Member/Date Anon, Marvin March 20, 2025 Name of Group Member/Date	Daray, Charles March 20, 2025 Name of Group Member/Date Mesinas, Michael William March 20, 2025 Name of Group Member/Date
Checked by:  Christman Sulapas Thesis/Capstone Project Adviser Date Signed:	Noted by: Thesis/Capstone Project Coordinator Date Signed:

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Figure 42.4 Accomplishment and Consultation Form (Week 9)

ACCOMPLISHMENT AND CONSULTATION FORM

INSTRUCTION: List all the activities, improvements, or accomplishments in your Thesis/Capstone Project Documentation and System/Prototype. This form may be reproduced as you go along with your thesis/capstone project. This form should be submitted to your Thesis/Capstone Project Adviser every week.

Thesis/Capstone Project Title:
Week Number: 11

ACTIVITY/ ACCOMPLISHMENT	REMARKS/ COMMENTS/ SUGGESTIONS/ DELIVERABLES and DUE DATE
checking of PRs Questions for Diagrams to make Clarification of Chapter 2	- Explain the methodology that you use.
Prepared by: Name of Group Member/Date Almendares, Marcus Name of Group Member/Date Anco, Marvin	Name of Group Member/Date Mesina, Michael Name of Group Member/Date William
Checked by: Sulapas, Christian (9-25) Name of Thesis/Capstone Project Adviser Thesis/Capstone Project Adviser Date Signed: mm/dd/yyyy	Noted by: Name of Thesis/Capstone Project Coordinator Thesis/Capstone Project Coordinator Date Signed: mm/dd/yyyy

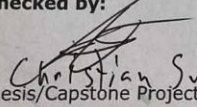
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Figure 42.5 Accomplishment and Consultation Form (Week 11)

ACCOMPLISHMENT AND CONSULTATION FORM

INSTRUCTION: List all the activities, improvements, or accomplishments in your Thesis/Capstone Project Documentation and System/Prototype. This form may be reproduced as you go along with your thesis/capstone project. This form should be submitted to your Thesis/Capstone Project Adviser every week.

Thesis/Capstone Project Title: Inventory and Appointment Management Application
Week Number: 14

ACTIVITY/ ACCOMPLISHMENT	REMARKS/ COMMENTS/ SUGGESTIONS/ DELIVERABLES and DUE DATE
• Revision on project context and purpose and description. • Added features on the scope • 2 Local sources added on the RRLS • Extended waterfall methodology • Requirements Analysis and documentation • Flowcharts • Use-case diagram • Updated prototype design	<i>(edit the</i> <i>- Remove the flow that</i> <i>has the decision for</i> <i>stock Stock level</i>
Prepared by: Almendares, John Marcus L. Name of Group Member/Date Ancog, Marvin Name of Group Member/Date	Daroy, Charles Name of Group Member/Date Mesina, Michael William Name of Group Member/Date
Checked by:  Christian Sulayman Thesis/Capstone Project Adviser Date Signed:	Noted by: Thesis/Capstone Project Coordinator Date Signed:

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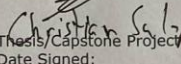
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Figure 42.6 Accomplishment and Consultation Form (Week 14)

ACCOMPLISHMENT AND CONSULTATION FORM

INSTRUCTION: List all the activities, improvements, or accomplishments in your Thesis/Capstone Project Documentation and System/Prototype. This form may be reproduced as you go along with your thesis/capstone project. This form should be submitted to your Thesis/Capstone Project Adviser every week.

Thesis/Capstone Project Title:
Week Number: 17

ACTIVITY/ ACCOMPLISHMENT	REMARKS/ COMMENTS/ SUGGESTIONS/ DELIVERABLES and DUE DATE
• Overall System Flowcharts • Use-case Diagrams (Admin & Staff) • Application Design Checking • Requirements Finalization	
Prepared by: Almendares, John Marcus L. / 05/13/25 Ancog, Marvin / 05/13/25 Checked by:  Christen Selvas Thesis/Capstone Project Adviser Date Signed:	Daroy, Charles Michael / 05/13/25 Mesina, Michael William / 05/13/25 Noted by: Thesis/Capstone Project Coordinator Date Signed:

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Figure 42.7 Accomplishment and Consultation Form (Week 17)

ACCOMPLISHMENT AND CONSULTATION FORM

INSTRUCTION: List all the activities, improvements, or accomplishments in your Thesis/Capstone Project Documentation and System/Prototype. This form may be reproduced as you go along with your thesis/capstone project. This form should be submitted to your Thesis/Capstone Project Adviser every week.

Thesis/Capstone Project Title:
Week Number: 17

ACTIVITY/ ACCOMPLISHMENT	REMARKS/ COMMENTS/ SUGGESTIONS/ DELIVERABLES and DUE DATE
• Data Flow Diagrams • Application Design • Revisions on Scope (Added archive section) • Appendices • Title Finalization - A Mobile Inventory and Procedure Management Application with Analytics for Tita Doc Pediatric and Familee Dental Clinic	
Prepared by: Almendares, John Marcus L. / 05/16/25 Ancog, Marvin / 05/16/25	Daroy, Charles Michael / 05/16/25 Mesina, Michael William / 05/16/25
Checked by: Thesis/Capstone Project Adviser Date Signed:	Noted by: Thesis/Capstone Project Coordinator Date Signed:

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Figure 42.8. Accomplishment and Consultation Form (Week 17)

ACCOMPLISHMENT AND CONSULTATION FORM

INSTRUCTION: List all the activities, improvements, or accomplishments in your Capstone Project Documentation and System/Prototype. This form may be reproduced and go along with your thesis/capstone project. This form should be submitted to your Thesis/Capstone Project Adviser every week.

Thesis/Capstone Project Title: _____
 Week Number: 18

ACTIVITY/ ACCOMPLISHMENT	REMARKS/ COMMENTS/ SUGGESTIONS/ DELIVERABLES and DUE DATE
Progress Report	- Item name should be alphanumeric - (Inventory) add batch number on the inventory - Remove Status: Low Stock from Sort by - Add function to search batch number number.
Prepared by: Almendares, John Marcus L. 08/05/2025 Ancog, Marvin 08/05/2025	Daroy, Charles Michael 08/05/2025 Mesina, Michael William 08/05/2025
Checked by:  Christian Sulapas Thesis/Capstone Project Adviser Date Signed: 08/05/2025	Noted by: Thesis/Capstone Project Coordinator Date Signed: 08/05/2025

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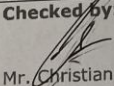
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Figure 42.9. Accomplishment and Consultation Form (Week 18)

ACCOMPLISHMENT AND CONSULTATION FORM

INSTRUCTION: List all the activities, improvements, or accomplishments in your Thesis/Capstone Project Documentation and System/Prototype. This form may be reproduced as you go along with your thesis/capstone project. This form should be submitted to your Thesis/Capstone Project Adviser every week.

Thesis/Capstone Project Title:
Week Number: 2

ACTIVITY/ ACCOMPLISHMENT	REMARKS/ COMMENTS/ SUGGESTIONS/ DELIVERABLES and DUE DATE
• Purchase Order Section • Bug fixes • Inventory Revisions	- Batch num Include on Inventory
Prepared by: Almendaras, John Marcus L. 08/15/2025	Daroy, Charles Michael 08/15/2025
Ancog, Marvin 08/15/2025	Mesina, Michael William 08/15/2025
Checked by:  Mr. Christian Sulapas Thesis/Capstone Project Adviser Date Signed: 08/15/2025	Noted by: Mr. Ron Allan Salem Thesis/Capstone Project Coordinator Date Signed: 08/15/2025

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Figure 42.10. Accomplishment and Consultation Form (Week 2)

ACCOMPLISHMENT AND CONSULTATION FORM

INSTRUCTION: List all the activities, improvements, or accomplishments in your Thesis/Capstone Project Documentation and System/Prototype. This form may be reproduced as you go along with your thesis/capstone project. This form should be submitted to your Thesis/Capstone Project Adviser every week.

Thesis/Capstone Project Title:
Week Number: 3

ACTIVITY / ACCOMPLISHMENT	REMARKS / COMMENTS / SUGGESTIONS / DELIVERABLES and DUE DATE
• Purchase Order Revisions ✓ • Stock Deduction section	- To Clear the functionality of Stock deduction.
Prepared by: Almendaras, John Marcus L. 08/22/2025 Ancog, Marvin 08/22/2025	Daroy, Charles Michael 08/22/2025 Mesina, Michael William 08/22/2025
Checked by: Mr. Christian Sulapas Thesis/Capstone Project Adviser Date Signed: 08/22/2025	Noted by: Mr. Ron Allan Salem Thesis/Capstone Project Coordinator Date Signed: 08/22/2025

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Figure 42.11. Accomplishment and Consultation Form (Week 3)

ACCOMPLISHMENT AND CONSULTATION FORM

INSTRUCTION: List all the activities, improvements, or accomplishments in your Thesis/Capstone Project Documentation and System/Prototype. This form may be reproduced as you go along with your thesis/capstone project. This form should be submitted to your Thesis/Capstone Project Adviser every week.

Thesis/Capstone Project Title:
Week Number: 7

ACTIVITY/ ACCOMPLISHMENT	REMARKS/ COMMENTS/ SUGGESTIONS/ DELIVERABLES and DUE DATE
- Activity Logs section - Notifications	- Stock Reverted notif
Prepared by: Almendares, John Marcus / 09/12/2025 Ancog, Marvin / 09/12/2025	Daroy, Charles Michael / 09/12/2025 Mesina, Michael William / 09/12/2025
Checked by:  Mr. Christian Sulapas Thesis/Capstone Project Adviser Date Signed: 09/12/2025	Noted by: Mr. Ron Salem Thesis/Capstone Project Coordinator Date Signed: 09/12/2025

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Figure 42.12. Accomplishment and Consultation Form (Week 7)

ACCOMPLISHMENT AND CONSULTATION FORM

INSTRUCTION: List all the activities, improvements, or accomplishments in your Thesis/Capstone Project Documentation and System/Prototype. This form may be reproduced as you go along with your thesis/capstone project. This form should be submitted to your Thesis/Capstone Project Adviser every week.

Thesis/Capstone Project Title:
Week Number: 10

ACTIVITY / ACCOMPLISHMENT	REMARKS/ COMMENTS/ SUGGESTIONS/ DELIVERABLES and DUE DATE
• Purchase Order Updates • Settings • Backup and Restore • Employee management • Change Password	
Prepared by: Almendares, John Marcus L. 09/26/2025 Ancog, Marvin 09/26/2025	Daroy, Charles Michael 09/26/2025 Mesina, Michael William 09/26/2025
Checked by: Mr. Christian Sulapas Thesis/Capstone Project Adviser Date Signed: 09/26/2025	Noted by: Mr. Ron Allan Salem Thesis/Capstone Project Coordinator Date Signed: 09/26/2025

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Figure 42.13. Accomplishment and Consultation Form (Week 10)

ACCOMPLISHMENT AND CONSULTATION FORM

INSTRUCTION: List all the activities, improvements, or accomplishments in your Thesis/Capstone Project Documentation and System/Prototype. This form may be reproduced as you go along with your thesis/capstone project. This form should be submitted to your Thesis/Capstone Project Adviser every week.

Thesis/Capstone Project Title:
Week Number: 13

ACTIVITY / ACCOMPLISHMENT	REMARKS / COMMENTS / SUGGESTIONS / DELIVERABLES and DUE DATE
• Tablet View • Alpha Testing Results • Automatic Backup and Restore • New Forgot Password Process • Updated Purchase Order • Beta test questions • User Acceptance Letter • Development (Paper) • Implementation Plan • 50% Conclusion • Role-based access	
Prepared by: Almendares, John Marcus L. 10/14/2025 Ancog, Marvin 10/14/2025	Daroy, Charles Michael 10/14/2025 Mesina, Michael William 10/14/2025
Checked by: Mr. Christian Sulapas Thesis/Capstone Project Adviser Date Signed: 10/14/2025	Noted by: Mr. Ron Allan Salem Thesis/Capstone Project Coordinator Date Signed: 10/14/2025

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Figure 42.14. Accomplishment and Consultation Form (Week 13)

APPENDIX D. ALPHA TESTER CURRICULUM VITAE

Ckyle Joshua B. Ancheta

BA Analyst I - SAP ABAP Developer

📍 Cainta, Rizal, Philippines

🌐 [linkedin.com/in/ckyle-joshua-ancheta](https://www.linkedin.com/in/ckyle-joshua-ancheta)

☎ 0976-309-6169

✉ ckylejoshuaancheta03@gmail.com

Work Experience

BA Analyst I - SAP ABAP

Robinsons Land Corporation

August 2023 – Present

- Created, built, maintained, and improved custom SAP applications for the organization, such as reports (ALV, Classical), SmartForms, Adobe Forms, and SapScripts.
- Worked with functional teams (MM, RE, FI) to turn business needs into technical solutions.
- Migrated and optimized legacy ECC ABAP custom programs to SAP S/4HANA.

Sales Department - IT Intern

Topbliss Solutions Inc.

April 2023 – June 2023

- Support the other team members and sales agents with technical assistance.
- Assist in addressing technical issues, ensuring the smooth running of computer systems, and diagnosing software and hardware problems.
- Contribute in the creation and maintenance of sales-related documents, such as presentations for product training and documentation of the sales process.

IT Technical Support Associate

WNO Electronics Trading

February 2022 – August 2022

- Responding to customer inquiries and requests for technical support via phone, email, chat, or in-person.
- Provided prompt and effective responses to client inquiries related to product functionality, troubleshooting, and best practices.

Core Skills

- **SAP ABAP:** Reports (ALV, Classical), SmartForms, Adobe Forms, SAPscript.
- **RICEFW Development:** Reports, Interfaces, Conversions, Enhancements, Forms, Workflows
- **S/4HANA Migration:** Custom code remediation, ECC object conversion.
- **Integration:** IDoc, BAPI, BADI
- **Tools & Utilities:** ABAP Workbench, Debugging, Performance Tuning, ABAP Test Cockpit (ATC)
- **SAP Modules Exposure:** MM, SD, FI, RE
- **Collaboration:** Agile/Scrum, functional-technical coordination, documentation

Education

Bachelor of Science in Information Technology

STI College Ortigas-Cainta

June 2018 – August 2023

Honors

Graduated Cum Laude

Certificates

Certificate of Completion

Six Sigma White Belt Certification May 2023

Certificate of Completion

JavaScript Essentials June 2022

ICT - Programming

Informatics College Cainta-Senior High School

June 2016 – March 2018

Achievements

- Best in Media Arts

Certificate of Completion

Web Development Basics June 2022

Certificate of Completion

AWS Siklab Pilipinas Cloud Computing Boot Camp 2019

CONTACT

anorienavarez@gmail.com
[Mar Anjorie Navarez | LinkedIn](#)
+639061692477
Antipolo City, Rizal

EDUCATION

BACHELOR OF SCIENCE IN Information Technology 2019 - Technological Institute of the Philippines – Quezon City

SKILLS

Technical Skills:

- Programming
- Testing
- Software Development
- Tools/Technologies:
 - Java, Springboot (Experienced)
 - HTML5, CSS3, JS, Bootstrap (Experienced)
 - Typescript, ReactJS (Basic)
 - Cloud Computing – Azure, AWS (Basic)
 - Docker Containers (Basic)
 - Databases - MSSQL, Oracle, MySQL, PostgreSQL (Basic)
 - IDE - Android Studio, Swift, Eclipse, Visual Studio (Basic)
 - Version Control – Git, Github (Basic)
 - Microsoft Office Suite

Soft Skills:

- Leadership
- Team works and collaboration
- People Management
- Creative
- Public Speaking
- Adaptability

CERTIFICATIONS

- AZ-900: Azure Fundamentals
- AI-900: Azure AI Fundamentals
- 1Z0-122-25: Oracle Cloud Infrastructure 2025 AI Foundations Associate

CONFERENCES/SEMINARS

- Childhope Philippines – TechTalk: Digital Safety (Speaker)
- STI College Ortigas-Cainta (ICT Week) – TechTalk: Digital Safety (Speaker)

Mar Anjorie Navarez



APPLICATION SYSTEMS ENGINEER

Application Systems Engineer proficient in software development, specializing in requirement analysis, designing, developing, testing and maintaining scalable applications. Currently expanding skills in full-stack development through an ongoing project. Co-pillar lead of the company's community organization in aiming to give impactful initiatives and actively participating in engagement activities such as event hosting, IT webinars as a speaker and program committee roles. Dedicated to apply my expertise in software engineering and strong communication skills to contribute innovative solutions and collaborate effectively within development teams.

WORK EXPERIENCE

APPLICATION SYSTEMS ENGINEER

Fujitsu WeServ Systems International Inc.
Taguig BGC, Philippines (Oct 2021 - Present)

- Collaborated in a 5-member team to develop a new pages and features to web application for a Japanese car company using JavaServer Pages.
- Authored and designed technical documents including system specification for a Japanese financial company.
- Developed and tested a banking web application using Java Springboot.
- Designed and tested an implemented file download automation system in C# Windows Application forms.
- Troubleshoot and resolve technical issues and improved tasks impediment.
- Provided project updates and status report to Japanese stakeholders through calls and emails.
- Worked independently and in close collaboration with senior engineers to deliver project milestones.

IT INSTRUCTOR (PART TIME)

STI College – Ortigas-Cainta
Ortigas Ave Ext, Cainta, Rizal (Aug 2024 – August 2025)

- Create a positive and supportive classroom environment.
- Deliver clear and effective instruction to students.
- Assess student progress through examination and other forms of evaluation.
- Communicate regularly to student about their performance.
- Instructing computer-related courses to tertiary students with demand of research and building industry-related project by the end of semester.

PROJECT-BASED FRONTEND DEVELOPER

Techfluent Corporation (Apr 2021 – Oct 2021)

- Responsible in developing concepts, graphic layouts for product illustrations, banners, company logo mockups and websites.
- Enhanced company website application for maximum speed and scalability.
- Maintained and improved client's e-commerce web application.

FRONTEND DEVELOPER (INTERNSHIP)

Philippine Government Electronic Procurement Services
Ortigas Center Pasig City (Dec 2018 - Mar 2019)

- Responsible in organizing an order web application workflow.
- Developed new user interface features using modern framework Bootstrap.
- Collaborated with backend developers to improved application usability.
- Ensures the technical feasibility of UI/UX designs.

EMIL PATRICK NAKPIL

Solution Engineering Manager, Asia

+639778505693 | enakpil@markem-imaje.com | www.linkedin.com/in/emil-patrick-nakpil | Pasig City

SUMMARY

Passionate Engineer with 14+ years professional experience. A leader with a strong technical foundation in hardware and software delivery. Skilled in project management, team leadership, and process optimization, with a proven ability to bridge technical solutions and business goals. Adept at mentoring teams, driving operational excellence, and delivering customer-focused results.

PROFESSIONAL EXPERIENCE

Solution Engineering Manager, Asia

2022 - Present

[Markem-Imaje](#)

Makati, Philippines

- Lead the Solution Engineering team in Asia, supporting global customers through effective planning, quality delivery, and on-time execution of software solutions.
- Oversee project activities to ensure technical excellence, alignment with customer requirements, and adherence to standards.
- Drive team development by enhancing technical competencies through projects, stretch assignments, and continuous learning.
- Foster collaboration within the team and with internal and external stakeholders to strengthen delivery and customer satisfaction.
- Successful implementation of cross-border configurations allowing Solution Engineers in Asia to configure software solutions for Europe and US.

Solution Engineer

2018 - 2022

[Markem-Imaje](#)

Makati, Philippines

- Configure and implement software solutions tailored to customer needs in the manufacturing industry, ensuring seamless integration with existing production systems.
- Collaborate closely with key customer stakeholders to analyze business requirements, identify process improvements, and deliver value-added solutions that enhance production efficiency.
- Leverage technical expertise in Markem-Imaje printers and in-depth understanding of manufacturing workflows to design robust and reliable configurations.
- Lead end-to-end project activities including data and requirements gathering, solution design, documentation, software configuration, testing, deployment, and user training.
- Ensure solutions meet quality standards, are delivered within scope and timeline, and support customer success through continuous improvement initiatives.

Presales Engineer

2016 - 2018

[Markem-Imaje](#)

Makati, Philippines

- Supported the growth of Markem-Imaje's business and presence by providing technical expertise, consultation, and solutions to partners and customers in the manufacturing sector.
- Collaborated closely with internal teams and partners to understand customer needs and deliver value-added solutions that enhanced production efficiency and reliability.
- Ensured partners were fully equipped to meet customer demands through product training, joint on-site visits, project planning, and implementation support.
- Acted as a key technical advisor, bridging customer requirements with appropriate solutions to drive satisfaction and strengthen long-term partnerships.



Software & Systems:

CoLOS (In-house app), MSSQL & Reporting Services, .NET Technologies, Automation Systems, Postman, SOAPUI, OPC, Power BI, JIRA, Salesforce

Hardware Expertise:

Marking & Coding Equipment, PLC, Manufacturing equipments.

Other:

System Integration, Documentation, Network Infrastructure.

Core Competencies:

Leadership & Team Development | Project Management | Communication | Problem Solving | Customer Focus | Continuous Improvement

Technical Sales Manager

2010 - 2015

PIC Enterprises Inc.

Pasig, Philippines

- Performed pre-installation, installation, commissioning, and end-user training of industrial printers within defined project timelines.
- Delivered high-quality technical support, ensuring timely resolution of customer issues and maintaining strong client relationships.
- Conducted new machine and refresher trainings for customers, enhancing equipment knowledge and operational efficiency.
- Supported technical and sales teams through training, tool preparation, and solution strategy development.
- Participated in client meetings and project biddings to recommend suitable printing technologies based on requirements.
- Managed the local repair center for two years, overseeing repairs and training repair engineers.
- Facilitated numerous customer training sessions, including large-scale programs for multinational companies.

EDUCATION

Bachelor of Science in Electronics and Communications Engineering

2004 - 2009

University of the East

Manila, Philippines

Electronics and Communications Engineering board passer

2009

Professional Regulation Commission

Manila, Philippines

APPENDIX E. PERSONAL TECHNICAL VITAE

Curriculum Vitae of

John Marcus L. Almendares

Marikina City

almendares.291447@ortigas-cainta.sti.edu.ph

09672743834

EDUCATIONAL BACKGROUND

Level	Inclusive Dates	Name of school/ Institution
Tertiary	August 2022 - Present	STI Ortigas-Cainta
Senior High School	August 2020 - June 2022	T.I.P Quezon City
High School	June 2016 - May 2020	Sta. Elena High School
Elementary	June 2010 - May 2016	Parang Elementary School

PROFESSIONAL OR VOLUNTEER EXPERIENCE

Inclusive Dates	Nature of Experience/ Job Title	Name and Address of Company or Organization
N/A	N/A	N/A

AFFILIATIONS

Inclusive Dates	Name of Organization	Position
January – May 2024	ALPHA	Documentation

SKILLS

SKILLS	Level of Competency	Date Acquired
Visual Paradigm	Intermediate	May 2025
Cybersecurity Knowledge	Intermediate	January 2025
Web-based programming (HTML, CSS, JavaScript)	Beginner	January 2025
Android Studio	Beginner	January 2025
Systems Administration	Intermediate	July 2023
Ubuntu (Bash)	Beginner	May 2023
Figma	Intermediate	September 2022
Programming Languages (Python, Java, C#)	Beginner	August 2022
Research-oriented skills	Intermediate	January 2021
Canva	Intermediate	August 2020
Microsoft Office Suite (Word, Excel, PowerPoint)	Intermediate	June 2016

TRAININGS, SEMINARS, OR WORKSHOPS ATTENDED

Inclusive Dates	Title of Training, Seminar, or Workshop
May 2024	ALPHA: DICT Seminar
May 2021	Mind Be Kind: A Stress Management Webinar for SHS grade 11
Apr 2021	HS-Career Orientation: Understanding my Career “Tips to Jumpstart your Career”

Curriculum Vitae of
Marvin B. Ancog
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09924082214

EDUCATIONAL BACKGROUND

Level	Inclusive Dates	Name of school/ Institution
Tertiary	August 2022 - Present	STI Ortigas-Cainta
Senior High School	August 2020 - June 2022	STI Ortigas-Cainta
High School	June 2016 - May 2020	Sta. Lucia High SchoolSan Juan National Highsc
Elementary	June 2010 - May 2016	Kabisig Elementary School

PROFESSIONAL OR VOLUNTEER EXPERIENCE

Inclusive Dates	Nature of Experience/ Job Title	Name and Address of Company or Organization
March 2021 - Present	Photographer/ Production Assistant	Creative Sequence

AFFILIATIONS

Inclusive Dates	Name of Organization	Position
August 2021 - Present	Aperture Media Club	Photographer

SKILLS

SKILLS	Level of Competency	Date Acquired
Research-oriented skills	Intermediate	January 2024
Programming Languages (Python, Java, C#)	Beginner	August 2023
Microsoft Office Suite (Word, Excel, PowerPoint)	Beginner	Jun 2023
Canva	Intermediate	August 2023
Figma	Intermediate	September 2023
Photography & Videography	Intermediate	January 2022

TRAININGS, SEMINARS, OR WORKSHOPS ATTENDED

Inclusive Dates	Title of Training, Seminar, or Workshop
May 2024	ALPHA Seminar: Cybersecurity

Curriculum Vitae of

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EDUCATIONAL BACKGROUND

Level	Inclusive Dates	Name of school/ Institution
Tertiary	August 2022 - Present	STI Ortigas-Cainta
Senior High School	August 2020 - June 2022	STI Ortigas-Cainta
High School	June 2016 - May 2020	Rich Golden Shower Montessori Center
Elementary	June 2010 - May 2016	San Beda University Rizal

PROFESSIONAL OR VOLUNTEER EXPERIENCE

Inclusive Dates	Nature of Experience/ Job Title	Name and Address of Company or Organization
N/A	N/A	N/A

AFFILIATIONS

Inclusive Dates	Name of Organization	Position
2018 - 2020	RGSMC Dance Troupe	Senior Member
2017 - 2020	RGSMC Taekwondo	Varsity Player
2012 - 2017	San Beda Red Jins	Varsity Player

SKILLS

SKILLS	Level of Competency	Date Acquired
Android Studio	Beginner	January 2025
Cybersecurity	Beginner	January 2025
Web-based programming (HTML, CSS, JavaScript)	Beginner	January 2025
Figma	Intermediate	April 2023
Ubuntu (Bash)	Beginner	May 2023
Systems Administration	Intermediate	July 2023
Programming Languages (Python, Java, C#)	Intermediate	August 2022
Unity	Beginner	June 2022
Microsoft Office Suite (Word, Excel, PowerPoint)	Intermediate	June 2017

TRAININGS, SEMINARS, OR WORKSHOPS ATTENDED

Inclusive Dates	Title of Training, Seminar, or Workshop
N/A	N/A

Curriculum Vitae of
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EDUCATIONAL BACKGROUND

Level	Inclusive Dates	Name of school/ Institution
Tertiary	August 2022 - Present	STI Ortigas-Cainta
Senior High School	August 2020 - Jun 2022	Harris Memorial College
High School	Jun 2016 - May 2020	Casimiro A. Ynares Sr. Memorial National High School
Elementary	Jun 2010 - May 2016	JEMS Fruit of Learning School

PROFESSIONAL OR VOLUNTEER EXPERIENCE

Inclusive Dates	Nature of Experience/ Job Title	Name and Address of Company or Organization
N/A	N/A	N/A

AFFILIATIONS

Inclusive Dates	Name of Organization	Position
N/A	N/A	N/A

SKILLS

SKILLS	Level of Competency	Date Acquired
Web-based programming (HTML, CSS, JavaScript)	Beginner	January 2025
Android Studio	Intermediate	January 2025
Flutter (Dart)	Intermediate	January 2025
Figma	Intermediate	November 2023
Systems Administration	Intermediate	July 2023
Ubuntu (Bash)	Beginner	May 2023
Microsoft Office (Word, Excel, PowerPoint)	Intermediate	June 2016

TRAININGS, SEMINARS, OR WORKSHOPS ATTENDED

Inclusive Dates	Title of Training, Seminar, or Workshop
May 2024	ALPHA Seminar: Cybersecurity